

THE CAFFARELLI-KOHN-NIRENBERG PARTIAL REGULARITY THEOREM: A SELF-CONTAINED PROOF WRITTEN FOR FORMALIZATION

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ABSTRACT. We give a complete and self-contained proof of the Caffarelli-Kohn-Nirenberg partial regularity theorem for suitable weak solutions of the three-dimensional incompressible Navier-Stokes equations with a force $f \in L^q$, $q > 5/2$: the singular set has vanishing one-dimensional parabolic Hausdorff measure. The exposition is written so that each statement can be transcribed into a single formal declaration. Every constant is named and its dependence is displayed, every cut-off function is written out, every passage to a limit is justified, and the external analytic inputs are isolated in an appendix with exact statements.

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INTRODUCTION

Leray's weak solutions of the incompressible Navier-Stokes equations in three space dimensions are global in time but are not known to be regular or unique. Partial regularity theory replaces the (open) question of full regularity by a quantitative bound on the size of the hypothetical singular set. Scheffer [16] initiated this programme, and Caffarelli, Kohn and Nirenberg [2] obtained the sharpest result known to date: for a *suitable* weak solution (one that additionally satisfies a localized form of the energy inequality) the set S of space-time points at which the velocity fails to be locally bounded satisfies $\mathcal{P}^1(S) = 0$, where $\mathcal{P}^1$ is the one-dimensional

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Hausdorff measure of the parabolic metric. Their hypothesis on the force was $f \in L^q$ with $q > 5/2$.

The proof in [2] rests on a two-step scheme that has since become standard. The first step is an ε -regularity theorem: if a suitable weak solution has small scale-invariant L^3 velocity energy and small scale-invariant $L^{3/2}$ pressure energy, together with a small force norm, on one parabolic cylinder, then the velocity has a Hölder representative on the half-cylinder with the same top centre. The second step is a covering argument: at every singular point, Theorem 2.9 gives $\limsup_{r \downarrow 0} r^{-1} \iint_{Q_r(z)} |\nabla u|^2 \geq \varepsilon_1^2$, and the finiteness of $\iint |\nabla u|^2$ then forces the singular set to be $\mathcal{P}^1$ -null. Two later treatments simplify the first step in different directions. Lin [12] replaced the explicit iteration of [2] by a blow-up (compactness plus linear decay) argument; Ladyzhenskaya and Seregin [8] and Colombo, De Lellis and Massaccesi [3] completed and corrected that argument. Kukavica [7] instead kept the direct iteration but organized it around a single monotone quantity. His regularity criterion uses local L^5 integrability and applies to divergence-free forces in L^2 , with an extension to L^q , $q > 5/3$, stated in his Theorem 2. This is a weaker notion of regularity than the Hölder conclusion considered here. A direct proof of the theorem, with a smallness criterion involving the vorticity component perpendicular to the velocity, is given by Wolf [23].

The present notes are written with a single purpose: to serve as the mathematical source of a machine-checked formalization. Accordingly, every constant is given a name ($C_1, C_2, \dots$) and is never reused; when a constant is absolute (that is, independent of the solution, of the cylinder, and of every parameter in the statement) we say so explicitly. The estimates used at later steps are stated explicitly and cross-referenced.

For background on suitable weak solutions, partial regularity and the harmonic-analysis tools used below we refer to [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 18, 19, 20, 21, 22].

Organization of the manuscript. Section 1 fixes the notation, defines suitable weak solutions with a force, and proves the two facts about the time variable that the definition does not literally contain: that $\partial_t u$ is a $L^{3/2}$ -in-time distribution of a definite order (Lemma 1.10), and that consequently u has a representative that is weakly continuous in time with values in L^2 (Lemma 1.11), as an optional strengthening. The energy estimates below use only the almost-everywhere-in-time local energy inequality (Lemma 1.13), obtained directly by time differentiation. Section 2 states the three theorems. Section 3 records the exact scaling and translation invariance and proves the interpolation inequalities. Section 4 decomposes the pressure and proves the pressure decay lemma. Section 5 contains the ε -regularity core, in the direct-iteration form of [7] and [11, §13.8–13.9], and proves Theorems 2.7 and 2.9. Section 6 derives Theorem 2.10 from Theorem 2.9 by a Vitali covering argument. Appendix A lists the external analytic inputs.

1. SETTING AND NOTATION

1.1. Ambient conventions. Throughout, space is $\mathbb{R}^3$ with the Euclidean norm $|\cdot|$, time is $\mathbb{R}$, and a space-time point is written $z = (x, t) \in \mathbb{R}^3 \times \mathbb{R}$. The viscosity is 1. Latin indices i, j, k run over $\{1, 2, 3\}$ and repeated indices are summed. We write $\partial_i = \partial/\partial x_i$, $\nabla = (\partial_1, \partial_2, \partial_3)$, $\Delta = \partial_i \partial_i$ (the spatial Laplacian), and $\partial_t = \partial/\partial t$. For a vector field u we write $\operatorname{div} u = \partial_i u_i$ and $(u \otimes u)_{ij} = u_i u_j$. Lebesgue measure on $\mathbb{R}^3$ and on $\mathbb{R}^3 \times \mathbb{R}$ is denoted dx and $dz = dx dt$; the measure of a measurable set E is $|E|$.

Convention 1.1. Constants are numbered $C_1, C_2, \dots$ in order of first appearance and are never reused with a different value. A constant is called *absolute* if it depends on nothing at

all; since the space dimension is fixed at 3 and the viscosity at 1, dimensional constants are absolute here. Any other dependence is displayed in the notation, as in $C_{17}(q)$.

Convention 1.2 (Representatives and finite local quantities). Spatial balls and vector norms in statements are Euclidean. A displayed gradient of a Sobolev function is its weak gradient, determined almost everywhere. Smooth compactly supported tests mean C^∞ tests, not real-analytic tests. Integrals of nonnegative functions may first be read in $[0, \infty]$; real-valued scale quantities are used only on cylinders whose closures are compactly contained in the solution domain. Local finiteness then follows from (S1) and the local force assumption. No infinite integral is assigned a finite numerical value.

Distributional identities are tested against every compactly supported smooth test. A slice identity initially obtained for each test outside its own null set is promoted, when needed, to one common null set by a countable dense test family and continuity of the pairings. Convolutions of nonsmooth data, and identities between their measurable representatives, are understood almost everywhere. Pointwise regularity estimates concern the continuous representative subsequently constructed, not arbitrary values on a null set.

1.2. Balls, cylinders, and the parabolic metric. For $x \in \mathbb{R}^3$ and $r > 0$ put $B_r(x) = \{y \in \mathbb{R}^3 : |y - x| < r\}$, so that $|B_r(x)| = \frac{4}{3}\pi r^3$. For $z = (x, t)$ the *parabolic cylinder of radius r with top centre z* is

$$Q_r(z) = B_r(x) \times (t - r^2, t], \quad \text{so} \quad |Q_r(z)| = \frac{4}{3}\pi r^5. \quad (1)$$

We abbreviate $B_r = B_r(0)$ and $Q_r = Q_r((0, 0))$. The interval $(t - r^2, t]$ is open at the bottom and closed at the top so that $z \in Q_r(z)$; since the top face is a null set, every integral below is unaffected by this choice.

The *parabolic metric* on $\mathbb{R}^3 \times \mathbb{R}$ is

$$d_{\text{par}}(z, z') = \max\{|x - x'|, |t - t'|^{1/2}\}, \quad z = (x, t), \quad z' = (x', t'). \quad (2)$$

Lemma 1.3. d_{par} is a metric on $\mathbb{R}^3 \times \mathbb{R}$, and the open d_{par} -ball of radius r about $z = (x, t)$ is

$$\mathfrak{B}_r(z) = \{z' : d_{\text{par}}(z', z) < r\} = B_r(x) \times (t - r^2, t + r^2). \quad (3)$$

Moreover $Q_r(z) \subset \mathfrak{B}_r(z)$ and $\mathfrak{B}_r(z) \subset Q_{2r}((x, t + r^2))$ for every z and every $r > 0$.

Proof. Symmetry and the vanishing of $d_{\text{par}}(z, z')$ exactly when $z = z'$ are immediate from the definition. For the triangle inequality, let z, z', z'' be given. Since $|\cdot|$ is a metric on $\mathbb{R}^3$, $|x - x''| \leq |x - x'| + |x' - x''| \leq d_{\text{par}}(z, z') + d_{\text{par}}(z', z'')$. For the time entry, the elementary inequality $(a + b)^{1/2} \leq a^{1/2} + b^{1/2}$, valid for $a, b \geq 0$ because $a + b \leq a + 2a^{1/2}b^{1/2} + b$, gives

$$|t - t''|^{1/2} \leq (|t - t'| + |t' - t''|)^{1/2} \leq |t - t'|^{1/2} + |t' - t''|^{1/2} \leq d_{\text{par}}(z, z') + d_{\text{par}}(z', z'').$$

Taking the maximum of the two bounds yields $d_{\text{par}}(z, z'') \leq d_{\text{par}}(z, z') + d_{\text{par}}(z', z'')$.

Identity (3) is a restatement of the definition: $d_{\text{par}}(z', z) < r$ holds if and only if $|x' - x| < r$ and $|t' - t| < r^2$. The inclusion $Q_r(z) \subset \mathfrak{B}_r(z)$ holds because $(t - r^2, t] \subset (t - r^2, t + r^2)$. Finally $Q_{2r}((x, t + r^2)) = B_{2r}(x) \times (t - 3r^2, t + r^2]$ contains $B_r(x) \times (t - r^2, t + r^2)$. $\square$

1.3. Averages. Two distinct averaging operations occur, and they must not be confused. For a locally integrable g on a space-time set, the *spatial average* over $B_r(x)$ is the function of time

$$[g]_{B_r(x)}(t) = \frac{1}{|B_r(x)|} \int_{B_r(x)} g(y, t) dy, \quad (4)$$

defined for almost every t ; and the *space-time average* over $Q_r(z)$ is the number

$$(g)_{Q_r(z)} = \frac{1}{|Q_r(z)|} \iint_{Q_r(z)} g(y, s) dy ds. \quad (5)$$

For a vector field these are applied componentwise. We also write $[g]_{B_r(x)}$ for the function $t \mapsto [g]_{B_r(x)}(t)$ when no confusion is possible.

1.4. Function spaces. Let $U \subset \mathbb{R}^3$ be open and bounded. We write $L^s(U)$, $1 \leq s \leq \infty$, for the usual Lebesgue spaces, $H^1(U) = W^{1,2}(U)$, and $H_0^2(U)$ for the closure of $C_c^\infty(U)$ in the norm $\|\varphi\|_{H^2(U)} = (\sum_{|a| \leq 2} \|\partial^a \varphi\|_{L^2(U)}^2)^{1/2}$. For vector fields we write $H_0^2(U; \mathbb{R}^3)$ and we set

$$Z_U = (H_0^2(U; \mathbb{R}^3))^*, \quad (6)$$

the topological dual, with the dual norm. For a time interval J and a Banach space X , $L^s(J; X)$ is the usual Bochner space. We abbreviate

$$\|g\|_{L^s(E)} = \left(\iint_E |g|^s \right)^{1/s}$$

for a space-time set E and $1 \leq s < \infty$.

We record the two Sobolev embeddings that will be used on H_0^2 . Because a function in $H_0^2(U)$ extends by zero to a function in $H^2(\mathbb{R}^3)$ of the same norm, the constants are absolute and do not depend on U ; see External Input A.4.

Lemma 1.4. *There is an absolute constant C_1 such that for every bounded open $U \subset \mathbb{R}^3$ and every $\varphi \in H_0^2(U; \mathbb{R}^3)$,*

$$\|\varphi\|_{L^\infty(U)} \leq C_1 \|\varphi\|_{H^2(U)}, \quad \|\nabla \varphi\|_{L^6(U)} \leq C_1 \|\varphi\|_{H^2(U)}. \quad (7)$$

In particular, for every $s \in [1, \infty]$, $\|\varphi\|_{L^s(U)} \leq C_1 |U|^{1/s} \|\varphi\|_{H^2(U)}$, and for every $s \in [1, 6]$, $\|\nabla \varphi\|_{L^s(U)} \leq C_1 |U|^{1/s-1/6} \|\varphi\|_{H^2(U)}$.

Proof. Let $\tilde{\varphi}$ denote the extension of φ to $\mathbb{R}^3$ by zero. Since $\varphi \in H_0^2(U; \mathbb{R}^3)$ is a limit in H^2 of elements of $C_c^\infty(U; \mathbb{R}^3)$, and the zero extension of such an element belongs to $C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ with unchanged H^2 norm, the map $\varphi \mapsto \tilde{\varphi}$ is an isometry of $H_0^2(U; \mathbb{R}^3)$ into $H^2(\mathbb{R}^3; \mathbb{R}^3)$. By External Input A.4(iii) there is an absolute C with $\|\tilde{\varphi}\|_{L^\infty(\mathbb{R}^3)} \leq C \|\tilde{\varphi}\|_{H^2(\mathbb{R}^3)}$, and by External Input A.4(i) applied to each $\partial_i \tilde{\varphi} \in H^1(\mathbb{R}^3)$ there is an absolute C' with $\|\nabla \tilde{\varphi}\|_{L^6(\mathbb{R}^3)} \leq C' \|\nabla \tilde{\varphi}\|_{H^1(\mathbb{R}^3)} \leq C' \|\tilde{\varphi}\|_{H^2(\mathbb{R}^3)}$. Take $C_1 = \max\{C, C'\}$ and restrict to U . The two consequences follow from Hölder's inequality on U : $\|\varphi\|_{L^s(U)} \leq |U|^{1/s} \|\varphi\|_{L^\infty(U)}$ and, for $s \leq 6$, $\|\nabla \varphi\|_{L^s(U)} \leq |U|^{1/s-1/6} \|\nabla \varphi\|_{L^6(U)}$. $\square$

1.5. Suitable weak solutions with a force. Throughout the paper $\Omega \subset \mathbb{R}^3$ is open, $I \subset \mathbb{R}$ is an open interval, and we write

$$\mathcal{O} = \Omega \times I \subset \mathbb{R}^3 \times \mathbb{R}.$$

Only local statements are made: no initial condition, no boundary condition, and no global energy inequality is imposed, and the existence of solutions in the class below is not discussed.

Definition 1.5 (Suitable weak solution). Let $q > 5/2$ and let $f \in L_{\text{loc}}^q(\mathcal{O}; \mathbb{R}^3)$. A pair (u, p) is a *suitable weak solution of the Navier-Stokes equations on $\mathcal{O}$ with force f* if the following four conditions hold.

(S1) (*Regularity class.*) For every open Ω' with $\overline{\Omega'} \subset \Omega$ compact and every interval J with $\overline{J} \subset I$ compact,

$$u \in L^\infty(J; L^2(\Omega'; \mathbb{R}^3)) \cap L^2(J; H^1(\Omega'; \mathbb{R}^3)), \quad p \in L^{3/2}(\Omega' \times J).$$

(S2) (*Incompressibility.*) $\operatorname{div} u = 0$ in $\mathcal{D}'(\mathcal{O})$, that is, $\iint_{\mathcal{O}} u \cdot \nabla \psi \, dz = 0$ for every $\psi \in C_c^\infty(\mathcal{O})$. No condition is imposed on $\operatorname{div} f$.

(S3) (*Momentum equation.*) For every $\varphi \in C_c^\infty(\mathcal{O}; \mathbb{R}^3)$,

$$\iint_{\mathcal{O}} \left[-u \cdot \partial_t \varphi - u_i u_j \partial_j \varphi_i + \nabla u : \nabla \varphi - p \operatorname{div} \varphi - f \cdot \varphi \right] dz = 0. \quad (8)$$

Here $\nabla u : \nabla \varphi = \partial_j u_i \partial_j \varphi_i$. Equation (8) is the distributional form of

$$\partial_t u + \operatorname{div}(u \otimes u) - \Delta u + \nabla p = f, \quad \operatorname{div} u = 0. \quad (9)$$

(S4) (*Local energy inequality.*) For every $\phi \in C_c^\infty(\mathcal{O})$ with $\phi \geq 0$,

$$2 \iint_{\mathcal{O}} |\nabla u|^2 \phi \, dz \leq \iint_{\mathcal{O}} \left[|u|^2 (\partial_t \phi + \Delta \phi) + (|u|^2 + 2p) u \cdot \nabla \phi + 2(f \cdot u) \phi \right] dz. \quad (10)$$

Remark 1.6 (The gradient is a datum). Condition (S1) asserts $u(\cdot, t) \in H^1(\Omega')$ for a.e. t , so the weak spatial gradient ∇u exists and is uniquely determined almost everywhere on $\mathcal{O}$. In a formalization it is convenient to carry ∇u as an explicit datum $\mathbf{G}$ satisfying $\iint \mathbf{G}_{ij} \chi = -\iint u_i \partial_j \chi$ for all $\chi \in C_c^\infty(\mathcal{O})$, rather than as a derived object; the a.e. uniqueness of $\mathbf{G}$ makes the two formulations equivalent. With that convention, all the quantities of §1.7 (in particular β , and hence θ of (112)) are functions of the tuple $(u, \mathbf{G}, p, f)$ and of (z, r) , and every statement below that mentions ∇u is to be read with $\mathbf{G}$ in its place. No mathematical content changes.

The sign conventions in (10) are those of [2], [12] and [7]: every term on the right carries the sign it acquires when $|u|^2 \phi$ is formally differentiated in time using (9) and integrated by parts, and the dissipation appears on the left with the factor 2. Formally, (10) is what one gets by multiplying (9) by $2u\phi$ and discarding the (nonnegative) defect measure.

Remark 1.7 (The force class). Following [2] we assume $f \in L^q$ with $q > 5/2$; unlike [7] and [11, Definition 13.4] we do *not* assume $\operatorname{div} f = 0$. Two comments.

(i) The exponent q enters only through the scale-invariant force functional λ of (24), for which $\lambda(z, r) \leq \|f\|_{L^q} r^\sigma$ with

$$\sigma = 3 - \frac{5}{q} > 1, \quad (11)$$

so that $\lambda(z, r) \rightarrow 0$ as $r \rightarrow 0$ at a rate strictly better than linear. The strict inequality $\sigma > 1$ is used in Lemma 4.6(d)–(e) and, through $\sigma > \varepsilon$, in Proposition 5.19; the threshold $q > 5/2$ reappears in Theorem 5.30 as the condition $\theta_0 > 5/2$ for the force slot. Kukavica [7, Theorem 2] shows that the decay statements of Sections 3 and 4 already hold for $q > 5/3$, where $\sigma > 0$ but possibly $\sigma \leq 1$. The endgame here uses $\sigma > 1$. Remark 1.8 shows that an L^2 force hypothesis does not suffice for this Hölder notion of regularity; the example does not establish sharpness of the exponent $5/2$.

(ii) We do *not* assume $\operatorname{div} f = 0$, in contrast with [7]. The hypothesis would matter at exactly two places: it would make the sum $p_7 + p_8$ vanish of the pressure decomposition of Proposition 4.4, and hence the term $C_{15} \kappa^{1/2} \lambda(z, \rho)^{1/2}$ in the pressure decay estimate (71) and the term $C_{32}(q)(r/\rho)^{3/2} \lambda^{3/2}$ in (82). Both terms are estimated in Lemma 4.6(d)–(e) and are carried through Section 5; in the scale iteration they contribute the term $C_{28} \kappa^{-3} \lambda(z, \rho)$ of (113), which is harmless because $\lambda(z, \rho) \leq \|f\|_{L^q} \rho^\sigma$ with $\sigma > 1 > \varepsilon$. In the endgame the force contribution to the pressure gradient is estimated in Lemma 5.26: the near force term is a

completed second Riesz transform of the localized force, and the annular term has a fixed-scale $L^{3/2}$ time majorant. Neither estimate requires $\operatorname{div} f = 0$.

Remark 1.8 (Why an L^2 force does not suffice: a shear flow). A planar shear flow gives a counterexample at force exponent 2. Write $x = (x', x_3)$ with $x' = (x_1, x_2)$, and let $w \in W_2^{2,1}(\mathbb{R}^2 \times (-1, 1))$ be a solution of a two-dimensional heat equation that is unbounded near the origin; such w exist because $W_2^{2,1}$ is the borderline Sobolev case in parabolic dimension 4, for instance a compactly supported cutoff of $(\log(1/\varrho))^{1/4}$ near the origin, with $\varrho = (|x'|^2 + |t|)^{1/2}$. Its second spatial derivatives and first time derivative are bounded by $C\varrho^{-2}(\log(1/\varrho))^{-3/4}$ near the origin. Their squares are integrable in parabolic dimension 4, since $\int_0^{1/2} r^{-1}(\log(1/r))^{-3/2} dr < \infty$. Put

$$u(x, t) = (0, 0, w(x', t)), \quad p \equiv 0, \quad f = (0, 0, \partial_t w - \Delta_{x'} w).$$

Then $\operatorname{div} u = 0$ and $\operatorname{div} f = 0$ (both fields depend only on (x', t) and point in the x_3 direction), $u \cdot \nabla u = u_3 \partial_3 u = 0$, $f \in L_{loc}^2$, $p \in L^{3/2}$, and the local energy inequality holds with equality. Since $\nabla w \in L_{loc}^4$ by the parabolic Sobolev embedding in dimension 4, one computes $\iint_{Q_r(0)} |\nabla u|^2 = o(r^3)$, so $\beta(0, r)^2 = r^{-1} \iint_{Q_r(0)} |\nabla u|^2 \rightarrow 0$ and the hypothesis (34) of Theorem 2.9 holds for every threshold. Yet u is unbounded at the origin, so the origin is not a regular point in the sense of Definition 2.2. Hence the conclusion of Theorem 2.9 is *false* for $f \in L^2$ with the Hölder (indeed with the L^∞) notion of regularity. Kukavica's Theorem 1 for $f \in L^2$ is stated for the weaker notion “ $u \in L^5$ on a neighbourhood” [7, p. 719], which this example does not contradict.

Remark 1.9 (Comparison with the sources). Definition 1.5 is the class used by Lin [12, §2] and by Ladyzhenskaya–Seregin [8, Definition 2.1], with a force added and with the domain restricted to a product $\Omega \times I$. Two features of our formulation should be noted. First, $p \in L_{loc}^{3/2}$ is an explicit hypothesis. We use the modern suitable-solution class just cited, without claiming that its pressure hypothesis is identical to the original definition in [2] or deriving it from weaker local data. Second, [8] and [7] state the local energy inequality directly in the pointwise-in-time form (19) below; we take the distributional form (10) as the definition and *prove* the pointwise form in Lemma 1.13, since the two are not literally the same statement and the derivation uses a time cutoff and Lebesgue differentiation.

1.6. The time derivative and weak continuity in time. We first derive a weakly continuous L^2 representative from (S3). This is a useful additional property of suitable solutions. The almost-every-time local energy inequality proved afterwards uses Lebesgue differentiation directly and does not require this representative. Fix, for the whole subsection, a ball $B = B_\rho(x_0)$ and a bounded open interval J with

$$\overline{B} \subset \Omega, \quad \overline{J} \subset I. \tag{12}$$

Lemma 1.10 (The time derivative lies in $L^{3/2}(J; Z_B)$). *Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force $f \in L_{loc}^q$, $q > 5/2$, and let B, J be as in (12). Then the distribution $\partial_t u$ on $B \times J$ belongs to $L^{3/2}(J; Z_B)$, where $Z_B = (H_0^2(B; \mathbb{R}^3))^*$ as in (6): there is a function $G \in L^{3/2}(J; Z_B)$ such that*

$$-\int_J \int_B u \cdot \varphi \theta'(t) dx dt = \int_J \langle G(t), \varphi \rangle \theta(t) dt \quad \text{for all } \varphi \in H_0^2(B; \mathbb{R}^3), \theta \in C_c^\infty(J), \tag{13}$$

and, with C_1 the constant of Lemma 1.4,

$$\begin{aligned} \|G\|_{L^{3/2}(J; Z_B)} &\leq \sqrt{3}C_1 \left[|J|^{1/6} |B|^{1/3} \|\nabla u\|_{L^2(B \times J)} + |J|^{1/15} |B|^{7/30} \|u\|_{L^{10/3}(B \times J)}^2 \right. \\ &\quad \left. + |B|^{1/6} \|p\|_{L^{3/2}(B \times J)} + |J|^{\frac{2}{3}-\frac{1}{q}} |B|^{1-\frac{1}{q}} \|f\|_{L^q(B \times J)} \right]. \end{aligned} \quad (14)$$

In particular, if $|B| \leq 1$ and $|J| \leq 1$ then

$$\|G\|_{L^{3/2}(J; Z_B)} \leq C_2 \left[\|\nabla u\|_{L^2(B \times J)} + \|u\|_{L^{10/3}(B \times J)}^2 + \|p\|_{L^{3/2}(B \times J)} + \|f\|_{L^q(B \times J)} \right], \quad (15)$$

where $C_2 = \sqrt{3}C_1$ is absolute.

Proof. Step 1: the candidate for G . For $t \in J$ define a linear functional on $H_0^2(B; \mathbb{R}^3)$ by

$$\langle G(t), \varphi \rangle = \int_B \left[u_i(x, t) u_j(x, t) \partial_j \varphi_i(x) - \nabla u(x, t) : \nabla \varphi(x) + p(x, t) \operatorname{div} \varphi(x) + f(x, t) \cdot \varphi(x) \right] dx. \quad (16)$$

By (S1) and $f \in L_{loc}^q$ each of the four integrands is in $L^1(B)$ for almost every $t \in J$, so $\langle G(t), \varphi \rangle$ is defined for a.e. t , and it is measurable in t for each fixed φ . Since $H_0^2(B; \mathbb{R}^3)$ is a separable Hilbert space, so is its dual Z_B ; hence, by Pettis' measurability theorem (External Input A.13), this weak measurability implies that $t \mapsto G(t)$ is strongly (Bochner) measurable, which is what membership in the Bochner space $L^{3/2}(J; Z_B)$ requires.

Step 2: the four bounds. Fix t for which all four integrands are integrable, and let $\varphi \in H_0^2(B; \mathbb{R}^3)$ with $\|\varphi\|_{H^2(B)} \leq 1$.

(a) *Dissipation.* By Cauchy-Schwarz and Lemma 1.4 (with $s = 2 \leq 6$, so $|B|^{1/2-1/6} \leq |B|^{1/3}$),

$$\left| \int_B \nabla u : \nabla \varphi \right| \leq \|\nabla u(\cdot, t)\|_{L^2(B)} \|\nabla \varphi\|_{L^2(B)} \leq C_1 |B|^{1/3} \|\nabla u(\cdot, t)\|_{L^2(B)}.$$

(b) *Transport.* Since $H_0^2(B) \not\hookrightarrow W^{1,\infty}(B)$ in three dimensions, the convective term must be paired against $\nabla \varphi \in L^6$ rather than against $\nabla \varphi \in L^\infty$. By Hölder with $\frac{5}{6} + \frac{1}{6} = 1$ and Lemma 1.4,

$$\left| \int_B u_i u_j \partial_j \varphi_i \right| \leq \|u(\cdot, t)\|_{L^{12/5}(B)}^2 \|\nabla \varphi\|_{L^6(B)} \leq C_1 \|u(\cdot, t)\|_{L^{12/5}(B)}^2,$$

where we used $\|u_i u_j\|_{L^{6/5}(B)} \leq \|u\|_{L^{12/5}(B)}^2$.

(c) *Pressure.* By Hölder with $\frac{2}{3} + \frac{1}{3} = 1$ and Lemma 1.4 with $s = 3 \leq 6$, using $|\operatorname{div} \varphi| \leq \sqrt{3} |\nabla \varphi|$,

$$\left| \int_B p \operatorname{div} \varphi \right| \leq \sqrt{3} \|p(\cdot, t)\|_{L^{3/2}(B)} \|\nabla \varphi\|_{L^3(B)} \leq \sqrt{3} C_1 |B|^{1/6} \|p(\cdot, t)\|_{L^{3/2}(B)}.$$

(d) *Force.* By Hölder with $\frac{1}{q} + \frac{1}{q'} = 1$ and Lemma 1.4,

$$\left| \int_B f \cdot \varphi \right| \leq \|f(\cdot, t)\|_{L^q(B)} \|\varphi\|_{L^{q'}(B)} \leq C_1 |B|^{1/q'} \|f(\cdot, t)\|_{L^q(B)}.$$

Adding,

$$\begin{aligned} \|G(t)\|_{Z_B} &\leq \sqrt{3} C_1 \left[|B|^{1/3} \|\nabla u(\cdot, t)\|_{L^2(B)} \right. \\ &\quad + \|u(\cdot, t)\|_{L^{12/5}(B)}^2 + |B|^{1/6} \|p(\cdot, t)\|_{L^{3/2}(B)} \\ &\quad \left. + |B|^{1/q'} \|f(\cdot, t)\|_{L^q(B)} \right]. \end{aligned}$$

Step 3: time integrability of each of the four terms. We show that each of the four functions of t on the right lies in $L^{3/2}(J)$.

(a') By (S1), $t \mapsto \|\nabla u(\cdot, t)\|_{L^2(B)}$ belongs to $L^2(J)$, and therefore to $L^{3/2}(J)$, with $\|\cdot\|_{L^{3/2}(J)} \leq |J|^{1/6} \|\cdot\|_{L^2(J)}$ by Hölder, since $\frac{1}{3/2} - \frac{1}{2} = \frac{1}{6}$.

(b) Interpolating $L^{12/5}$ between L^2 and L^6 with $\frac{5}{12} = \frac{3/4}{2} + \frac{1/4}{6}$ gives

$$\|u\|_{L^{12/5}(B)} \leq \|u\|_{L^2(B)}^{3/4} \|u\|_{L^6(B)}^{1/4},$$

so

$$\|u(\cdot, t)\|_{L^{12/5}(B)}^2 \leq \|u(\cdot, t)\|_{L^2(B)}^{3/2} \|u(\cdot, t)\|_{L^6(B)}^{1/2}.$$

By (S1) and External Input A.4(ii), $t \mapsto \|u(\cdot, t)\|_{L^6(B)}$ lies in $L^2(J)$ and $t \mapsto \|u(\cdot, t)\|_{L^2(B)}$ lies in $L^\infty(J)$; hence the product lies in $L^4(J) \subset L^{3/2}(J)$. Equivalently, and this is the form recorded in (14), the interpolation $L^\infty(J; L^2) \cap L^2(J; L^6) \subset L^{10/3}(B \times J)$ (External Input A.4(iv)) gives

$$\|u\|_{L^{12/5}(B)}^2 \leq \|u\|_{L^{10/3}(B)}^2 |B|^{2(5/12-3/10)}$$

and $t \mapsto \|u(\cdot, t)\|_{L^{10/3}(B)}^2 \in L^{5/3}(J) \subset L^{3/2}(J)$.

(c) $t \mapsto \|p(\cdot, t)\|_{L^{3/2}(B)}$ lies in $L^{3/2}(J)$ by (S1), with $L^{3/2}(J)$ -norm equal to $\|p\|_{L^{3/2}(B \times J)}$. This is the term that fixes the exponent $3/2$ in the statement: none of the other three terms forces an exponent below $3/2$, and this one is exactly $3/2$.

(d) $t \mapsto \|f(\cdot, t)\|_{L^q(B)}$ lies in $L^q(J)$ with norm $\|f\|_{L^q(B \times J)}$, and $q > 5/2 > 3/2$, so it lies in $L^{3/2}(J)$ with $\|\cdot\|_{L^{3/2}(J)} \leq |J|^{2/3-1/q} \|f\|_{L^q(B \times J)}$.

Collecting the four contributions gives (14): the first is $C_1 |B|^{1/3} |J|^{1/6} \|\nabla u\|_{L^2(B \times J)}$ by (a) and (a'); the second is $C_1 |B|^{7/30} |J|^{1/15} \|u\|_{L^{10/3}(B \times J)}^2$, since

$$\|u(\cdot, t)\|_{L^{12/5}(B)}^2 \leq |B|^{2(\frac{5}{12}-\frac{3}{10})} \|u(\cdot, t)\|_{L^{10/3}(B)}^2$$

with $2(\frac{5}{12} - \frac{3}{10}) = \frac{7}{30}$ and

$$\|\cdot\|_{L^{3/2}(J)} \leq |J|^{\frac{2}{3}-\frac{3}{5}} \|\cdot\|_{L^{5/3}(J)}$$

with $\frac{2}{3} - \frac{3}{5} = \frac{1}{15}$; the third is

$$\sqrt{3} C_1 |B|^{1/6} \|p\|_{L^{3/2}(B \times J)}$$

by (c); and the fourth is

$$C_1 |B|^{1/q'} |J|^{\frac{2}{3}-\frac{1}{q}} \|f\|_{L^q(B \times J)}$$

by (d), with $1/q' = 1 - 1/q$. In particular $G \in L^{3/2}(J; Z_B)$, and (15) follows by discarding the factors $|B|^a, |J|^b \leq 1$.

Step 4: G represents $\partial_t u$. Let $\varphi \in C_c^\infty(B; \mathbb{R}^3)$ and $\theta \in C_c^\infty(J)$, and apply (S3) with test field $(x, t) \mapsto \theta(t)\varphi(x)$, which belongs to $C_c^\infty(\mathcal{O}; \mathbb{R}^3)$ by (12). Since $\partial_t(\theta\varphi) = \theta'\varphi$ and all spatial derivatives fall on φ , (8) reads

$$-\int_J \int_B u \cdot \varphi \theta' dx dt = \int_J \theta(t) \int_B \left[u_i u_j \partial_j \varphi_i - \nabla u : \nabla \varphi + p \operatorname{div} \varphi + f \cdot \varphi \right] dx dt = \int_J \langle G(t), \varphi \rangle \theta(t) dt,$$

which is (13) for $\varphi \in C_c^\infty(B; \mathbb{R}^3)$. Both sides are continuous in φ with respect to the $H^2(B)$ norm (the left side because

$$\begin{aligned} & \left| \int_J \int_B u \cdot \varphi \theta' \right| \\ & \leq \|\theta'\|_{L^\infty} \|u\|_{L^1(B \times J)} \|\varphi\|_{L^\infty(B)} \quad , \\ & \leq C_1 \|\theta'\|_{L^\infty} \|u\|_{L^1(B \times J)} \|\varphi\|_{H^2(B)} \end{aligned}$$

the right side by Step 2) and $C_c^\infty(B; \mathbb{R}^3)$ is dense in $H_0^2(B; \mathbb{R}^3)$ by definition. Hence (13) holds for all $\varphi \in H_0^2(B; \mathbb{R}^3)$. $\square$

Lemma 1.11 (Weak continuity in time). *Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force $f \in L_{loc}^q$, $q > 5/2$, and let B, J be as in (12). Then there is a set $N \subset J$ of Lebesgue measure zero and a map $\bar{u} : \bar{J} \rightarrow L^2(B; \mathbb{R}^3)$ such that*

- (i) $\bar{u}(t) = u(\cdot, t)$ in $L^2(B; \mathbb{R}^3)$ for every $t \in J \setminus N$;
- (ii) $\sup_{t \in \bar{J}} \|\bar{u}(t)\|_{L^2(B)} \leq \|u\|_{L^\infty(J; L^2(B))} =: M_{B,J}$;
- (iii) for every $w \in L^2(B; \mathbb{R}^3)$ the function $t \mapsto \int_B \bar{u}(t) \cdot w \, dx$ is continuous on $\bar{J}$.

We say that u belongs to $C(\bar{J}; L_w^2(B))$ and we always work with the representative $\bar{u}$.

Proof. Step 1: a continuous Z_B -valued representative. By Lemma 1.10, $G \in L^{3/2}(J; Z_B) \subset L^1(J; Z_B)$, so $t \mapsto \int_{t_1}^t G(s) \, ds$ is a continuous Z_B -valued function on $\bar{J}$ for any fixed $t_1 \in J$. Set $v(t) = \int_{t_1}^t G(s) \, ds \in Z_B$. Regarding $u(\cdot, t) \in L^2(B)$ as an element of Z_B through $\langle u(\cdot, t), \varphi \rangle = \int_B u(\cdot, t) \cdot \varphi$ (a bounded map by Lemma 1.4), identity (13) says that the Z_B -valued distribution $u - v$ on J has vanishing time derivative, i.e. $\int_J (u(\cdot, t) - v(t)) \theta'(t) \, dt = 0$ in Z_B for every $\theta \in C_c^\infty(J)$. By the vector-valued du Bois-Reymond lemma (External Input A.14) applied in the Banach space Z_B , there is $\xi \in Z_B$ with $u(\cdot, t) = \xi + v(t)$ in Z_B for a.e. $t \in J$; let N contain the null set where this fails and the null set where the L^2 bound $\|u(\cdot, t)\|_{L^2(B)} \leq M_{B,J}$ fails. Define $\bar{u}(t) = \xi + v(t) \in Z_B$ for all $t \in \bar{J}$; then $\bar{u} : \bar{J} \rightarrow Z_B$ is continuous and agrees with $u(\cdot, t)$ for $t \in J \setminus N$. This is (i), once we know $\bar{u}(t) \in L^2$.

Step 2: $\bar{u}(t) \in L^2(B)$ with the bound (ii). Let $t_0 \in \bar{J}$ and choose $t_k \in J \setminus N$ with $t_k \rightarrow t_0$; this is possible because $J \setminus N$ has full measure in J and $\bar{J}$ is the closure of J . By (S1), $\|\bar{u}(t_k)\|_{L^2(B)} = \|u(\cdot, t_k)\|_{L^2(B)} \leq M_{B,J}$ for every k , by the choice of N , and $\bar{u}(t_k) \rightarrow \bar{u}(t_0)$ in Z_B by Step 1. We claim that a sequence $g_k \in L^2(B; \mathbb{R}^3)$ with $\|g_k\|_{L^2} \leq M$ that converges in Z_B to some $g \in Z_B$ satisfies $g \in L^2(B; \mathbb{R}^3)$, $\|g\|_{L^2} \leq M$, and $g_k \rightharpoonup g$ weakly in L^2 . Indeed, for $\varphi \in C_c^\infty(B; \mathbb{R}^3)$, $|\langle g, \varphi \rangle| = \lim_k |\int_B g_k \cdot \varphi| \leq M \|\varphi\|_{L^2(B)}$, so g extends to a bounded functional on the L^2 -closure of $C_c^\infty(B; \mathbb{R}^3)$, namely $L^2(B; \mathbb{R}^3)$, of norm at most M ; by the Riesz representation theorem $g \in L^2(B; \mathbb{R}^3)$ with $\|g\|_{L^2} \leq M$. For weak convergence, let $w \in L^2(B; \mathbb{R}^3)$ and $\varepsilon > 0$, and choose $\varphi \in C_c^\infty(B; \mathbb{R}^3)$ with $\|w - \varphi\|_{L^2(B)} \leq \varepsilon/(4M + 1)$. Then

$$\left| \int_B (g_k - g) \cdot w \right| \leq \left| \int_B (g_k - g) \cdot (w - \varphi) \right| + \left| \int_B (g_k - g) \cdot \varphi \right| \leq 2M \frac{\varepsilon}{4M + 1} + |\langle g_k - g, \varphi \rangle|,$$

and the last term tends to 0 because $g_k \rightarrow g$ in Z_B and $\varphi \in H_0^2(B; \mathbb{R}^3)$. Hence

$$\limsup_k \left| \int_B (g_k - g) \cdot w \right| \leq \varepsilon/2 < \varepsilon.$$

Applying the claim with $g_k = \bar{u}(t_k)$, $g = \bar{u}(t_0)$, $M = M_{B,J}$ gives $\bar{u}(t_0) \in L^2(B; \mathbb{R}^3)$ with $\|\bar{u}(t_0)\|_{L^2(B)} \leq M_{B,J}$, which is (ii).

Step 3: weak continuity (iii). Let $t_0 \in \bar{J}$ and let $t_k \rightarrow t_0$ in $\bar{J}$ be arbitrary. By Step 2 each $\bar{u}(t_k) \in L^2(B)$ with norm at most $M_{B,J}$, and $\bar{u}(t_k) \rightarrow \bar{u}(t_0)$ in Z_B by Step 1. The claim of Step 2 gives $\bar{u}(t_k) \rightharpoonup \bar{u}(t_0)$ weakly in $L^2(B; \mathbb{R}^3)$, i.e. $\int_B \bar{u}(t_k) \cdot w \rightarrow \int_B \bar{u}(t_0) \cdot w$ for every $w \in L^2(B; \mathbb{R}^3)$. Since $t_k \rightarrow t_0$ was arbitrary, (iii) follows. $\square$

Lemma 1.12 (A common null set for the divergence-free condition). *Let $u \in L^1_{loc}(\mathcal{O}; \mathbb{R}^3)$ satisfy $\operatorname{div} u = 0$ in $\mathcal{D}'(\mathcal{O})$, let $B = B_\rho(x_0)$ with $\bar{B} \subset \Omega$, let J be an interval with $\bar{J} \subset I$, and let $E \subset B$ be compact. Then there is a Lebesgue-null set $N \subset J$ such that*

$$\int_B u(y, s) \cdot \nabla \psi(y) dy = 0 \quad \text{for every } s \in J \setminus N \text{ and every } \psi \in C^1_c(B) \text{ with } \operatorname{supp} \psi \subset E. \quad (17)$$

Proof. Let E' be a compact neighbourhood of E in B . Restriction of a function and its three first derivatives embeds $C^1_E = \{\psi \in C^1(\mathbb{R}^3) : \operatorname{supp} \psi \subset E\}$ isometrically into $C(E')^4$. Since E' is compact metrizable, this space and its subspaces are separable. Let $\{\psi_k\}_{k \in \mathbb{N}} \subset C^\infty_c(B)$ with $\operatorname{supp} \psi_k \subset E'$ be a family whose C^1 closure contains C^1_E . To construct it, mollify a countable dense subset of C^1_E at a countable sequence of sufficiently small radii. The supports then lie in E' , and the mollifications converge in C^1 (External Input A.15). Testing (S2) with $\psi_k(x)\theta(t)$, $\theta \in C^\infty_c(J)$, gives $\int_J \theta(s) (\int_B u(\cdot, s) \cdot \nabla \psi_k) ds = 0$ for all θ , hence a null set $N_k \subset J$ off which $\int_B u(\cdot, s) \cdot \nabla \psi_k = 0$. Let $N = (\bigcup_k N_k) \cup \{s : \|u(\cdot, s)\|_{L^1(B)} = \infty\}$, a null set by Fubini. For $s \notin N$ and $\psi \in C^1_E$, choose $\psi_k \rightarrow \psi$ in C^1 ; then $|\int_B u(\cdot, s) \cdot \nabla(\psi - \psi_k)| \leq \|u(\cdot, s)\|_{L^1(B)} \|\nabla \psi - \nabla \psi_k\|_{L^\infty} \rightarrow 0$, which gives (17). $\square$

Lemma 1.13 (Pointwise-in-time local energy inequality). *Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force $f \in L^q_{loc}$, $q > 5/2$. Let $\phi \in C^\infty_c(\mathcal{O})$ with $\phi \geq 0$, and let $J = (t_-, t_+)$ be a bounded open interval with $\bar{J} \subset I$ and*

$$\phi(\cdot, s) = 0 \quad \text{for all } s \leq t_- + \varepsilon' \quad \text{for some } \varepsilon' > 0, \quad (18)$$

so that in particular $\operatorname{supp} \phi \subset \Omega \times (t_-, \infty)$. Then for almost every $t \in J$,

$$\begin{aligned} \int_\Omega |u(x, t)|^2 \phi(x, t) dx + 2 \int_{t_-}^t \int_\Omega |\nabla u|^2 \phi dx ds \\ \leq \int_{t_-}^t \int_\Omega \left[|u|^2 (\partial_s \phi + \Delta \phi) + (|u|^2 + 2p) u \cdot \nabla \phi + 2(f \cdot u) \phi \right] dx ds. \end{aligned} \quad (19)$$

Proof. Fix $t \in J$ and let $K = \operatorname{supp} \phi$, a compact subset of $\mathcal{O}$. Cover its spatial projection by finitely many balls whose closures lie in Ω , and let B denote their union. Choose an open interval J' with $\bar{J}' \subset I$, $K \subset B \times J'$ and $\bar{J} \subset J'$. The local integrability bounds used below hold on each ball and hence on their finite union. All spatial integrals below may be restricted to B , since the test function vanishes outside it.

Step 1: a Lipschitz cut-off in time. For $0 < h < t - t_-$ let $\chi_h : \mathbb{R} \rightarrow [0, 1]$ be the piecewise affine function

$$\chi_h(s) = \begin{cases} 1, & s \leq t - h, \\ \frac{t - s}{h}, & t - h < s < t, \\ 0, & s \geq t. \end{cases} \quad (20)$$

Then χ_h is Lipschitz with $\chi'_h = -h^{-1} \mathbf{1}_{(t-h, t)}$ almost everywhere. Fix $\zeta \in C^\infty_c((-1, 1))$ with $\zeta \geq 0$ and $\int_{\mathbb{R}} \zeta = 1$, put $\zeta_\varepsilon(s) = \varepsilon^{-1} \zeta(s/\varepsilon)$, and set $\chi_{h,\varepsilon} = \chi_h * \zeta_\varepsilon$, which is smooth, takes values in $[0, 1]$, equals 1 for $s \leq t - h - \varepsilon$ and 0 for $s \geq t + \varepsilon$. For ε small enough that

$t + \varepsilon < t_+$, the function $\phi \chi_{h,\varepsilon}$ belongs to $C_c^\infty(\mathcal{O})$ and is nonnegative, so (10) applies to it. Using $\partial_s(\phi \chi_{h,\varepsilon}) = (\partial_s \phi) \chi_{h,\varepsilon} + \phi \chi'_{h,\varepsilon}$ and $\Delta(\phi \chi_{h,\varepsilon}) = (\Delta \phi) \chi_{h,\varepsilon}$, inequality (10) becomes

$$\begin{aligned} 2 \iint |\nabla u|^2 \phi \chi_{h,\varepsilon} - \iint |u|^2 \phi \chi'_{h,\varepsilon} \\ \leq \iint \left[|u|^2 (\partial_s \phi + \Delta \phi) + (|u|^2 + 2p) u \cdot \nabla \phi + 2(f \cdot u) \phi \right] \chi_{h,\varepsilon}. \end{aligned} \quad (21)$$

Step 2: $\varepsilon \rightarrow 0$ at fixed h . Each integrand in (21) other than the one containing χ' is in $L^1(B \times J')$: indeed $|\nabla u|^2 \phi \in L^1$ by (S1), $|u|^2 (\partial_s \phi + \Delta \phi) \in L^1$ because $u \in L^{10/3}(B \times J') \subset L^2$, $|u|^2 u \cdot \nabla \phi \in L^1$ because $u \in L^{10/3} \subset L^3$, $p u \cdot \nabla \phi \in L^1$ by Hölder with $\frac{2}{3} + \frac{1}{3} = 1$ and $p \in L^{3/2}$, $u \in L^3$, and $(f \cdot u) \phi \in L^1$ by Hölder with $f \in L^q$, $q > 5/2$, and $u \in L^{10/3}$ (note $\frac{1}{q} + \frac{3}{10} < \frac{2}{5} + \frac{3}{10} < 1$). Since $0 \leq \chi_{h,\varepsilon} \leq 1$ and $\chi_{h,\varepsilon} \rightarrow \chi_h$ pointwise away from the two corner points as $\varepsilon \rightarrow 0$, dominated convergence applies to all these terms. For the remaining term, $\chi'_{h,\varepsilon} = \chi'_h * \zeta_\varepsilon$ is bounded by h^{-1} and supported in $(t-h-\varepsilon, t+\varepsilon)$, and $\chi'_{h,\varepsilon} \rightarrow \chi'_h$ in $L^1(\mathbb{R})$; since $s \mapsto \int |u(x, s)|^2 \phi(x, s) dx$ is in $L^\infty(J')$ by (S1), we get $\iint |u|^2 \phi \chi'_{h,\varepsilon} \rightarrow -h^{-1} \int_{t-h}^t \int |u|^2 \phi dx ds$. Hence

$$\begin{aligned} 2 \int_{t-}^t \int |\nabla u|^2 \phi \chi_h + \frac{1}{h} \int_{t-h}^t \int |u|^2 \phi dx ds \\ \leq \int_{t-}^t \int \left[|u|^2 (\partial_s \phi + \Delta \phi) + (|u|^2 + 2p) u \cdot \nabla \phi + 2(f \cdot u) \phi \right] \chi_h. \end{aligned} \quad (22)$$

Step 3: $h \rightarrow 0$. On the right of (22) we have $0 \leq \chi_h \leq \mathbf{1}_{(-\infty, t)}$ and $\chi_h \rightarrow \mathbf{1}_{(-\infty, t)}$ pointwise, so dominated convergence gives the right side of (19). In the first term on the left, χ_h increases to $\mathbf{1}_{(-\infty, t)}$ and the integrand is nonnegative, so monotone convergence gives $2 \int_{t-}^t \int |\nabla u|^2 \phi$. It remains to prove

$$\int |u(x, t)|^2 \phi(x, t) dx \leq \liminf_{h \rightarrow 0^+} \frac{1}{h} \int_{t-h}^t \int |u(x, s)|^2 \phi(x, s) dx ds. \quad (23)$$

Write $\Lambda(s) = \int |u(x, s)|^2 \phi(x, s) dx$. This function is locally integrable by (S1). At every one-sided Lebesgue point t of Λ ,

$$\lim_{h \downarrow 0} \frac{1}{h} \int_{t-h}^t \Lambda(s) ds = \Lambda(t).$$

Almost every t has this property. This gives (23) (in fact equality) at precisely the times required by the statement. The remaining terms converge by the preceding integrability and monotone-convergence arguments. $\square$

Remark 1.14. At the Lebesgue times used above the average converges to the slice energy. Only the corresponding lower bound is needed for the left side of (19). Every subsequent energy estimate uses an essential supremum or an integral in time, so it needs no assertion at an exceptional time. The earlier weak-time-continuity discussion provides a stronger optional representative convention, not an input to this route.

The local product-domain formulation loses no generality for local regularity on an open spacetime set: cover that set by relatively compact products $\Omega' \times J$. All weak-equation and energy pairings are integrable on the support of a test, by (S1), local interpolation, and the force assumption.

1.7. Scale-invariant quantities. Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force f , let $z = (x, t) \in \mathcal{O}$ and let $r > 0$ be such that $\overline{Q_r(z)} \subset \mathcal{O}$. The five quantities we work with are those of Kukavica [7, §2]:

$$\begin{aligned} \alpha(z, r) &= \left(\frac{1}{r} \operatorname{ess\,sup}_{s \in (t-r^2, t)} \int_{B_r(x)} |u(y, s)|^2 dy \right)^{1/2}, & \beta(z, r) &= \left(\frac{1}{r} \iint_{Q_r(z)} |\nabla u|^2 dz' \right)^{1/2}, \\ \gamma(z, r) &= \left(\frac{1}{r^2} \iint_{Q_r(z)} |u|^3 dz' \right)^{1/3}, & \delta(z, r) &= \left(\frac{1}{r^2} \iint_{Q_r(z)} |p|^{3/2} dz' \right)^{1/3}, \end{aligned} \quad (24)$$

$$\lambda(z, r) = r^\sigma \left(\iint_{Q_r(z)} |f|^q dz' \right)^{1/q} = r^\sigma \|f\|_{L^q(Q_r(z))}, \quad \sigma = 3 - \frac{5}{q} > 1. \quad (25)$$

When the base point is the origin we write $\alpha(r) = \alpha((0, 0), r)$, and so on. All five are dimensionless, in the precise sense of Lemma 3.3: each is unchanged when (u, p, f) is replaced by the rescaled triple (35) and r by r/μ . For $q = 2$ one has $\sigma = 1/2$ and $\lambda(z, r) = (r \iint_{Q_r(z)} |f|^2)^{1/2}$, which is the force functional of [7, §2].

Lemma 1.15 (Dictionary to the Lin, CKN and Ladyzhenskaya–Seregin notation). *Put*

$$\begin{aligned} A(z, r) &= \frac{1}{r} \operatorname{ess\,sup}_{s \in (t-r^2, t)} \int_{B_r(x)} |u|^2, & B(z, r) &= \frac{1}{r} \iint_{Q_r(z)} |\nabla u|^2, & C(z, r) &= \frac{1}{r^2} \iint_{Q_r(z)} |u|^3, \\ D(z, r) &= \frac{1}{r^2} \iint_{Q_r(z)} |p|^{3/2}, & E_q(z, r) &= \lambda(z, r). \end{aligned} \quad (26)$$

Then

$$\alpha = A^{1/2}, \quad \beta = B^{1/2}, \quad \gamma = C^{1/3}, \quad \delta = D^{1/3}, \quad \lambda = E_q. \quad (27)$$

Proof. Each identity is the definition (24) with the outer exponent removed, compared with (26). $\square$

Remark 1.16 (Letters used differently by different authors). A, B, C, D are Lin's notation [12, §3], which is also that of [2]. Ladyzhenskaya–Seregin [8, §5] and Tsai [21, §6.3] write $E(r)$ for what is here $B(z, r)$, i.e. for the scale-invariant Dirichlet energy; we never use the letter E alone, precisely to avoid this collision, and we write E_q only as an alias for λ when comparing with [21]. Tsai [21, §6.3] also uses $\tilde{C}$ for the velocity excess, which agrees with (28) below.

We shall also use the two *excess* (oscillation) quantities, in which the velocity is measured against its space-time mean and the pressure against its spatial mean at each time,

$$\tilde{C}(z, r) = \frac{1}{r^2} \iint_{Q_r(z)} |u - (u)_{Q_r(z)}|^3 dz', \quad \tilde{D}(z, r) = \frac{1}{r^2} \iint_{Q_r(z)} |p(y, s) - [p]_{B_r(x)}(s)|^{3/2} dy ds, \quad (28)$$

the *drift functional*

$$\Psi(z, r) = r |(u)_{Q_r(z)}|, \quad (29)$$

and the *excess* of (u, p) on $Q_r(z)$, in the normalization of [21, (6.14)],

$$\begin{aligned} \varphi(z, r) &= \tilde{C}(z, r)^{1/3} + \tilde{D}(z, r)^{2/3} \\ &= \left(\frac{1}{r^2} \iint_{Q_r(z)} |u - (u)_{Q_r(z)}|^3 \right)^{1/3} + \left(\frac{1}{r^2} \iint_{Q_r(z)} |p - [p]_{B_r(x)}|^{3/2} \right)^{2/3}. \end{aligned} \quad (30)$$

Both φ and Ψ are dimensionless (Lemma 3.3).

Remark 1.17 (Why the two means are different). In (30) the velocity is compared with a *space-time* mean and the pressure with a *spatial* mean at each time. These choices serve different purposes.

The pressure of a suitable weak solution is determined only up to an additive function of time: if (u, p) is a suitable weak solution with force f and $c \in L_{loc}^{3/2}(I)$, then $(u, p + c)$ is one as well, since $\nabla c = 0$ and $c u \cdot \nabla \phi$ integrates to 0 against $\operatorname{div} u = 0$ after an integration by parts. Hence no statement about p can be invariant unless a spatial mean is subtracted at each time.

The velocity excess uses a space-time mean because the parabolic Campanato characterization of Hölder continuity (External Input A.9) measures oscillation in both space and time. A spatial mean alone removes all information about a spatially constant, time-dependent velocity. This observation explains the choice of excess; it is not a counterexample to regularity under the full suitable-solution hypotheses.

Lemma 1.18 (Comparison of the excess with γ, δ). *For every z and every admissible r ,*

$$\tilde{C}(z, r) \leq 8C(z, r), \quad \tilde{D}(z, r) \leq 2^{3/2}D(z, r), \quad \varphi(z, r) \leq 2\gamma(z, r) + 2\delta(z, r)^2, \quad (31)$$

and $\Psi(z, r) \leq C_3 \gamma(z, r)$ with $C_3 = (4\pi/3)^{-1/3}$, an absolute constant.

Proof. Jensen's inequality, applied to $\tau \mapsto \tau^3$ with the normalized Lebesgue measure on $Q_r(z)$, gives

$$|(u)_{Q_r(z)}|^3 \leq (|Q_r(z)|^{-1} \iint |u|)^3 \leq |Q_r(z)|^{-1} \iint |u|^3,$$

hence

$$\iint_{Q_r(z)} |(u)_{Q_r(z)}|^3 \leq \iint_{Q_r(z)} |u|^3.$$

Combining with $|a - b|^3 \leq 4(|a|^3 + |b|^3)$ gives $\tilde{C} \leq 4(C + C) = 8C$. The same argument with $\tau \mapsto \tau^{3/2}$, applied for each fixed s on $B_r(x)$, gives

$$\int_{B_r(x)} |[p]_{B_r(x)}(s)|^{3/2} dy \leq \int_{B_r(x)} |p(\cdot, s)|^{3/2} dy,$$

and with $|a - b|^{3/2} \leq 2^{1/2}(|a|^{3/2} + |b|^{3/2})$ this gives $\tilde{D} \leq 2^{1/2} \cdot 2D = 2^{3/2}D$. Taking the $1/3$ and $2/3$ powers, $\tilde{C}^{1/3} \leq 8^{1/3}C^{1/3} = 2\gamma$ and $\tilde{D}^{2/3} \leq (2^{3/2})^{2/3}D^{2/3} = 2\delta^2$, which is the third inequality. Finally, by Jensen again,

$$\begin{aligned} & |(u)_{Q_r(z)}| \\ & \leq (|Q_r(z)|^{-1} \iint |u|^3)^{1/3}, \\ & = \left(\frac{4\pi}{3}\right)^{-1/3} r^{-5/3} (r^2 \gamma^3)^{1/3} \\ & = \left(\frac{4\pi}{3}\right)^{-1/3} r^{-1} \gamma \end{aligned}$$

so $\Psi = r |(u)_{Q_r(z)}| \leq \left(\frac{4\pi}{3}\right)^{-1/3} \gamma$. □

Remark 1.19 (Normalizations that are *not* used). Lin [12] writes $v_\theta = \theta^{-5} \iint_{Q_\theta} v$ and $P_\theta(t) = \theta^{-3} \int_{B_\theta} P$. These differ from the genuine averages (4), (5) by the fixed factors $|Q_1| = \frac{4\pi}{3}$ and $|B_1| = \frac{4\pi}{3}$. We use only genuine averages: every mean in this paper divides by the measure of the set it is taken over. This matters for the Campanato step (External Input A.9), which is stated for genuine averages, and for the Poincaré and Jensen inequalities, which are false with a mismatched normalizing factor.

2. STATEMENTS

2.1. Regular and singular points.

Definition 2.1 (Parabolic Hölder spaces). Let $E \subset \mathbb{R}^3 \times \mathbb{R}$ and $\gamma \in (0, 1]$. A function $w : E \rightarrow \mathbb{R}^m$ belongs to $C^{\gamma, \gamma/2}(E)$ if it is bounded and

$$[w]_{C^{\gamma, \gamma/2}(E)} = \sup_{z \neq z' \in E} \frac{|w(z) - w(z')|}{d_{\text{par}}(z, z')^\gamma} < \infty,$$

with d_{par} as in (2). We set $\|w\|_{C^{\gamma, \gamma/2}(E)} = \|w\|_{L^\infty(E)} + [w]_{C^{\gamma, \gamma/2}(E)}$.

Definition 2.2 (Regular and singular points). Let (u, p) be a suitable weak solution on $\mathcal{O} = \Omega \times I$ with force f . A point $z_0 \in \mathcal{O}$ is a *regular point* of u if there exist an open set $\mathcal{N}$ with $z_0 \in \mathcal{N} \subset \mathcal{O}$, an exponent $\gamma \in (0, 1]$, and a function $w \in C^{\gamma, \gamma/2}(\mathcal{N})$ such that $u = w$ almost everywhere in $\mathcal{N}$. Otherwise z_0 is a *singular point*. The *singular set* is

$$S = S(u) = \{z_0 \in \mathcal{O} : z_0 \text{ is a singular point of } u\}.$$

Lemma 2.3. S is relatively closed in $\mathcal{O}$.

Proof. Let $R = \mathcal{O} \setminus S$ be the set of regular points and let $z_0 \in R$, with $\mathcal{N}, \gamma, w$ as in Definition 2.2. Every $z_1 \in \mathcal{N}$ is then regular, witnessed by the same $\mathcal{N}, \gamma$ and w . Hence $\mathcal{N} \subset R$, so R is open in $\mathcal{O}$ and S is relatively closed. $\square$

Definition 2.4 (Parabolic Hausdorff measure). For $\alpha \geq 0$, $\delta > 0$ and $E \subset \mathbb{R}^3 \times \mathbb{R}$ put

$$\begin{aligned} \mathcal{P}_\delta^\alpha(E) &= \inf \left\{ \sum_{j \in J} r_j^\alpha : J \subset \mathbb{N}, \quad E \subset \bigcup_{j \in J} \mathfrak{B}_{r_j}(z_j), \quad 0 < r_j \leq \delta \ (j \in J) \right\}, \\ \mathcal{P}^\alpha(E) &= \lim_{\delta \rightarrow 0^+} \mathcal{P}_\delta^\alpha(E) = \sup_{\delta > 0} \mathcal{P}_\delta^\alpha(E), \end{aligned}$$

where empty and finite J are allowed, an empty sum is zero, and $\mathfrak{B}_r(z)$ is the d_{par} -ball (3). For $\alpha = 0$ each nonempty covering ball has cost 1. The same index convention is used for the cylinder gauge below. Thus $\mathcal{P}^\alpha$ is the *spherical* α -dimensional Hausdorff measure of the metric space $(\mathbb{R}^3 \times \mathbb{R}, d_{\text{par}})$: the gauge is the radius of a covering ball rather than the diameter of an arbitrary covering set (External Input A.8). The two differ by a factor between 1 and 2^α and in particular have the same null sets, which is all that is used.

Remark 2.5 (Gauge independence). Definition 2.4 uses the *radius* of a covering ball as gauge. The Hausdorff measure of $(\mathbb{R}^3 \times \mathbb{R}, d_{\text{par}})$ built, as in the usual measure-theoretic convention, from covers by arbitrary sets E_j of small d_{par} -diameter and the gauge $(\text{diam}_{d_{\text{par}}} E_j)^\alpha$, denoted by $\mathcal{P}_{\text{diam}}^\alpha$, satisfies

$$\mathcal{P}_{\text{diam}}^\alpha(E) \leq 2^\alpha \mathcal{P}^\alpha(E) \quad \text{and} \quad \mathcal{P}^\alpha(E) \leq 2^\alpha \mathcal{P}_{\text{diam}}^\alpha(E) \quad (32)$$

for every E . Indeed, a ball $\mathfrak{B}_r(z)$ has $\text{diam}_{d_{\text{par}}} \leq 2r$, which gives the first inequality; conversely a non-empty set E_j of diameter d_j containing a point z_j satisfies $E_j \subset \mathfrak{B}_{r_j}(z_j)$ for every $r_j > d_j$, and letting $r_j \downarrow d_j$ gives the second, in fact with factor 1 for $\alpha > 0$: choose the radii with arbitrarily small summable excess in $\sum_j r_j^\alpha$, including arbitrarily small positive radii when $d_j = 0$. At $\alpha = 0$ both measures are counting measure: a finite set of n points requires n sufficiently small covering sets, and every infinite set contains finite subsets of arbitrarily large cardinality. Empty covering sets are discarded at zero cost in the diameter convention. Thus both comparisons include $\alpha = 0$ and $E = \emptyset$. In particular the two have exactly the same null sets, and Theorem 2.10 (an assertion that a set is $\mathcal{P}^1$ -null) is unaffected by the choice of gauge. The same applies to the cylinder gauge of Lemma 2.6.

Lemma 2.6 (Cylinders and parabolic balls give the same null sets). *Define $\widehat{\mathcal{P}}^\alpha$ exactly as in Definition 2.4 but with the covering sets $\mathfrak{B}_{r_j}(z_j)$ replaced by cylinders $Q_{r_j}(z_j)$. Then for every E and every $\alpha \geq 0$,*

$$\mathcal{P}^\alpha(E) \leq \widehat{\mathcal{P}}^\alpha(E) \leq 2^\alpha \mathcal{P}^\alpha(E).$$

In particular $\mathcal{P}^\alpha(E) = 0$ if and only if $\widehat{\mathcal{P}}^\alpha(E) = 0$.

Proof. If $E \subset \bigcup_j Q_{r_j}(z_j)$ with $r_j \leq \delta$ then $E \subset \bigcup_j \mathfrak{B}_{r_j}(z_j)$ by Lemma 1.3, with the same radii; hence $\mathcal{P}_\delta^\alpha(E) \leq \widehat{\mathcal{P}}_\delta^\alpha(E)$. Conversely, if $E \subset \bigcup_j \mathfrak{B}_{r_j}(z_j)$ with $z_j = (x_j, t_j)$ and $r_j \leq \delta$, then by Lemma 1.3 $\mathfrak{B}_{r_j}(z_j) \subset Q_{2r_j}((x_j, t_j + r_j^2))$, so E is covered by cylinders of radii $2r_j \leq 2\delta$ and $\widehat{\mathcal{P}}_{2\delta}^\alpha(E) \leq \sum_j (2r_j)^\alpha = 2^\alpha \sum_j r_j^\alpha$. Taking the infimum and then $\delta \rightarrow 0^+$ gives the second inequality. $\square$

2.2. The three theorems.

Theorem 2.7 (ε -regularity, L^3 form). *For every $q > 5/2$ there exist $\varepsilon_0 = \varepsilon_0(q) > 0$, $\gamma_0 = \gamma_0(q) \in (0, \frac{2}{3}]$ and $C_4 = C_4(q) < \infty$ with the following property. Let (u, p) be a suitable weak solution on $\mathcal{O} = \Omega \times I$, with $\overline{Q_1} \subset \mathcal{O}$, and force $f \in L^q(Q_1; \mathbb{R}^3)$, and suppose that, with the notation (24),*

$$\iint_{Q_1} (|u|^3 + |p|^{3/2} + |f|^q) dz \leq \varepsilon_0, \quad \text{equivalently} \quad \gamma(1)^3 + \delta(1)^3 + \lambda(1)^q \leq \varepsilon_0. \quad (33)$$

Then u agrees almost everywhere on $Q_{1/2}$ with a function $w \in C^{\gamma_0, \gamma_0/2}(\overline{Q_{1/2}}; \mathbb{R}^3)$ satisfying $\|w\|_{C^{\gamma_0, \gamma_0/2}(\overline{Q_{1/2}})} \leq C_4$. In particular every point of the open cylinder $B_{1/2}(0) \times (-\frac{1}{4}, 0)$ is a regular point of u .

Remark 2.8 (The top face). Definition 2.2 requires an *open* neighbourhood, whereas $Q_{1/2}$ contains its top face $\{t = 0\}$; Theorem 2.7 therefore does not by itself declare the points of that face regular. This costs nothing, because Theorem 2.9 is not deduced from Theorem 2.7 by applying it at z_0 : it is proved in §5.6 from Proposition 5.22, whose conclusion (123) holds at every centre of the *open* d_{par} -ball $\mathfrak{B}_{r_2}(z_0)$, and from Theorem 5.30, which produces a Hölder representative on an open d_{par} -ball. The upper-semicontinuity Lemma 5.20 is exactly what makes the passage to later base points legitimate.

Theorem 2.9 (ε -regularity, gradient form). *For every $q > 5/2$ there exists $\varepsilon_1 = \varepsilon_1(q) > 0$ with the following property. Let (u, p) be a suitable weak solution on $\mathcal{O} = \Omega \times I$ with force $f \in L_{\text{loc}}^q(\mathcal{O}; \mathbb{R}^3)$, and let $z_0 \in \mathcal{O}$. Suppose that the following inequality holds, with both limsups taken in $[0, \infty]$:¹*

$$\limsup_{r \rightarrow 0^+} \beta(z_0, r)^2 = \limsup_{r \rightarrow 0^+} \frac{1}{r} \iint_{Q_r(z_0)} |\nabla u|^2 dz < \varepsilon_1^2. \quad (34)$$

Then z_0 is a regular point of u .

Theorem 2.10 (Caffarelli-Kohn-Nirenberg). *Let (u, p) be a suitable weak solution on $\mathcal{O} = \Omega \times I$ with force $f \in L_{\text{loc}}^q(\mathcal{O}; \mathbb{R}^3)$ for some $q > 5/2$. Then the singular set S satisfies $\mathcal{P}^1(S) = 0$.*

Theorems 2.7 and 2.9 are proved in §5.6; Theorem 2.10 is deduced from Theorem 2.9 in Section 6.

¹Suitability and interior geometry make each sufficiently small cylinder mass finite, so the squared real β expression agrees with the nonnegative integral expression.

Remark 2.11 (Strictness in (34)). The inequality in (34) is strict, as in [8, Theorem 2.2] and [7, Theorem 1]; the proof produces a threshold ε_1 and needs $B(z_0, r) < \varepsilon_1^2$ for all sufficiently small r , which is what the strict limsup inequality supplies. For $0 < \varepsilon'_1 < \varepsilon_1$, the nonstrict hypothesis $\limsup_{r \downarrow 0} B(z_0, r) \leq (\varepsilon'_1)^2 < \varepsilon_1^2$ suffices.

3. INVARIANCES AND INTERPOLATION

3.1. Translation and parabolic rescaling.

Definition 3.1. Let $z_0 = (x_0, t_0) \in \mathbb{R}^3 \times \mathbb{R}$ and $\mu > 0$. For functions u, p, f defined on $\mathcal{O} = \Omega \times I$ set

$$\begin{aligned} u^{\mu, z_0}(y, s) &= \mu u(x_0 + \mu y, t_0 + \mu^2 s), \\ p^{\mu, z_0}(y, s) &= \mu^2 p(x_0 + \mu y, t_0 + \mu^2 s), \\ f^{\mu, z_0}(y, s) &= \mu^3 f(x_0 + \mu y, t_0 + \mu^2 s), \end{aligned} \tag{35}$$

defined on $\mathcal{O}^{\mu, z_0} = (\mu^{-1}(\Omega - x_0)) \times (\mu^{-2}(I - t_0))$.

Lemma 3.2 (The class is invariant). *Let $q > 5/2$. If (u, p) is a suitable weak solution on $\mathcal{O}$ with force $f \in L^q_{loc}(\mathcal{O}; \mathbb{R}^3)$, then for every z_0 and every $\mu > 0$ the pair $(u^{\mu, z_0}, p^{\mu, z_0})$ is a suitable weak solution on $\mathcal{O}^{\mu, z_0}$ with force f^{μ, z_0} .*

Proof. Throughout write $u^\mu = u^{\mu, z_0}$ and let $T(y, s) = (x_0 + \mu y, t_0 + \mu^2 s)$, so that $u^\mu = \mu(u \circ T)$, and T maps $\mathcal{O}^{\mu, z_0}$ onto $\mathcal{O}$ with Jacobian μ^5 in (y, s) .

(S1). If $\Omega'' \times J''$ has compact closure in $\mathcal{O}^{\mu, z_0}$ then $T(\Omega'' \times J'')$ has compact closure in $\mathcal{O}$, and

$$\int_{\Omega''} |u^\mu(y, s)|^2 dy = \mu^2 \mu^{-3} \int_{T_x(\Omega'')} |u(\cdot, t_0 + \mu^2 s)|^2 dx = \mu^{-1} \int_{T_x(\Omega'')} |u|^2 dx,$$

where $T_x(y) = x_0 + \mu y$; by the same change of variables,

$$\iint |\nabla_y u^\mu|^2 dy ds = \mu^2 \mu^2 \mu^{-5} \iint |\nabla u|^2 dz = \mu^{-1} \iint |\nabla u|^2,$$

$$\iint |u^\mu|^2 = \mu^{-3} \iint |u|^2,$$

$$\iint |p^\mu|^{3/2} = \mu^3 \mu^{-5} \iint |p|^{3/2} = \mu^{-2} \iint |p|^{3/2}$$

and $\iint |f^\mu|^q = \mu^{3q-5} \iint |f|^q$. All are finite, so (S1) holds for (u^μ, p^μ) .

(S2) and (S3). Given $\psi \in C_c^\infty(\mathcal{O}^{\mu, z_0})$, put $\hat{\psi} = \psi \circ T^{-1} \in C_c^\infty(\mathcal{O})$. Then $\nabla_y \psi = \mu(\nabla \hat{\psi}) \circ T$ and $\iint u^\mu \cdot \nabla_y \psi dy ds = \mu \cdot \mu \cdot \mu^{-5} \iint u \cdot \nabla \hat{\psi} dz = \mu^{-3} \iint u \cdot \nabla \hat{\psi} dz = 0$, which is (S2). For (S3), let $\varphi \in C_c^\infty(\mathcal{O}^{\mu, z_0}; \mathbb{R}^3)$ and put $\hat{\varphi} = \varphi \circ T^{-1}$. Using $\partial_s \varphi = \mu^2(\partial_t \hat{\varphi}) \circ T$, $\nabla_y \varphi = \mu(\nabla \hat{\varphi}) \circ T$, one computes the five terms of (8) for $(u^\mu, p^\mu, f^\mu, \varphi)$:

$$\begin{aligned} \iint u^\mu \cdot \partial_s \varphi &= \mu \cdot \mu^2 \cdot \mu^{-5} \iint u \cdot \partial_t \hat{\varphi} = \mu^{-2} \iint u \cdot \partial_t \hat{\varphi}, \\ \iint u_i^\mu u_j^\mu \partial_j \varphi_i &= \mu^2 \cdot \mu \cdot \mu^{-5} \iint u_i u_j \partial_j \hat{\varphi}_i = \mu^{-2} \iint u_i u_j \partial_j \hat{\varphi}_i, \\ \iint \nabla_y u^\mu : \nabla_y \varphi &= \mu^2 \cdot \mu \cdot \mu^{-5} \iint \nabla u : \nabla \hat{\varphi} = \mu^{-2} \iint \nabla u : \nabla \hat{\varphi}, \\ \iint p^\mu \operatorname{div}_y \varphi &= \mu^2 \cdot \mu \cdot \mu^{-5} \iint p \operatorname{div} \hat{\varphi} = \mu^{-2} \iint p \operatorname{div} \hat{\varphi}, \\ \iint f^\mu \cdot \varphi &= \mu^3 \cdot \mu^{-5} \iint f \cdot \hat{\varphi} = \mu^{-2} \iint f \cdot \hat{\varphi}. \end{aligned}$$

Every term carries the same factor μ^{-2} , so (8) for (u^μ, p^μ, f^μ) is μ^{-2} times (8) for (u, p, f) with test field $\widehat{\phi}$, hence holds.

(S4). Let $\phi \in C_c^\infty(\mathcal{O}^{\mu, z_0})$, $\phi \geq 0$, and $\widehat{\phi} = \phi \circ T^{-1} \geq 0$. Then $\partial_s \phi + \Delta_y \phi = \mu^2 ((\partial_t \widehat{\phi} + \Delta \widehat{\phi}) \circ T)$, so

$$\begin{aligned} \iint |\nabla_y u^\mu|^2 \phi &= \mu^{-1} \iint |\nabla u|^2 \widehat{\phi}, \\ \iint |u^\mu|^2 (\partial_s \phi + \Delta_y \phi) &= \mu^2 \mu^2 \mu^{-5} \iint |u|^2 (\partial_t \widehat{\phi} + \Delta \widehat{\phi}), \\ \iint |u^\mu|^2 u^\mu \cdot \nabla_y \phi &= \mu^3 \mu \mu^{-5} \iint |u|^2 u \cdot \nabla \widehat{\phi}, \\ \iint p^\mu u^\mu \cdot \nabla_y \phi &= \mu^2 \mu \mu \mu^{-5} \iint p u \cdot \nabla \widehat{\phi}, \\ \iint (f^\mu \cdot u^\mu) \phi &= \mu^3 \mu \mu^{-5} \iint (f \cdot u) \widehat{\phi}. \end{aligned}$$

Each of the five quantities equals μ^{-1} times the corresponding quantity for $(u, p, f, \widehat{\phi})$. Since $\mu^{-1} > 0$, inequality (10) for (u, p, f) with test function $\widehat{\phi}$ is equivalent to (10) for (u^μ, p^μ, f^μ) with test function ϕ . $\square$

Lemma 3.3 (Transformation of the scale-invariant quantities). *Let (u, p, f) , z_0 , μ be as in Lemma 3.2, let $z = (x, t)$, and let $\zeta = (\mu^{-1}(x - x_0), \mu^{-2}(t - t_0))$ be the corresponding point for the rescaled solution. Then for every $r > 0$ with $\overline{Q_{\mu r}(z)} \subset \mathcal{O}$,*

$$\begin{aligned} A(\zeta, r; u^{\mu, z_0}) &= A(z, \mu r; u), & B(\zeta, r; u^{\mu, z_0}) &= B(z, \mu r; u), & C(\zeta, r; u^{\mu, z_0}) &= C(z, \mu r; u), \\ D(\zeta, r; p^{\mu, z_0}) &= D(z, \mu r; p), & E_q(\zeta, r; f^{\mu, z_0}) &= E_q(z, \mu r; f), & \Psi(\zeta, r; u^{\mu, z_0}) &= \Psi(z, \mu r; u), \end{aligned} \tag{36}$$

and likewise $\widetilde{C}(\zeta, r) = \widetilde{C}(z, \mu r)$, $\widetilde{D}(\zeta, r) = \widetilde{D}(z, \mu r)$ and $\varphi(\zeta, r) = \varphi(z, \mu r)$.

Proof. All six identities are the same computation. The map $T(y, s) = (x_0 + \mu y, t_0 + \mu^2 s)$ carries $Q_r(\zeta)$ onto $Q_{\mu r}(z)$ with $dz = \mu^5 dy ds$ and carries $B_r(\cdot)$ onto $B_{\mu r}(\cdot)$ with $dx = \mu^3 dy$. Hence, writing $v = u^{\mu, z_0}$,

$$\begin{aligned} A(\zeta, r; v) &= \frac{1}{r} \operatorname{ess\,sup}_s \int_{B_r(\cdot)} |v|^2 dy \\ &= \frac{1}{r} \mu^2 \mu^{-3} \operatorname{ess\,sup} \int_{B_{\mu r}(\cdot)} |u|^2 dx \\ &= \frac{1}{\mu r} \operatorname{ess\,sup} \int_{B_{\mu r}(\cdot)} |u|^2 dx = A(z, \mu r; u), \\ B(\zeta, r; v) &= \frac{1}{r} \mu^2 \mu^2 \mu^{-5} \iint_{Q_{\mu r}(z)} |\nabla u|^2 = \frac{1}{\mu r} \iint_{Q_{\mu r}(z)} |\nabla u|^2 = B(z, \mu r; u), \\ C(\zeta, r; v) &= \frac{1}{r^2} \mu^3 \mu^{-5} \iint_{Q_{\mu r}(z)} |u|^3 = \frac{1}{(\mu r)^2} \iint_{Q_{\mu r}(z)} |u|^3 = C(z, \mu r; u), \\ D(\zeta, r; p^{\mu, z_0}) &= \frac{1}{r^2} \mu^3 \mu^{-5} \iint_{Q_{\mu r}(z)} |p|^{3/2} = D(z, \mu r; p), \\ E_q(\zeta, r; f^{\mu, z_0}) &= r^{3-\frac{5}{q}} (\mu^{3q} \mu^{-5})^{1/q} \left(\iint_{Q_{\mu r}(z)} |f|^q \right)^{1/q} = (\mu r)^{3-\frac{5}{q}} \|f\|_{L^q(Q_{\mu r}(z))} = E_q(z, \mu r; f). \end{aligned}$$

For Ψ , the space-time mean transforms as $(v)_{Q_r(\zeta)} = \mu (u)_{Q_{\mu r}(z)}$ (the Jacobian cancels against the normalization), so $\Psi(\zeta, r; v) = r\mu \left| (u)_{Q_{\mu r}(z)} \right| = \Psi(z, \mu r; u)$. The same transformation of the mean, together with the computation for C , gives $\tilde{C}(\zeta, r) = \tilde{C}(z, \mu r)$; and since $[p^{\mu, z_0}]_{B_r(\cdot)}(s) = \mu^2 [p]_{B_{\mu r}(\cdot)}(t_0 + \mu^2 s)$, the computation for D gives $\tilde{D}(\zeta, r) = \tilde{D}(z, \mu r)$. Adding the appropriate powers yields the statement for φ . $\square$

3.2. Interpolation on balls and cylinders. We use the Sobolev and Poincaré inequalities on a ball in the following scale-explicit form; see External Input A.4 and External Input A.5.

Lemma 3.4 (Sobolev and Poincaré on a ball). *There is an absolute constant C_5 such that for every $r > 0$, every $x \in \mathbb{R}^3$ and every $v \in H^1(B_r(x))$,*

$$\|v - [v]_{B_r(x)}\|_{L^6(B_r(x))} \leq C_5 \|\nabla v\|_{L^2(B_r(x))}, \quad (37)$$

$$\|v\|_{L^6(B_r(x))} \leq C_5 \left(\|\nabla v\|_{L^2(B_r(x))} + r^{-1} \|v\|_{L^2(B_r(x))} \right), \quad (38)$$

and, for every $g \in W^{1,1}(B_r(x))$,

$$\|g - [g]_{B_r(x)}\|_{L^1(B_r(x))} \leq C_5 r \|\nabla g\|_{L^1(B_r(x))}. \quad (39)$$

Proof. Inequality (37) for $r = 1$, $x = 0$ is External Input A.5(ii); the general case follows from the change of variables $v(y) = \psi((y - x)/r)$, under which $\|v - [v]_{B_r(x)}\|_{L^6(B_r(x))} = r^{1/2} \|\psi - [\psi]_{B_1}\|_{L^6(B_1)}$ and $\|\nabla v\|_{L^2(B_r(x))} = r^{1/2} \|\nabla \psi\|_{L^2(B_1)}$, so that both sides carry the same power $r^{1/2}$ and the constant is unchanged. Inequality (39) is External Input A.5(i) rescaled in the same way: both sides carry the factor r^3 after the substitution, once the factor r on the right is accounted for. For (38), write $v = (v - [v]_{B_r(x)}) + [v]_{B_r(x)}$ and use (37) together with

$$\begin{aligned} \|[v]_{B_r(x)}\|_{L^6(B_r(x))} &= |B_r(x)|^{1/6} |[v]_{B_r(x)}| \\ &\leq |B_r(x)|^{1/6} |B_r(x)|^{-1/2} \|v\|_{L^2(B_r(x))} = \left(\frac{4\pi}{3}\right)^{-1/3} r^{-1} \|v\|_{L^2(B_r(x))}, \end{aligned}$$

where the middle step is the Cauchy–Schwarz inequality. Enlarging the constant to cover all three inequalities gives C_5 . $\square$

Lemma 3.5 (Interpolation on a ball). *Let $2 \leq \ell \leq 6$ and set*

$$a = a(\ell) = \frac{3}{4}(\ell - 2) \in [0, 3], \quad \text{so that} \quad \frac{\ell}{2} - a = \frac{6 - \ell}{4} \in [0, 1]. \quad (40)$$

There is an absolute constant C_6 such that for every $r > 0$, every $x \in \mathbb{R}^3$ and every $v \in H^1(B_r(x))$,

$$\int_{B_r(x)} |v|^\ell dy \leq C_6 \left(\int_{B_r(x)} |\nabla v|^2 \right)^a \left(\int_{B_r(x)} |v|^2 \right)^{\frac{\ell}{2} - a} + \frac{C_6}{r^{2a}} \left(\int_{B_r(x)} |v|^2 \right)^{\frac{\ell}{2}}. \quad (41)$$

Proof. Write $B = B_r(x)$. Let $\theta \in [0, 1]$ be defined by $\frac{1}{\ell} = \frac{1 - \theta}{2} + \frac{\theta}{6}$, that is $\theta = 3\left(\frac{1}{2} - \frac{1}{\ell}\right) = \frac{3(\ell - 2)}{2\ell}$; this is in $[0, 1]$ precisely because $2 \leq \ell \leq 6$. The Hölder interpolation inequality between L^2 and L^6 gives $\|v\|_{L^\ell(B)} \leq \|v\|_{L^2(B)}^{1 - \theta} \|v\|_{L^6(B)}^\theta$, hence

$$\int_B |v|^\ell = \|v\|_{L^\ell(B)}^\ell \leq \|v\|_{L^2(B)}^{\ell(1 - \theta)} \|v\|_{L^6(B)}^{\ell\theta}. \quad (42)$$

The two exponents are

$$\ell\theta = \frac{3(\ell-2)}{2} = 2a, \quad \ell(1-\theta) = \ell - \frac{3(\ell-2)}{2} = \frac{6-\ell}{2} = 2\left(\frac{\ell}{2} - a\right).$$

Inserting (38) into (42) and using $(s_1 + s_2)^{2a} \leq 2^{2a}(s_1^{2a} + s_2^{2a}) \leq 64(s_1^{2a} + s_2^{2a})$ for $s_1, s_2 \geq 0$ and $0 \leq a \leq 3$,

$$\begin{aligned} \int_B |v|^\ell &\leq 64 C_5^{2a} \|v\|_{L^2(B)}^{2(\frac{\ell}{2}-a)} \left(\|\nabla v\|_{L^2(B)}^{2a} + r^{-2a} \|v\|_{L^2(B)}^{2a} \right) \\ &= 64 C_5^{2a} \left[\left(\int_B |\nabla v|^2 \right)^a \left(\int_B |v|^2 \right)^{\frac{\ell}{2}-a} + r^{-2a} \left(\int_B |v|^2 \right)^{\frac{\ell}{2}} \right]. \end{aligned}$$

Since $0 \leq a \leq 3$, the factor C_5^{2a} is bounded by $\max\{1, C_5^6\}$; take $C_6 = 64 \max\{1, C_5^6\}$, which is absolute. $\square$

Remark 3.6 (Two instances, and a defect in the source). The two cases used below are

$$\ell = 3, \quad a = \frac{3}{4} : \quad \int_{B_r(x)} |v|^3 \leq C_6 \left(\int |\nabla v|^2 \right)^{3/4} \left(\int |v|^2 \right)^{3/4} + \frac{C_6}{r^{3/2}} \left(\int |v|^2 \right)^{3/2}, \quad (43)$$

$$\ell = \frac{10}{3}, \quad a = 1 : \quad \int_{B_r(x)} |v|^{10/3} \leq C_6 \left(\int |\nabla v|^2 \right) \left(\int |v|^2 \right)^{2/3} + \frac{C_6}{r^2} \left(\int |v|^2 \right)^{5/3}. \quad (44)$$

Integrating (44) in time over $(t-r^2, t)$ and using (S1) gives the embedding $L^\infty L^2 \cap L^2 H^1 \subset L^{10/3}$ of External Input A.4(iv).

The inequality [12, (2.1)] is printed with the exponents a and $\ell/2 - a$ interchanged, i.e. with $(\int |\nabla v|^2)^{\ell/2-a} (\int |v|^2)^a$ in place of the first term of (41). That form agrees with (41) when $\ell = 3$, the only case Lin uses, but it is false for every $\ell \in (3, 6]$. Indeed, take $r = 1$, fix $\psi \in C_c^\infty(B_1)$ with $\psi \not\equiv 0$, and set $v_\varepsilon(y) = \varepsilon^{-3/2} \psi(y/\varepsilon)$ for $0 < \varepsilon < 1$. Then

$$\int_{B_1} |v_\varepsilon|^2 = \int_{B_1} |\psi|^2, \quad \int_{B_1} |\nabla v_\varepsilon|^2 = \varepsilon^{-2} \int_{B_1} |\nabla \psi|^2, \quad \int_{B_1} |v_\varepsilon|^\ell = \varepsilon^{3-\frac{3\ell}{2}} \int_{B_1} |\psi|^\ell,$$

so the printed right side is $O(\varepsilon^{-2(\ell/2-a)}) + O(1) = O(\varepsilon^{(\ell-6)/2}) + O(1)$ while the left side is of exact order $\varepsilon^{3-3\ell/2}$; since $3 - \frac{3\ell}{2} < \min\{\frac{\ell-6}{2}, 0\}$ for $\ell > 3$, the printed inequality fails as $\varepsilon \rightarrow 0$. With the exponents as in (41) the first term is of order $\varepsilon^{-2a} = \varepsilon^{3-3\ell/2}$, matching the left side exactly.

Lemma 3.7 (Interpolation on cylinders). *There is an absolute constant C_7 with the following property. Let (u, p) satisfy (S1), let $z = (x, t)$ and let $0 < r \leq \rho$ with $\overline{Q_\rho(z)} \subset \mathcal{O}$. Then*

$$C(z, r) \leq C_7 \left[\left(\frac{r}{\rho} \right)^3 A(z, \rho)^{3/2} + \left(\frac{\rho}{r} \right)^3 A(z, \rho)^{3/4} B(z, \rho)^{3/4} \right]. \quad (45)$$

Equivalently, in Kukavica's variables, $\gamma(z, r)^3 \leq C_7 [(r/\rho)^3 \alpha(z, \rho)^3 + (\rho/r)^3 \alpha(z, \rho)^{3/2} \beta(z, \rho)^{3/2}]$.

Proof. Write $A = A(z, \rho)$, $B = B(z, \rho)$, $B_r = B_r(x)$, $B_\rho = B_\rho(x)$, and for $s \in (t - \rho^2, t)$ put

$$Y(s) = \int_{B_r} |u(y, s)|^2 dy, \quad G(s) = \int_{B_\rho} |\nabla u(y, s)|^2 dy.$$

All statements below hold for almost every such s ; we suppress ‘‘a.e.’’.

Step 1: a bound for $Y(s)$. Split

$$\int_{B_r} |u|^2 = \int_{B_r} \left(|u|^2 - [u^2]_{B_\rho}(s) \right) + |B_r| [u^2]_{B_\rho}(s) \leq \int_{B_\rho} \left| |u|^2 - [u^2]_{B_\rho}(s) \right| + \left(\frac{r}{\rho} \right)^3 \int_{B_\rho} |u|^2,$$

where we used $B_r \subset B_\rho$ in the first integral and $|B_r|/|B_\rho| = (r/\rho)^3$ in the second. Apply (39) to $g = |u(\cdot, s)|^2$, which lies in $W^{1,1}(B_\rho)$ with $|\nabla g| \leq 2|u||\nabla u|$ because $u(\cdot, s) \in H^1(B_\rho)$ and $u(\cdot, s) \in L^4(B_\rho)$ by (41) with $q = 4$:

$$\int_{B_\rho} \left| |u|^2 - [|u|^2]_{B_\rho} \right| \leq 2C_5 \rho \int_{B_\rho} |u| |\nabla u| \leq 2C_5 \rho \|u(\cdot, s)\|_{L^2(B_\rho)} G(s)^{1/2}.$$

Since $\|u(\cdot, s)\|_{L^2(B_\rho)}^2 \leq \rho A$ by definition of A ,

$$Y(s) \leq 2C_5 \rho^{3/2} A^{1/2} G(s)^{1/2} + \frac{r^3}{\rho^2} A. \quad (46)$$

Step 2: the pointwise-in-time L^3 bound. Apply (43) on B_r :

$$\int_{B_r} |u|^3 \leq C_6 \left(\int_{B_r} |\nabla u|^2 \right)^{3/4} Y(s)^{3/4} + \frac{C_6}{r^{3/2}} Y(s)^{3/2}.$$

In the first term use

$$\int_{B_r} |\nabla u|^2 \leq G(s), \quad Y(s) \leq \int_{B_\rho} |u|^2 \leq \rho A,$$

obtaining $C_6 \rho^{3/4} A^{3/4} G(s)^{3/4}$. In the second term insert (46) and use $(s_1 + s_2)^{3/2} \leq 2^{3/2}(s_1^{3/2} + s_2^{3/2})$:

$$\frac{C_6}{r^{3/2}} Y(s)^{3/2} \leq \frac{2^{3/2} C_6}{r^{3/2}} \left[(2C_5)^{3/2} \rho^{9/4} A^{3/4} G(s)^{3/4} + \frac{r^{9/2}}{\rho^3} A^{3/2} \right].$$

Hence, with $C_8 = C_6(1 + 2^{3/2}(2C_5)^{3/2} + 2^{3/2})$,

$$\int_{B_r} |u(\cdot, s)|^3 \leq C_8 \left(\rho^{3/4} + \frac{\rho^{9/4}}{r^{3/2}} \right) A^{3/4} G(s)^{3/4} + C_8 \frac{r^3}{\rho^3} A^{3/2}. \quad (47)$$

Step 3: integration in time. Integrate (47) over $s \in (t - r^2, t)$. For the first term use Hölder with exponents $\frac{4}{3}$ and 4 and then $\int_{t-r^2}^t G(s) ds = \rho B$:

$$\int_{t-r^2}^t G(s)^{3/4} ds \leq (r^2)^{1/4} \left(\int_{t-r^2}^t G(s) ds \right)^{3/4} \leq r^{1/2} (\rho B)^{3/4}.$$

The second term contributes $C_8 r^5 \rho^{-3} A^{3/2}$. Therefore

$$\begin{aligned} \iint_{Q_r(z)} |u|^3 &\leq C_8 \left(\rho^{3/4} + \frac{\rho^{9/4}}{r^{3/2}} \right) A^{3/4} r^{1/2} \rho^{3/4} B^{3/4} \\ &\quad + C_8 \frac{r^5}{\rho^3} A^{3/2} \\ &= C_8 \left(\rho^{3/2} r^{1/2} + \frac{\rho^3}{r} \right) A^{3/4} B^{3/4} + C_8 \frac{r^5}{\rho^3} A^{3/2}. \end{aligned}$$

Dividing by r^2 ,

$$C(z, r) \leq C_8 \left[\left(\frac{\rho}{r} \right)^{3/2} + \left(\frac{\rho}{r} \right)^3 \right] A^{3/4} B^{3/4} + C_8 \left(\frac{r}{\rho} \right)^3 A^{3/2},$$

and since $r \leq \rho$ gives $(\rho/r)^{3/2} \leq (\rho/r)^3$, the claim follows with $C_7 = 2C_8$. $\square$

Corollary 3.8 (Gagliardo–Nirenberg in Kukavica’s variables). *With C_7 as in Lemma 3.7, for every z and every $\rho > 0$ with $\overline{Q_\rho(z)} \subset \mathcal{O}$,*

$$\gamma(z, \rho) \leq C_9 \alpha(z, \rho)^{1/2} \beta(z, \rho)^{1/2} + C_9 \alpha(z, \rho), \quad C_9 = (2C_7)^{1/3}, \quad (48)$$

The same cylinder interpolation estimate also gives

$$C(z, \rho) \leq 2C_7 [A(z, \rho)^{3/2} + A(z, \rho)^{3/4} B(z, \rho)^{3/4}].$$

Proof. Take $r = \rho$ in (45): $C(z, \rho) \leq C_7 [A^{3/2} + A^{3/4} B^{3/4}]$. Using $\gamma = C^{1/3}$, $\alpha = A^{1/2}$, $\beta = B^{1/2}$ from Lemma 1.15 and $(s_1 + s_2)^{1/3} \leq s_1^{1/3} + s_2^{1/3}$,

$$\gamma(z, \rho) = C(z, \rho)^{1/3} \leq C_7^{1/3} (A^{1/2} + A^{1/4} B^{1/4}) = C_7^{1/3} (\alpha + \alpha^{1/2} \beta^{1/2}).$$

Since $C_7^{1/3} \leq (2C_7)^{1/3} = C_9$, (48) follows. $\square$

Lemma 3.9 (Monotonicity in the radius). *Let $z \in \mathcal{O}$ and $0 < \rho_1 < \rho_2$ with $\overline{Q_{\rho_2}(z)} \subset \mathcal{O}$. Then*

$$\begin{aligned} \alpha(z, \rho_1) &\leq \left(\frac{\rho_2}{\rho_1}\right)^{1/2} \alpha(z, \rho_2), & \beta(z, \rho_1) &\leq \left(\frac{\rho_2}{\rho_1}\right)^{1/2} \beta(z, \rho_2), \\ \gamma(z, \rho_1) &\leq \left(\frac{\rho_2}{\rho_1}\right)^{2/3} \gamma(z, \rho_2), & \delta(z, \rho_1)^2 &\leq \left(\frac{\rho_2}{\rho_1}\right)^{4/3} \delta(z, \rho_2)^2, \end{aligned} \quad (49)$$

and $\lambda(z, \rho_1) \leq (\rho_1/\rho_2)^\sigma \lambda(z, \rho_2)$.

Proof. All five follow from $Q_{\rho_1}(z) \subset Q_{\rho_2}(z)$, $B_{\rho_1}(x) \subset B_{\rho_2}(x)$ and $(t - \rho_1^2, t) \subset (t - \rho_2^2, t)$, which make every integral in (24) monotone in the radius, combined with the explicit power of r in the prefactor. For instance

$$\begin{aligned} &\alpha(z, \rho_1)^2 \\ &= \rho_1^{-1} \operatorname{ess\,sup}_{(t-\rho_1^2, t)} \int_{B_{\rho_1}(x)} |u|^2 \\ &\leq \rho_1^{-1} \operatorname{ess\,sup}_{(t-\rho_2^2, t)} \int_{B_{\rho_2}(x)} |u|^2; \\ &= (\rho_2/\rho_1) \alpha(z, \rho_2)^2 \end{aligned}$$

enlarging the domain of integration in the same way in the three remaining integrals gives $\beta(z, \rho_1)^2 \leq (\rho_2/\rho_1) \beta(z, \rho_2)^2$, $\gamma(z, \rho_1)^3 \leq (\rho_2/\rho_1)^2 \gamma(z, \rho_2)^3$ and $\delta(z, \rho_1)^3 \leq (\rho_2/\rho_1)^2 \delta(z, \rho_2)^3$; raising the last two to the powers $1/3$ and $2/3$ gives the stated forms. For λ , $\|f\|_{L^q(Q_{\rho_1}(z))} \leq \|f\|_{L^q(Q_{\rho_2}(z))}$ gives

$$\lambda(z, \rho_1) \leq \rho_1^\sigma \|f\|_{L^q(Q_{\rho_2}(z))} = (\rho_1/\rho_2)^\sigma \lambda(z, \rho_2).$$

$\square$

Remark 3.10. Kukavica [7, (2.23) and the line above it] states the fourth inequality of (49) as $\delta(\rho_1) \leq (\rho_2/\rho_1)^{4/3} \delta(\rho_2)$. The exponent $4/3$ belongs to δ^2 , not to δ : the correct statements are $\delta(\rho_1) \leq (\rho_2/\rho_1)^{2/3} \delta(\rho_2)$ and $\delta(\rho_1)^2 \leq (\rho_2/\rho_1)^{4/3} \delta(\rho_2)^2$. Since $\rho_2/\rho_1 \geq 1$ the printed inequality is weaker than the truth, so the argument of [7] is unaffected.

4. THE PRESSURE

Remark 4.1 (Operator and source conventions). The pressure identities below use weak derivatives and completed L^p Riesz operators. One may first prove the Newtonian derivative adjoint identity on compact smooth tests by heat-kernel subordination and then pass to nonsmooth compact sources by density. Exterior kernel representations are justified on sets separated from the source support. Local potential integrability and growth at infinity identify the harmonic residual by Liouville's theorem; they do not turn a singular Hessian kernel into an everywhere absolutely integrable convolution.

The singly centred tensor $-u_i(u_j - c_j)$ used in the pressure-decay estimate is distinct from the doubly centred tensor $-(u_i - c_i)(u_j - c_j)$ used in Proposition 4.11. For the latter, subtract the two distributional pairings and use incompressibility and Hessian symmetry to cancel the constant terms. The nine scalar Riesz estimates give the factor $9C_{11}$. A source obtained by differentiating the velocity tensor contains no force term: the force potentials p_7, p_8 are added separately.

Throughout this section (u, p) is a suitable weak solution on $\mathcal{O}$ with force $f \in L^q_{loc}$, $q > 5/2$, and $z_0 = (x_0, t_0) \in \mathcal{O}$ and $\rho > 0$ are such that $\overline{Q_\rho(z_0)} \subset \mathcal{O}$. We write $B_\rho = B_\rho(x_0)$, $J_\rho = (t_0 - \rho^2, t_0)$, and

$$\kappa = \frac{r}{\rho} \in (0, \frac{1}{2}] \quad (50)$$

for the two radii $0 < r \leq \rho/2$ appearing in the estimates. The symbol κ keeps this meaning up to and including Lemma 5.17, where it is still a free parameter; from Convention 5.18 onwards it denotes the *fixed* absolute constant defined there, and the functional θ of (112) is understood with that fixed value.

4.1. The spatial Poisson equation for the pressure.

Lemma 4.2 (Δp at almost every time). *For almost every $t \in J_\rho$ and every $\psi \in C_c^\infty(B_\rho)$,*

$$\int_{B_\rho} p(y, t) \Delta \psi(y) dy = \int_{B_\rho} U_{ij}(y, t) \partial_i \partial_j \psi(y) dy - \int_{B_\rho} f(y, t) \cdot \nabla \psi(y) dy, \quad (51)$$

where

$$U_{ij}(y, t) = -u_i(y, t)(u_j(y, t) - [u_j]_{B_\rho}(t)). \quad (52)$$

Equivalently, $\Delta p = \partial_i \partial_j U_{ij} + \operatorname{div} f$ in $\mathcal{D}'(B_\rho)$ for a.e. $t \in J_\rho$.

Proof. Step 1: a null set that works for all ψ . Let $\mathcal{S} \subset C_c^\infty(B_\rho)$ be a countable set that is dense in $C_c^\infty(B_\rho)$ for the C^2 norm together with uniform control of supports, for instance the set of finite rational-coefficient combinations of a fixed countable family of bump functions. It suffices to prove (51) for each fixed $\psi \in \mathcal{S}$ outside a null set N_ψ depending on ψ , since then (51) holds for all $\psi \in \mathcal{S}$ outside $N = \bigcup_{\psi \in \mathcal{S}} N_\psi$, and both sides of (51) are continuous in ψ for the C^2 norm on a fixed compact support (each integrand is in $L^1(B_\rho)$ for a.e. t).

Step 2: the mean of u against a gradient. Applying (S2) to $\psi(x)\theta(t)$ with $\theta \in C_c^\infty(J_\rho)$ gives $\int_{J_\rho} \theta(t) \left(\int_{B_\rho} u(\cdot, t) \cdot \nabla \psi \right) dt = 0$ for all such θ , hence

$$\int_{B_\rho} u(y, t) \cdot \nabla \psi(y) dy = 0 \quad \text{for a.e. } t \in J_\rho. \quad (53)$$

Step 3: testing the momentum equation with a gradient. Fix $\psi \in \mathcal{S}$ and $\theta \in C_c^\infty(J_\rho)$, and apply (S3) with $\varphi(x, t) = \theta(t)\nabla\psi(x)$, which lies in $C_c^\infty(\mathcal{O}; \mathbb{R}^3)$. The five terms of (8) are:

$$\iint u \cdot \partial_t \varphi = \int \theta'(t) \left(\int_{B_\rho} u \cdot \nabla \psi \right) dt = 0$$

by (53);

$$\iint \nabla u : \nabla \varphi = \int \theta(t) \left(\int_{B_\rho} \partial_j u_i \partial_j \partial_i \psi \right) dt = \int \theta(t) \left(- \int_{B_\rho} u_i \partial_i (\Delta \psi) \right) dt = 0,$$

where the middle equality is an integration by parts in x_j , licit because $u(\cdot, t) \in H^1(B_\rho)$ and $\psi \in C_c^\infty(B_\rho)$, and the last equality is (53) applied to $\Delta \psi \in C_c^\infty(B_\rho)$; and

$$\begin{aligned} \iint u_i u_j \partial_j \varphi_i &= \int \theta \int u_i u_j \partial_i \partial_j \psi, \\ \iint p \operatorname{div} \varphi &= \int \theta \int p \Delta \psi, \quad \iint f \cdot \varphi = \int \theta \int f \cdot \nabla \psi. \end{aligned}$$

Thus (8) reads $\int \theta(t) [- \int u_i u_j \partial_i \partial_j \psi - \int p \Delta \psi - \int f \cdot \nabla \psi] dt = 0$ for every $\theta \in C_c^\infty(J_\rho)$, whence for a.e. t

$$\int_{B_\rho} p \Delta \psi = - \int_{B_\rho} u_i u_j \partial_i \partial_j \psi - \int_{B_\rho} f \cdot \nabla \psi. \quad (54)$$

Step 4: replacing u_j by $u_j - [u_j]_{B_\rho}(t)$. Let $c \in \mathbb{R}^3$ be a constant vector. Then, for a.e. t ,

$$\int_{B_\rho} u_i c_j \partial_i \partial_j \psi = c_j \int_{B_\rho} u_i \partial_i (\partial_j \psi) = 0$$

by (53) applied to $\partial_j \psi \in C_c^\infty(B_\rho)$. Taking $c = [u]_{B_\rho}(t)$, which is a constant vector for each fixed t , turns $-\int u_i u_j \partial_i \partial_j \psi$ into $-\int u_i (u_j - c_j) \partial_i \partial_j \psi = \int U_{ij} \partial_i \partial_j \psi$. Substituting into (54) gives (51). $\square$

4.2. The cut-off and the decomposition.

Lemma 4.3 (The spatial cut-off). *There is a family of constants $C_{10}(k)$, $k \in \mathbb{N}_0$, absolute in the sense that they depend on k only, with the following property. For every x_0 and every $\rho > 0$ there is $\eta = \eta_{x_0, \rho} \in C_c^\infty(\mathbb{R}^3)$ with*

$$0 \leq \eta \leq 1, \quad \eta \equiv 1 \text{ on } B_{13\rho/20}(x_0), \quad \operatorname{supp} \eta \subset B_{3\rho/4}(x_0), \quad |\partial^a \eta| \leq C_{10}(|a|)\rho^{-|a|} \text{ on } \mathbb{R}^3. \quad (55)$$

Moreover $\nabla \eta$ and all higher derivatives of η vanish outside the annulus

$$\mathcal{A}_\rho = B_{3\rho/4}(x_0) \setminus \overline{B_{13\rho/20}(x_0)}, \quad (56)$$

and for every $x \in B_r(x_0)$ with $r \leq \rho/2$ and every $y \in \mathcal{A}_\rho$ one has $|x - y| \geq \frac{3\rho}{20}$.

Proof. Let $\zeta \in C_c^\infty(B_1)$ be a fixed nonnegative radial mollifier with $\int_{\mathbb{R}^3} \zeta = 1$, and for $\varepsilon > 0$ put $\zeta_\varepsilon(y) = \varepsilon^{-3} \zeta(y/\varepsilon)$. Since $\operatorname{supp} \zeta$ is compact in B_1 , choose $a_* < 1$ with $\operatorname{supp} \zeta \subset \overline{B_{a_*}}$. The convolution below then has support in $\overline{B_{(7/10+a_*/20)\rho}(x_0)}$, which is a compact subset of $B_{3\rho/4}(x_0)$. Set $\eta = \mathbf{1}_{B_{7\rho/10}(x_0)} * \zeta_{\rho/20}$. Then $0 \leq \eta \leq 1$; if $|y - x_0| \leq \frac{7\rho}{10} - \frac{\rho}{20} = \frac{13\rho}{20}$ then the whole ball $B_{\rho/20}(y)$ lies in $B_{7\rho/10}(x_0)$ and $\eta(y) = 1$; if $|y - x_0| \geq \frac{7\rho}{10} + \frac{\rho}{20} = \frac{3\rho}{4}$ then $B_{\rho/20}(y) \cap B_{7\rho/10}(x_0) = \emptyset$ and $\eta(y) = 0$. Also $\partial^a \eta = \mathbf{1}_{B_{7\rho/10}(x_0)} * \partial^a \zeta_{\rho/20}$, so $|\partial^a \eta| \leq \|\partial^a \zeta_{\rho/20}\|_{L^1} = (20/\rho)^{|a|} \|\partial^a \zeta\|_{L^1(\mathbb{R}^3)}$, giving (55) with $C_{10}(k) = 20^k \max_{|a|=k} \|\partial^a \zeta\|_{L^1(\mathbb{R}^3)}$. Since η is constant on $B_{13\rho/20}(x_0)$ and on the complement of $\overline{B_{3\rho/4}(x_0)}$, all its derivatives of order ≥ 1 vanish off $\mathcal{A}_\rho$. Finally if $|x - x_0| < r \leq \rho/2$ and $|y - x_0| \geq \frac{13\rho}{20}$ then $|x - y| \geq \frac{13\rho}{20} - \frac{\rho}{2} = \frac{3\rho}{20}$. $\square$

Proposition 4.4 (Local decomposition of the pressure). *Let η be as in Lemma 4.3 and let $N(y) = -\frac{1}{4\pi|y|}$ be the Newtonian kernel of $\mathbb{R}^3$, so that $\Delta N = \delta_0$. Then for almost every $t \in J_\rho$ the identity*

$$\Delta(\eta p) = \partial_i \partial_j (\eta U_{ij}) + (\partial_i \partial_j \eta) U_{ij} - \partial_j (U_{ij} \partial_i \eta) - \partial_i (U_{ij} \partial_j \eta) - p \Delta \eta + 2 \partial_j ((\partial_j \eta) p) + \eta \operatorname{div} f \quad (57)$$

holds in $\mathcal{D}'(\mathbb{R}^3)$, all functions being extended by zero outside B_ρ , and consequently

$$\eta p = p_1 + p_2 + p_3 + p_4 + p_5 + p_6 + p_7 + p_8 \quad \text{a.e. in } \mathbb{R}^3, \quad (58)$$

where, with R_i the i -th Riesz transform of $\mathbb{R}^3$ (External Input A.1),

$$\begin{aligned} p_1 &= -R_i R_j (\eta U_{ij}), & p_2 &= N * ((\partial_i \partial_j \eta) U_{ij}), \\ p_3 &= -\partial_j N * (U_{ij} \partial_i \eta), & p_4 &= -\partial_i N * (U_{ij} \partial_j \eta), \\ p_5 &= -N * (p \Delta \eta), & p_6 &= 2 \partial_j N * ((\partial_j \eta) p), \\ p_7 &= \partial_j N * (\eta f_j), & p_8 &= -N * ((\partial_j \eta) f_j). \end{aligned} \quad (59)$$

If, in addition, $\operatorname{div} f = 0$ locally in $\mathcal{D}'(\mathcal{O})$, then $p_7 + p_8 = 0$ and the last two terms may be omitted. This scalar-test condition is not part of Definition 1.5; it imposes no whole-space divergence condition on a zero extension of f .

Proof. Step 1: the identity (57). Fix t for which Lemma 4.2 holds. For $\psi \in C_c^\infty(\mathbb{R}^3)$ we have $\eta \psi \in C_c^\infty(B_\rho)$ by (55), so all pairings below are legitimate. Three elementary distributional identities are used. First, for a scalar distribution T and smooth η ,

$$\Delta(\eta T) = \eta \Delta T + 2 \partial_j \eta \partial_j T + T \Delta \eta. \quad (60)$$

Second, for a matrix field U ,

$$\eta \partial_i \partial_j U_{ij} = \partial_i \partial_j (\eta U_{ij}) + (\partial_i \partial_j \eta) U_{ij} - \partial_j (U_{ij} \partial_i \eta) - \partial_i (U_{ij} \partial_j \eta), \quad (61)$$

which is verified by expanding the right side: $\partial_i \partial_j (\eta U_{ij}) = \eta \partial_i \partial_j U_{ij} + \partial_i \eta \partial_j U_{ij} + \partial_j \eta \partial_i U_{ij} + (\partial_i \partial_j \eta) U_{ij}$, while $\partial_j (U_{ij} \partial_i \eta) = \partial_j U_{ij} \partial_i \eta + U_{ij} \partial_j \partial_i \eta$ and $\partial_i (U_{ij} \partial_j \eta) = \partial_i U_{ij} \partial_j \eta + U_{ij} \partial_i \partial_j \eta$; substituting and cancelling gives (61). Third, $2 \partial_j \eta \partial_j p = 2 \partial_j ((\partial_j \eta) p) - 2 p \Delta \eta$. Now combine: by (60) and Lemma 4.2,

$$\Delta(\eta p) = \eta (\partial_i \partial_j U_{ij} + \operatorname{div} f) + 2 \partial_j ((\partial_j \eta) p) - 2 p \Delta \eta + p \Delta \eta,$$

and inserting (61) yields exactly (57).

Step 2: the representation (58). The function $\eta p(\cdot, t)$ belongs to $L^{3/2}(\mathbb{R}^3)$ and has compact support in $\overline{B_{3\rho/4}(x_0)}$. We apply External Input A.3 with

$$G_{ij} = \eta U_{ij}, \quad H_j = -U_{ij} \partial_i \eta - U_{ji} \partial_i \eta + 2(\partial_j \eta) p + \eta f_j, \quad K = (\partial_i \partial_j \eta) U_{ij} - p \Delta \eta - (\partial_j \eta) f_j.$$

All three have compact support in $\overline{B_{3\rho/4}(x_0)}$; $G \in L^{3/2}$ because $U(\cdot, t) \in L^{3/2}(B_\rho)$ by Hölder ($|U| \leq |u| \left| u - [u]_{B_\rho} \right|$ with $u(\cdot, t) \in L^2 \cap L^6$), $H \in L^m$ with $m = \frac{5}{4} \in (1, 3)$, and $K \in L^{m'}$ with $m' = \frac{5}{4} \in (1, 3/2)$: indeed each of $U(\cdot, t)$, $p(\cdot, t)$ and $f(\cdot, t)$ lies in $L^{3/2}(B_\rho)$ (for f because $q > 5/2 > 3/2$), all the densities are supported in the bounded set $\overline{B_{3\rho/4}(x_0)}$, and $L^{3/2}(B_\rho) \subset L^{5/4}(B_\rho)$ by Hölder. The exponent $5/4$ lies strictly inside the admissible range of External Input A.3. The identity $\Delta(\eta p) = \partial_i \partial_j G_{ij} + \partial_j H_j + K$ is (57), and External Input A.3 then gives $\eta p = -R_i R_j G_{ij} + \partial_j N * H_j + N * K$, which is (58) with the terms grouped as in (59).

Step 3: the solenoidal case. If $\operatorname{div} f = 0$ in $\mathcal{D}'(\mathcal{O})$ then, arguing as in Step 2 of the proof of Lemma 4.2, $\int_{B_\rho} f(\cdot, t) \cdot \nabla \psi = 0$ for a.e. t and every $\psi \in C_c^\infty(B_\rho)$, i.e. $\operatorname{div} f(\cdot, t) = 0$ in $\mathcal{D}'(B_\rho)$. Hence the last term of (57) vanishes, and since $\eta \operatorname{div} f = \operatorname{div}(\eta f) - (\nabla \eta) \cdot f$ we get $\partial_j N * (\eta f_j) - N * ((\partial_j \eta) f_j) = N * (\eta \operatorname{div} f) = 0$, i.e. $p_7 + p_8 = 0$. $\square$

4.3. Bounds for the eight terms. Throughout this subsection $0 < r \leq \rho/2$, $\kappa = r/\rho$, $B_r = B_r(x_0)$, and for a.e. $t \in J_\rho$ we abbreviate

$$\begin{aligned} \mathbf{u}(t) &= \|u(\cdot, t)\|_{L^2(B_\rho)}, & \mathbf{g}(t) &= \|\nabla u(\cdot, t)\|_{L^2(B_\rho)}, \\ \mathbf{p}(t) &= \|p(\cdot, t)\|_{L^{3/2}(B_\rho)}, & \mathbf{f}(t) &= \|f(\cdot, t)\|_{L^q(B_\rho)}. \end{aligned} \quad (62)$$

By (24),

$$\operatorname{ess\,sup}_{J_\rho} \mathbf{u} \leq \rho^{1/2} \alpha(\rho), \quad \|\mathbf{g}\|_{L^2(J_\rho)} = \rho^{1/2} \beta(\rho), \quad \|\mathbf{p}\|_{L^{3/2}(J_\rho)} = \rho^{4/3} \delta(\rho)^2, \quad \|\mathbf{f}\|_{L^q(J_\rho)} = \rho^{-\sigma} \lambda(\rho), \quad (63)$$

where $\alpha(\rho) = \alpha(z_0, \rho)$ and so on. We use three instances of Hölder's inequality in time, valid for any measurable $g \geq 0$ on J_ρ and any $0 < r \leq \rho/2$:

$$\left(\int_{t_0-r^2}^{t_0} g^{3/2} \right)^{2/3} \leq r^{1/3} \|g\|_{L^2(J_\rho)}, \quad \left(\int_{t_0-r^2}^{t_0} g^{3/2} \right)^{2/3} \leq \|g\|_{L^{3/2}(J_\rho)}, \quad (64)$$

$$\left(\int_{t_0-r^2}^{t_0} g^{3/2} \right)^{2/3} \leq r^{2(\frac{2}{3}-\frac{1}{q})} \|g\|_{L^q(J_\rho)} \quad (q \geq \frac{3}{2}). \quad (65)$$

The first follows from Hölder with exponents $\frac{4}{3}$ and 4 (so that the measure factor is $(r^2)^{2/3-1/2} = r^{1/3}$), the third from Hölder with exponents $\frac{2q}{3}$ and its conjugate; the second is monotonicity of the integral.

Lemma 4.5 (Algebraic bounds on U). *Let U be as in (52), with its Euclidean matrix norm $|U| = (\sum_{i,j} U_{ij}^2)^{1/2}$. For almost every $t \in J_\rho$,*

$$\|U(\cdot, t)\|_{L^{3/2}(B_\rho)} \leq C_5 \mathbf{u}(t) \mathbf{g}(t), \quad \|U(\cdot, t)\|_{L^1(B_\rho)} \leq \left(\frac{4\pi}{3}\right)^{1/3} C_5 \rho \mathbf{u}(t) \mathbf{g}(t). \quad (66)$$

Proof. Pointwise $|U| \leq |u| |u - [u]_{B_\rho}|$. By Hölder with $\frac{2}{3} = \frac{1}{2} + \frac{1}{6}$ and (37),

$$\|U(\cdot, t)\|_{L^{3/2}(B_\rho)} \leq \|u(\cdot, t)\|_{L^2(B_\rho)} \|u(\cdot, t) - [u]_{B_\rho}(t)\|_{L^6(B_\rho)} \leq C_5 \mathbf{u}(t) \mathbf{g}(t).$$

For the L^1 bound, Hölder with $1 = \frac{2}{3} + \frac{1}{3}$ gives

$$\|U\|_{L^1(B_\rho)} \leq |B_\rho|^{1/3} \|U\|_{L^{3/2}(B_\rho)} = \left(\frac{4\pi}{3}\right)^{1/3} \rho \|U\|_{L^{3/2}(B_\rho)}.$$

□

All space-time norms of the pressure terms below use jointly measurable representatives. For the singular term, the localized tensor is a strongly measurable function of time with values in the separable space $L^{3/2}$; the bounded spatial Riesz operator preserves this measurability and admits a jointly measurable representative. The remaining localized potentials are handled by measurable truncated kernel integrals and their a.e. limits. In particular, their spatial power integrals and the slice norms and majorants used below are measurable functions of time, so that Fubini and the time Hölder inequalities apply.

Lemma 4.6 (Term-by-term bounds). *There is an absolute constant C_{12} , and a constant $C_{13} = C_{13}(q)$, such that for every $0 < r \leq \rho/2$, with $\kappa = r/\rho$,*

$$\frac{1}{r^{4/3}} \|p_1\|_{L^{3/2}(Q_r(z_0))} \leq C_{12} \kappa^{-1} \alpha(\rho) \beta(\rho), \quad (67)$$

$$\frac{1}{r^{4/3}} \sum_{k=2}^4 \|p_k\|_{L^{3/2}(Q_r(z_0))} \leq C_{12} \kappa \alpha(\rho) \beta(\rho), \quad (68)$$

$$\frac{1}{r^{4/3}} \sum_{k=5}^6 \|p_k\|_{L^{3/2}(Q_r(z_0))} \leq C_{12} \kappa^{2/3} \delta(\rho)^2, \quad (69)$$

$$\frac{1}{r^{4/3}} \sum_{k=7}^8 \|p_k\|_{L^{3/2}(Q_r(z_0))} \leq C_{13} \kappa \lambda(\rho). \quad (70)$$

Proof. Throughout, c denotes a quantity bounded by an absolute constant times $1 + C_5 + C_{10}(1) + C_{10}(2) + C_{11}$, where C_{11} is the constant of External Input A.1; at the end we let C_{12} be the largest of the finitely many such quantities.

(a) *The term p_1 .* By External Input A.1 applied for each fixed t to each of the nine functions $\eta U_{ij}(\cdot, t) \in L^{3/2}(\mathbb{R}^3)$,

$$\|p_1(\cdot, t)\|_{L^{3/2}(\mathbb{R}^3)} \leq C_{11} \sum_{i,j} \|\eta U_{ij}(\cdot, t)\|_{L^{3/2}(\mathbb{R}^3)} \leq 9C_{11} \|U(\cdot, t)\|_{L^{3/2}(B_\rho)} \leq 9C_{11} C_5 \mathbf{u}(t) \mathbf{g}(t),$$

using $0 \leq \eta \leq 1$, $\text{supp } \eta \subset B_\rho$ and Lemma 4.5. Restricting to B_r , raising to the power $3/2$, integrating over $(t_0 - r^2, t_0)$ and using the first inequality of (64) with $g = \mathbf{u} \mathbf{g}$ together with (63),

$$\begin{aligned} \|p_1\|_{L^{3/2}(Q_r(z_0))} &\leq 9C_{11} C_5 r^{1/3} \|\mathbf{u} \mathbf{g}\|_{L^2(J_\rho)} \\ &\leq 9C_{11} C_5 r^{1/3} (\text{ess sup } \mathbf{u}) \|\mathbf{g}\|_{L^2(J_\rho)} \leq 9C_{11} C_5 r^{1/3} \rho \alpha(\rho) \beta(\rho). \end{aligned}$$

Dividing by $r^{4/3}$ gives (67).

(b) *The terms p_2, p_3, p_4 .* The densities $(\partial_i \partial_j \eta) U_{ij}$, $U_{ij} \partial_i \eta$ and $U_{ij} \partial_j \eta$ vanish outside the annulus $\mathcal{A}_\rho$ of (56), and for $x \in B_r$, $y \in \mathcal{A}_\rho$ we have $|x - y| \geq \frac{3\rho}{20}$ by Lemma 4.3, hence

$$|N(x - y)| = \frac{1}{4\pi |x - y|} \leq \frac{5}{3\pi \rho} \leq \frac{2}{\rho}, \quad |\nabla N(x - y)| = \frac{1}{4\pi |x - y|^2} \leq \frac{100}{9\pi \rho^2} \leq \frac{4}{\rho^2}.$$

By Cauchy–Schwarz over the indices, $\sum_{i,j} |\partial_i \partial_j \eta| |U_{ij}| \leq 3C_{10}(2) \rho^{-2} |U|$ and $\sum_j |\sum_i U_{ij} \partial_i \eta| \leq 3C_{10}(1) \rho^{-1} |U|$. Therefore, for $x \in B_r$,

$$|p_2(x, t)| \leq \frac{2}{\rho} \cdot \frac{3C_{10}(2)}{\rho^2} \|U(\cdot, t)\|_{L^1(B_\rho)}, \quad |p_3(x, t)| + |p_4(x, t)| \leq 2 \cdot \frac{4}{\rho^2} \cdot \frac{3C_{10}(1)}{\rho} \|U(\cdot, t)\|_{L^1(B_\rho)},$$

so by Lemma 4.5

$$\sum_{k=2}^4 |p_k(x, t)| \leq c \rho^{-2} \mathbf{u}(t) \mathbf{g}(t).$$

Hence

$$\|p_k(\cdot, t)\|_{L^{3/2}(B_r)} \leq |B_r|^{2/3} \|p_k(\cdot, t)\|_{L^\infty(B_r)} \leq c r^2 \rho^{-2} \mathbf{u}(t) \mathbf{g}(t),$$

and by (64) and (63),

$$\sum_{k=2}^4 \|p_k\|_{L^{3/2}(Q_r(z_0))} \leq c \frac{r^2}{\rho^2} r^{1/3} \rho \alpha(\rho) \beta(\rho) = c \frac{r^{7/3}}{\rho} \alpha(\rho) \beta(\rho),$$

and dividing by $r^{4/3}$ gives (68).

(c) *The terms p_5, p_6 .* The densities $p \Delta \eta$ and $(\partial_j \eta)p$ also vanish outside $\mathcal{A}_\rho$, so the same kernel bounds give, for $x \in B_r$,

$$\begin{aligned} & |p_5(x, t)| + |p_6(x, t)| \\ & \leq \left(\frac{2}{\rho} \cdot \frac{3C_{10}(2)}{\rho^2} + 2 \cdot \frac{4}{\rho^2} \cdot \frac{3C_{10}(1)}{\rho} \right) \|p(\cdot, t)\|_{L^1(B_\rho)} \\ & \leq c \rho^{-3} \cdot \left(\frac{4\pi}{3} \right)^{1/3} \rho \mathbf{p}(t) = c \rho^{-2} \mathbf{p}(t), \end{aligned}$$

where $\|p\|_{L^1(B_\rho)} \leq |B_\rho|^{1/3} \mathbf{p}$ by Hölder. Then $\|p_5(\cdot, t) + p_6(\cdot, t)\|_{L^{3/2}(B_r)} \leq c r^2 \rho^{-2} \mathbf{p}(t)$, and by the second inequality of (64) and (63),

$$\sum_{k=5}^6 \|p_k\|_{L^{3/2}(Q_r(z_0))} \leq c \frac{r^2}{\rho^2} \|\mathbf{p}\|_{L^{3/2}(J_\rho)} = c \frac{r^2}{\rho^2} \rho^{4/3} \delta(\rho)^2 = c r^2 \rho^{-2/3} \delta(\rho)^2,$$

and dividing by $r^{4/3}$ gives $c (r/\rho)^{2/3} \delta(\rho)^2$, which is (69).

(d) *The term p_7 .* By External Input A.2 with $\frac{1}{15} = \frac{2}{5} - \frac{1}{3}$, for a.e. t , $\|p_7(\cdot, t)\|_{L^{15}(\mathbb{R}^3)} \leq C'_{11} \sum_{j=1}^3 \|\eta f_j(\cdot, t)\|_{L^{5/2}(\mathbb{R}^3)} \leq 3C'_{11} \|f(\cdot, t)\|_{L^{5/2}(B_\rho)}$. By Hölder on B_ρ (legitimate since $q > 5/2$), $\|f(\cdot, t)\|_{L^{5/2}(B_\rho)} \leq |B_\rho|^{\frac{2}{5}-\frac{1}{q}} \mathbf{f}(t) = c \rho^{\frac{6}{5}-\frac{3}{q}} \mathbf{f}(t)$, and by Hölder on B_r ,

$$\|p_7(\cdot, t)\|_{L^{3/2}(B_r)} \leq |B_r|^{\frac{2}{3}-\frac{1}{15}} \|p_7(\cdot, t)\|_{L^{15}(\mathbb{R}^3)} \leq c r^{9/5} \rho^{\frac{6}{5}-\frac{3}{q}} \mathbf{f}(t).$$

Applying (65) and (63),

$$\|p_7\|_{L^{3/2}(Q_r(z_0))} \leq c r^{9/5} \rho^{\frac{6}{5}-\frac{3}{q}} r^{\frac{4}{3}-\frac{2}{q}} \rho^{-\sigma} \lambda(\rho) = c r^{\frac{47}{15}-\frac{2}{q}} \rho^{-\frac{9}{5}+\frac{2}{q}-\frac{4}{3}+\frac{4}{3}} \lambda(\rho),$$

where we used $\frac{6}{5} - \frac{3}{q} - \sigma = \frac{6}{5} - \frac{3}{q} - 3 + \frac{5}{q} = -\frac{9}{5} + \frac{2}{q}$. Dividing by $r^{4/3}$ and recalling $\frac{47}{15} - \frac{4}{3} = \frac{27}{15} = \frac{9}{5}$,

$$\frac{1}{r^{4/3}} \|p_7\|_{L^{3/2}(Q_r(z_0))} \leq c \kappa^{\frac{9}{5}-\frac{2}{q}} \lambda(\rho) \leq c \kappa \lambda(\rho),$$

the last step because $q > 5/2$ gives $\frac{9}{5} - \frac{2}{q} > \frac{9}{5} - \frac{4}{5} = 1$ and $\kappa \leq \frac{1}{2} < 1$.

(e) *The term p_8 .* Its density $(\partial_j \eta) f_j$ is supported in $\mathcal{A}_\rho$, so for $x \in B_r$,

$$|p_8(x, t)| \leq \frac{2}{\rho} \cdot \frac{3C_{10}(1)}{\rho} \|f(\cdot, t)\|_{L^1(B_\rho)} \leq c \rho^{-2} |B_\rho|^{1-\frac{1}{q}} \mathbf{f}(t) = c \rho^{1-\frac{3}{q}} \mathbf{f}(t).$$

Hence $\|p_8(\cdot, t)\|_{L^{3/2}(B_r)} \leq c r^2 \rho^{1-\frac{3}{q}} \mathbf{f}(t)$ and, by (65),

$$\frac{1}{r^{4/3}} \|p_8\|_{L^{3/2}(Q_r(z_0))} \leq c r^{2-\frac{4}{3}+\frac{4}{3}-\frac{2}{q}} \rho^{1-\frac{3}{q}-\sigma} \lambda(\rho) = c r^{2-\frac{2}{q}} \rho^{-2+\frac{2}{q}} \lambda(\rho) = c \kappa^{2-\frac{2}{q}} \lambda(\rho) \leq c \kappa \lambda(\rho),$$

because $q > 5/2$ gives $2 - \frac{2}{q} > \frac{6}{5} > 1$. Adding (d) and (e) and taking $C_{13}(q)$ to be the resulting constant proves (70). $\square$

4.4. The pressure decay lemma.

Theorem 4.7 (Pressure decay). *There is an absolute constant C_{14} and a constant $C_{15} = C_{15}(q)$ such that for every suitable weak solution as above, every z_0 , every ρ with $Q_\rho(z_0) \subset \mathcal{O}$, and every $0 < r \leq \rho/2$, writing $\kappa = r/\rho$,*

$$\delta(z_0, r) \leq C_{14} \kappa^{-1/2} \alpha(z_0, \rho)^{1/2} \beta(z_0, \rho)^{1/2} + C_{14} \kappa^{1/3} \delta(z_0, \rho) + C_{15} \kappa^{1/2} \lambda(z_0, \rho)^{1/2}. \quad (71)$$

If moreover $\operatorname{div} f = 0$ then the last term may be omitted; that hypothesis is never used below.

Proof. Since $r \leq \rho/2 < \frac{13\rho}{20}$, Lemma 4.3 gives $\eta \equiv 1$ on B_r , so $p = \eta p$ on $Q_r(z_0)$ and (58) gives $p = \sum_{k=1}^8 p_k$ there. By definition of δ ,

$$\delta(z_0, r)^2 = \left(\frac{1}{r^2} \iint_{Q_r(z_0)} |p|^{3/2} \right)^{2/3} = \frac{1}{r^{4/3}} \|p\|_{L^{3/2}(Q_r(z_0))} \leq \frac{1}{r^{4/3}} \sum_{k=1}^8 \|p_k\|_{L^{3/2}(Q_r(z_0))},$$

by the triangle inequality in $L^{3/2}(Q_r(z_0))$. Lemma 4.6 and $\kappa \leq \frac{1}{2}$, which gives $\kappa \leq \kappa^{-1}$, yield

$$\delta(z_0, r)^2 \leq 2C_{12} \kappa^{-1} \alpha(\rho) \beta(\rho) + C_{12} \kappa^{2/3} \delta(\rho)^2 + C_{13} \kappa \lambda(\rho).$$

Taking square roots and using $(s_1 + s_2 + s_3)^{1/2} \leq s_1^{1/2} + s_2^{1/2} + s_3^{1/2}$ for $s_i \geq 0$ gives (71) with $C_{14} = (2C_{12})^{1/2}$ and $C_{15} = C_{13}^{1/2}$. Under the additional hypothesis $\operatorname{div} f = 0$ one has $p_7 + p_8 = 0$ by Proposition 4.4 and the term with C_{13} is absent. $\square$

4.5. The harmonic part and its interior estimates.

Corollary 4.8 (Calderón–Zygmund part, harmonic part, force part). *Set*

$$p_{\text{CZ}} = p_1 = -R_i R_j (\eta U_{ij}), \quad p_{\text{har}} = p_2 + p_3 + p_4 + p_5 + p_6, \quad p_{\text{f}} = p_7 + p_8, \quad (72)$$

so that $p = p_{\text{CZ}} + p_{\text{har}} + p_{\text{f}}$ on $B_{13\rho/20}(x_0)$ for a.e. $t \in J_\rho$, with $p_{\text{f}} = 0$ when $\operatorname{div} f = 0$ (which Definition 1.5 does not assume). Then for a.e. $t \in J_\rho$ the function $p_{\text{har}}(\cdot, t)$ is harmonic on $B_{13\rho/20}(x_0)$, and for every multi-index a there is a constant $C_{16}(|a|)$, depending on $|a|$ and on $C_{10}(1), C_{10}(2)$ only, with

$$\sup_{x \in B_{\rho/2}(x_0)} |\partial^a p_{\text{har}}(x, t)| \leq C_{16}(|a|) \rho^{-2-|a|} \left(\alpha(z_0, \rho)^2 + \|p(\cdot, t)\|_{L^{3/2}(B_\rho)} \right). \quad (73)$$

The exponent depends on $|a|$ and not only on its bound: replacing $\rho^{-2-|a|}$ by ρ^{-2-k} for all $|a| \leq k$ would be false for $\rho > 1$, as the suitable weak solution $u \equiv 0$, $p \equiv 1$, $f \equiv 0$ shows, for which $p_{\text{har}} \equiv 1$ on $B_{\rho/2}(x_0)$ while the right side with $k = 1$ is $O(\rho^{-1})$.

Proof. Each of $p_2, \dots, p_6$ is of the form $N * g$ or $\partial_j N * g$ with g supported in $\overline{\mathcal{A}_\rho} \subset \mathbb{R}^3 \setminus B_{13\rho/20}(x_0)$ and $g(\cdot, t) \in L^1(\mathbb{R}^3)$. For x in the open set $B_{13\rho/20}(x_0)$ the kernel $y \mapsto N(x - y)$ is smooth and harmonic in x on $\operatorname{supp} g$, and $|\partial_x^a N(x - y)| \leq c_a |x - y|^{-1-|a|}$ is integrable against $g(\cdot, t) dy$ locally uniformly in x ; hence differentiation under the integral sign is licit and $\Delta_x(N * g)(x) = \int \Delta_x N(x - y) g(y) dy = 0$ there. The same applies to $\partial_j N * g$. So $p_{\text{har}}(\cdot, t)$ is harmonic on $B_{13\rho/20}(x_0)$.

For (73), let $x \in B_{\rho/2}(x_0)$ and $y \in \mathcal{A}_\rho$, so $|x - y| \geq \frac{3\rho}{20}$ by Lemma 4.3. Differentiating under the integral, $|\partial^a p_{\text{har}}(x, t)|$ is bounded by $c_a \rho^{-1-|a|}$ times the total mass of the densities, one extra factor ρ^{-1} being present for p_3, p_4, p_6 because the kernel is $\partial_j N$; together with $|\partial \eta| \leq C_{10}(1) \rho^{-1}$, $|\partial^2 \eta| \leq C_{10}(2) \rho^{-2}$ this gives

$$|\partial^a p_{\text{har}}(x, t)| \leq c_a (C_{10}(1) + C_{10}(2)) \rho^{-3-|a|} \left(\|U(\cdot, t)\|_{L^1(B_\rho)} + \|p(\cdot, t)\|_{L^1(B_\rho)} \right).$$

Now $\|p(\cdot, t)\|_{L^1(B_\rho)} \leq (\frac{4\pi}{3})^{1/3} \rho \|p(\cdot, t)\|_{L^{3/2}(B_\rho)}$ by Hölder, and

$$\|U(\cdot, t)\|_{L^1(B_\rho)} \leq \|u(\cdot, t)\|_{L^2(B_\rho)} \|u(\cdot, t) - [u]_{B_\rho}(t)\|_{L^2(B_\rho)} \leq 2 \|u(\cdot, t)\|_{L^2(B_\rho)}^2 \leq 2\rho \alpha(z_0, \rho)^2,$$

where the middle step uses that $g \mapsto g - [g]_{B_\rho}$ has operator norm at most 2 on $L^2(B_\rho)$ (indeed at most 1, by orthogonality), and the last is (63). Substituting and absorbing the numeric factors into $C_{16}(|a|)$ gives (73). $\square$

Lemma 4.9 (Interior estimates for harmonic functions in $L^{3/2}$). *There is an absolute constant C_{17} with the following property. Let h be harmonic on $B_\rho(x_0)$ with $h \in L^{3/2}(B_\rho(x_0))$. Then*

$$\|h\|_{L^\infty(B_{3\rho/4}(x_0))} \leq C_{17} \rho^{-2} \|h\|_{L^{3/2}(B_\rho(x_0))}, \quad (74)$$

$$\|\nabla h\|_{L^\infty(B_{\rho/2}(x_0))} \leq C_{17} \rho^{-3} \|h\|_{L^{3/2}(B_\rho(x_0))}, \quad (75)$$

and for every $0 < r \leq \rho/2$,

$$\frac{1}{r^2} \int_{B_r(x_0)} |h|^{3/2} \leq C_{17} \frac{r}{\rho} \cdot \frac{1}{\rho^2} \int_{B_\rho(x_0)} |h|^{3/2}, \quad (76)$$

$$\frac{1}{r^2} \int_{B_r(x_0)} |h - [h]_{B_r(x_0)}|^{3/2} \leq C_{17} \left(\frac{r}{\rho}\right)^{5/2} \cdot \frac{1}{\rho^2} \int_{B_\rho(x_0)} |h|^{3/2}. \quad (77)$$

Proof. We use the classical smooth harmonic representative of h . If h is initially given as an $L^{3/2}$ function with $\Delta h = 0$ distributionally, Weyl's lemma gives an a.e.-equal smooth harmonic representative on the ball. All supremum and gradient estimates below refer to this representative; its $L^{3/2}$ norm agrees with that of h .

(74). Let $x \in B_{3\rho/4}(x_0)$. Then $B_{\rho/4}(x) \subset B_\rho(x_0)$, and the mean value property (External Input A.6) gives $h(x) = |B_{\rho/4}(x)|^{-1} \int_{B_{\rho/4}(x)} h$. Hence, by Hölder with $1 = \frac{2}{3} + \frac{1}{3}$,

$$|h(x)| \leq |B_{\rho/4}(x)|^{-1} |B_\rho(x_0)|^{1/3} \|h\|_{L^{3/2}(B_\rho(x_0))} = \left(\frac{4\pi}{3}\right)^{-2/3} 4^3 \rho^{-2} \|h\|_{L^{3/2}(B_\rho(x_0))}.$$

(75). Each $\partial_i h$ is harmonic on $B_\rho(x_0)$, and for $x \in B_{\rho/2}(x_0)$ the gradient estimate (External Input A.6) on the ball $B_{\rho/4}(x) \subset B_{3\rho/4}(x_0)$ gives $|\nabla h(x)| \leq \frac{3}{\rho/4} \|h\|_{L^\infty(B_{3\rho/4}(x_0))}$; combining with (74) proves (75).

(76). For $r \leq \rho/2$,

$$\begin{aligned} & \int_{B_r(x_0)} |h|^{3/2} \\ & \leq |B_r(x_0)| \|h\|_{L^\infty(B_{3\rho/4}(x_0))}^{3/2} \\ & \leq \frac{4\pi}{3} r^3 (C'_{17} \rho^{-2})^{3/2} \left(\int_{B_\rho(x_0)} |h|^{3/2} \right) \end{aligned}$$

by (74), where C'_{17} is the constant there. Dividing by r^2 gives $r\rho^{-3}$ times $\int_{B_\rho(x_0)} |h|^{3/2}$, i.e. (76).

(77). For $x \in B_r(x_0)$, $|h(x) - [h]_{B_r(x_0)}| \leq \sup_{y, y' \in B_r(x_0)} |h(y) - h(y')| \leq 2r \|\nabla h\|_{L^\infty(B_{\rho/2}(x_0))}$, since $B_r(x_0) \subset B_{\rho/2}(x_0)$ is convex. Hence, by (75),

$$\int_{B_r(x_0)} |h - [h]_{B_r(x_0)}|^{3/2} \leq \frac{4\pi}{3} r^3 (2r C'_{17} \rho^{-2})^{3/2} \int_{B_\rho(x_0)} |h|^{3/2} = c r^{9/2} \rho^{-9/2} \int_{B_\rho(x_0)} |h|^{3/2},$$

and dividing by r^2 gives $c(r/\rho)^{5/2} \rho^{-2} \int_{B_\rho(x_0)} |h|^{3/2}$. $\square$

4.6. The oscillation form of the pressure estimate. The estimates of Lemma 4.6 measure the velocity through $\alpha\beta$. For the scale iteration one also needs a form in which the velocity enters only through its oscillation. Define, for $Q_\rho(z_0) \subset \mathcal{O}$,

$$\widehat{C}(z_0, \rho) = \frac{1}{\rho^2} \iint_{Q_\rho(z_0)} |u(y, s) - [u]_{B_\rho}(s)|^3 dy ds. \quad (78)$$

Lemma 4.10 (Δp with the doubly centred nonlinearity). *For a.e. $t \in J_\rho$, $\Delta p = \partial_i \partial_j \widehat{U}_{ij} + \operatorname{div} f$ in $\mathcal{D}'(B_\rho)$, where*

$$\widehat{U}_{ij}(y, t) = -(u_i(y, t) - [u_i]_{B_\rho}(t))(u_j(y, t) - [u_j]_{B_\rho}(t)). \quad (79)$$

Proof. Write $c = [u]_{B_\rho}(t) \in \mathbb{R}^3$, a constant vector at fixed t . Expanding, $(u_i - c_i)(u_j - c_j) = u_i u_j - c_i u_j - c_j u_i + c_i c_j$. For $\psi \in C_c^\infty(B_\rho)$ and a.e. t , $\int c_i u_j \partial_i \partial_j \psi = c_i \int u_j \partial_j (\partial_i \psi) = 0$ and $\int c_j u_i \partial_i \partial_j \psi = c_j \int u_i \partial_i (\partial_j \psi) = 0$ by (53), while $\int c_i c_j \partial_i \partial_j \psi = 0$ because ψ has compact support. Hence $\int u_i u_j \partial_i \partial_j \psi = \int (u_i - c_i)(u_j - c_j) \partial_i \partial_j \psi$, and substituting into (54) gives the claim. $\square$

Proposition 4.11 (Oscillation form of the pressure estimate). *There is an absolute constant C_{18} such that for every z_0 and $\rho > 0$ with $\overline{Q_\rho(z_0)} \subset \mathcal{O}$ and every $0 < r \leq \rho/2$, the following hold.*

(i) (Pointwise in time, when $\operatorname{div} f = 0$.) *For a.e. $t \in J_\rho$,*

$$\frac{1}{r^2} \int_{B_r(x_0)} |p(y, t)|^{3/2} dy \leq C_{18} \left(\frac{\rho}{r}\right)^2 \frac{1}{\rho^2} \int_{B_\rho} |u(y, t) - [u]_{B_\rho}(t)|^3 dy + C_{18} \frac{r}{\rho} \cdot \frac{1}{\rho^2} \int_{B_\rho} |p(y, t)|^{3/2} dy. \quad (80)$$

(ii-a) (Integrated in time, when $\operatorname{div} f = 0$.) *If $\operatorname{div} f = 0$ in $\mathcal{D}'(\mathcal{O})$ then*

$$D(z_0, r) \leq C_{18} \left(\frac{\rho}{r}\right)^2 \widehat{C}(z_0, \rho) + C_{18} \frac{r}{\rho} D(z_0, \rho). \quad (81)$$

(ii-b) (Integrated in time, general force.) *Without any hypothesis on $\operatorname{div} f$, with $C_{32}(q) = 2^{1/2}(C_{18} + C_{13}(q)^{3/2})$,*

$$D(z_0, r) \leq C_{32}(q) \left[\left(\frac{\rho}{r}\right)^2 \widehat{C}(z_0, \rho) + \frac{r}{\rho} D(z_0, \rho) + \left(\frac{r}{\rho}\right)^{3/2} \lambda(z_0, \rho)^{3/2} \right]. \quad (82)$$

Proof. Run the decomposition of Proposition 4.4 with $\widehat{U}$ of (79) in place of U , which is legitimate by Lemma 4.10. In part (i) we assume $\operatorname{div} f = 0$, so that $p_7 + p_8 = 0$ and $p = \sum_{k=1}^6 p_k$ on $B_{13\rho/20}(x_0) \supset B_r(x_0)$; part (ii) adds the contribution of $p_7 + p_8$ at the end. Write $w(\cdot, t) = u(\cdot, t) - [u]_{B_\rho}(t)$, so that $|\widehat{U}| \leq |w|^2$ and $\|\widehat{U}(\cdot, t)\|_{L^{3/2}(B_\rho)} \leq \|w(\cdot, t)\|_{L^3(B_\rho)}^2$.

(i). For p_1 , External Input A.1 gives $\|p_1(\cdot, t)\|_{L^{3/2}(\mathbb{R}^3)} \leq 9C_{11} \|w(\cdot, t)\|_{L^3(B_\rho)}^2$, hence

$$\int_{B_r(x_0)} |p_1(\cdot, t)|^{3/2} \leq (9C_{11})^{3/2} \int_{B_\rho} |w(\cdot, t)|^3,$$

and dividing by r^2 yields the first term of (80) with the factor $(\rho/r)^2$. For p_2, p_3, p_4 , the far-field bounds of Lemma 4.6(b) with U replaced by $\widehat{U}$ give, for $x \in B_r(x_0)$,

$$\sum_{k=2}^4 |p_k(x, t)| \leq c \rho^{-3} \|\widehat{U}(\cdot, t)\|_{L^1(B_\rho)} \leq c \rho^{-3} |B_\rho|^{1/3} \|w(\cdot, t)\|_{L^3(B_\rho)}^2 = c \rho^{-2} \|w(\cdot, t)\|_{L^3(B_\rho)}^2,$$

so $\int_{B_r(x_0)} (\sum_{k=2}^4 |p_k|)^{3/2} \leq c r^3 \rho^{-3} \int_{B_\rho} |w(\cdot, t)|^3$, and dividing by r^2 gives $c(r/\rho) \rho^{-2} \int_{B_\rho} |w|^3 \leq c(\rho/r)^2 \rho^{-2} \int_{B_\rho} |w|^3$. For p_5, p_6 , Lemma 4.6(c) gives $|p_5| + |p_6| \leq c \rho^{-2} \|p(\cdot, t)\|_{L^{3/2}(B_\rho)}$ on $B_r(x_0)$, so

$$\int_{B_r(x_0)} (|p_5| + |p_6|)^{3/2} \leq c r^3 \rho^{-3} \int_{B_\rho} |p(\cdot, t)|^{3/2},$$

and dividing by r^2 gives $c(r/\rho)\rho^{-2} \int_{B_\rho} |p|^{3/2}$, the second term of (80). Summing the three groups, using $(\sum_k |p_k|)^{3/2} \leq 3^{1/2} \sum_k |p_k|^{3/2}$ for three groups, and absorbing numeric factors into C_{18} proves (i).

(ii-a). Integrate (80) over $t \in (t_0 - r^2, t_0)$ and enlarge the two time integrals on the right from $(t_0 - r^2, t_0)$ to $J_\rho = (t_0 - \rho^2, t_0)$, which is legitimate since both integrands are nonnegative. This gives exactly (81).

(ii-b). For a general force write $p = (\sum_{k=1}^6 p_k) + (p_7 + p_8)$ on $B_{13\rho/20}(x_0)$ and use $|a + b|^{3/2} \leq 2^{1/2}(|a|^{3/2} + |b|^{3/2})$. The first group is bounded exactly as in (ii-a), and by Lemma 4.6(d)–(e)

$$\frac{1}{r^2} \iint_{Q_r(z_0)} |p_7 + p_8|^{3/2} = \left(\frac{1}{r^{4/3}} \|p_7 + p_8\|_{L^{3/2}(Q_r(z_0))} \right)^{3/2} \leq (C_{13}(q) \kappa \lambda(z_0, \rho))^{3/2},$$

which gives (82) with $C_{32}(q) = 2^{1/2}(C_{18} + C_{13}(q)^{3/2})$. $\square$

Remark 4.12 (The pressure-decay step in Lin's proof). Lin [12, Lemma 3.4] uses a coefficient C_{θ_0} depending on the lower scale ratio θ_0 , not the small factor $C\theta_0$. His Corollary 3.5 restricts $r > \theta_0\rho$, so the factor $(\rho/r)^2$ multiplying the velocity oscillation in (80) can be absorbed into C_{θ_0} . The velocity term of the corollary is therefore consistent with that estimate.

The gap in the deduction from the printed lemma concerns the harmonic pressure term. The useful unnormalized estimate retains the factor $(r/\rho)^3$ in that term, supplied by the harmonic interior estimate. After division by r^2 this becomes $(r/\rho)\rho^{-2} \int_{B_\rho} |p|^{3/2}$, as in the corollary. Without the cubic factor, direct rescaling gives $(\rho/r)^2\rho^{-2} \int_{B_\rho} |p|^{3/2}$ instead. Thus the lemma as printed is too weak to imply its corollary; the missing cubic decay belongs to the harmonic term. We use (81), retaining the explicit scale ratios throughout.

5. THE ε -REGULARITY CORE

This section proves Theorem 2.7 and Theorem 2.9. The architecture is that of Lemarié-Rieusset [11, §13.8–13.9], which combines the scale iteration of Kukavica [7] with the parabolic Morrey/potential endgame of O'Leary [14]. There are four steps.

- Step 1** (§5.2) The local energy inequality and the pressure decomposition of Section 4 give a pair of coupled decay estimates for α, β, δ at two scales. This is [11, Lemma 13.3, (13.30)–(13.31), pp. 430–434]; we obtain it with Kukavica's backward-Gaussian test function.
- Step 2** (§5.3) Iterating these gives, on a neighbourhood of a point at which $\limsup_{r \rightarrow 0} \beta(z_0, r)$ is small, the parabolic Morrey bounds $\mathbf{1}_Q u \in \mathcal{M}^{3, \tau_2}$, $\mathbf{1}_Q \nabla u \in \mathcal{M}^{2, \tau_3}$, $\mathbf{1}_Q p \in \mathcal{M}^{3/2, \tau_p}$ with $\tau_2 > 5$. We use Kukavica's Lemma 3; compare [11, Lemma 13.4, pp. 435–439]. The pressure exponents differ, as explained in Remark 5.24.
- Step 3** (§5.4) A representation of ϕu by parabolic Riesz potentials, together with the Adams–Hedberg inequality, improves the Morrey exponent of u in finitely many rounds. This is [11, Lemma 13.5, pp. 439–441].
- Step 4** (§5.5) Once the exponents are large enough, the same representation and the parabolic Campanato lemma give Hölder continuity. This is [11, Lemma 13.6 and Proposition 13.4].

Convention 5.1 (Standing choices in this section). Throughout Section 5:

$$\begin{aligned} q > \frac{5}{2}, \quad \sigma = 3 - \frac{5}{q} > 1, \quad \varepsilon = \frac{2}{5}, \quad \tau_2 = \frac{5}{1-\varepsilon} = \frac{25}{3}, \\ \frac{1}{\tau_3} = \frac{1}{\tau_2} + \frac{1}{5}, \quad \tau_3 = \frac{25}{8}, \quad \tau_p = \frac{25}{8}. \end{aligned} \quad (83)$$

The value $\varepsilon = 2/5$ is admissible because $\varepsilon < \min\{2/3, \sigma\}$ (Proposition 5.22), and it gives $\tau_2 = 25/3$ and $\tau_3 = 25/8$. Steps 3 and 4 use these actual values in the product and potential exponent identities, not merely $\tau_2 > 5$ and $\tau_3 > 5/2$. The pressure exponent $\tau_p = 25/8$ is recorded for completeness but is *not* consumed downstream: the pressure enters Steps 3 and 4 through fixed-scale $L^{3/2}$ bounds for the harmonic remainder in Lemma 5.26. See Remark 5.24. The pressure is taken in $L_{loc}^{3/2}$, i.e. the exponent called q_0 in [11] is $q_0 = 3/2$; see Remark 5.24.

5.1. Parabolic Morrey spaces and Riesz potentials.

Definition 5.2 (Parabolic Morrey space). For $1 \leq P < \infty$ and $P \leq \tau < \infty$ and $g \in L_{loc}^P(\mathbb{R}^3 \times \mathbb{R})$ put

$$\|g\|_{\mathcal{M}^{P,\tau}} = \sup_{z \in \mathbb{R}^3 \times \mathbb{R}} \sup_{r > 0} r^{-5(\frac{1}{P} - \frac{1}{\tau})} \left(\iint_{\mathfrak{B}_r(z)} |g|^P dz' \right)^{1/P}, \quad (84)$$

with $\mathfrak{B}_r(z)$ the d_{par} -ball (3), and $\mathcal{M}^{P,\tau} = \{g : \|g\|_{\mathcal{M}^{P,\tau}} < \infty\}$.

The number 5 is the homogeneous dimension of $(\mathbb{R}^3 \times \mathbb{R}, d_{\text{par}})$: by Lemma 1.3, $|\mathfrak{B}_r(z)| = \frac{8\pi}{3}r^5$. Note $L^\tau \subset \mathcal{M}^{P,\tau}$ and $\mathcal{M}^{\tau,\tau} = L^\tau$.

Lemma 5.3 (Cylinders, one-sided or two-sided, give the same space). *Let $1 \leq P \leq \tau < \infty$ and put, for g measurable,*

$$\begin{aligned} N_1(g) &= \sup_{z,r} r^{-5(\frac{1}{P} - \frac{1}{\tau})} \left(\iint_{Q_r(z)} |g|^P \right)^{1/P}, \\ N_2(g) &= \sup_{z,r} r^{-5(\frac{1}{P} - \frac{1}{\tau})} \left(\iint_{\mathfrak{B}_r(z)} |g|^P \right)^{1/P} = \|g\|_{\mathcal{M}^{P,\tau}}. \end{aligned}$$

Then

$$N_1(g) \leq N_2(g) \leq 2^{5(\frac{1}{P} - \frac{1}{\tau})} N_1(g). \quad (85)$$

Proof. The first inequality is $Q_r(z) \subset \mathfrak{B}_r(z)$ (Lemma 1.3). For the second, fix $z = (x, t)$ and $r > 0$ and set $z' = (x, t + r^2)$. By Lemma 1.3, $\mathfrak{B}_r(z) \subset Q_{2r}(z')$, so $\iint_{\mathfrak{B}_r(z)} |g|^P \leq \iint_{Q_{2r}(z')} |g|^P \leq N_1(g)^P (2r)^{5P(\frac{1}{P} - \frac{1}{\tau})}$. Taking P -th roots and dividing by $r^{5(\frac{1}{P} - \frac{1}{\tau})}$ gives the claim. $\square$

Remark 5.4 (One-sided cylinders are what the energy inequality produces). The local energy inequality is naturally an estimate on the one-sided cylinders $Q_r(z) = B_r(x) \times (t - r^2, t]$, and the hypothesis of Theorem 2.9 is stated on these cylinders. The original [2, Proposition 2] instead uses a shifted backward cylinder; we do not identify its cylinder convention with this one. Lemarié-Rieusset [11, §13.8] uses the two-sided cylinders $(t - r^2, t + r^2) \times B_r(x)$ throughout, so his hypothesis is formally stronger than ours. Lemma 5.3 shows the difference is immaterial for Morrey norms, and Lemma 5.23 below converts the one-sided bounds produced by Step 2 into the two-sided form used by Steps 3 and 4: the point is that Step 2 produces its bounds simultaneously at *every* centre in a neighbourhood, and $\mathfrak{B}_r(z) \subset Q_{\sqrt{2}r}((x, t + r^2))$.

Remark 5.5 (Normalization of local norms). The metric ball has volume $(8\pi/3)r^5$; the backward cylinder has volume $(4\pi/3)r^5$. Thus lowering the local integrability exponent from P to P' contributes the volume of the unit domain to the power $1/P' - 1/P$, using the domain of

integration. These factors, and the comparison in (85), cannot be replaced by literal equality of ball and cylinder norms.

Definition 5.6 (Parabolic Riesz potential). For $a \in (0, 5)$ and $g \in L^1_{loc}(\mathbb{R}^3 \times \mathbb{R})$ set

$$(I_a g)(z) = \iint_{\mathbb{R}^3 \times \mathbb{R}} \frac{|g(w)|}{d_{\text{par}}(z, w)^{5-a}} dw, \quad z \in \mathbb{R}^3 \times \mathbb{R}. \quad (86)$$

External input 5.7 (Parabolic Hardy–Littlewood maximal theorem). For $g \in L^1_{loc}(\mathbb{R}^3 \times \mathbb{R})$ put

$$(\mathcal{M}g)(z) = \sup_{r>0} \frac{1}{|\mathfrak{B}_r(z)|} \iint_{\mathfrak{B}_r(z)} |g| dw.$$

Then $\mathcal{M}g$ is lower semicontinuous, hence measurable. The doubling property $|\mathfrak{B}_{2r}(z)| = 2^5 |\mathfrak{B}_r(z)|$ and the Vitali covering lemma give the weak $(1, 1)$ bound $|\{\mathcal{M}g > \varsigma\}| \leq C\varsigma^{-1} \|g\|_{L^1}$; the trivial L^∞ bound and Marcinkiewicz interpolation then give, for every $P \in (1, \infty)$, a constant $C_{19}(P)$ with $\|\mathcal{M}g\|_{L^P} \leq C_{19}(P) \|g\|_{L^P}$. The same construction yields the Lebesgue differentiation theorem on $(\mathbb{R}^3 \times \mathbb{R}, d_{\text{par}}, dz)$, used in Lemma 5.11. Reference: [18, Ch. I]; the metric measure space $(\mathbb{R}^3 \times \mathbb{R}, d_{\text{par}}, dz)$ is of homogeneous type.

Lemma 5.8 (Hedberg’s inequality). Let $1 \leq P \leq \tau < \infty$ and $0 < a < 5/\tau$. Set

$$\varkappa = \frac{a\tau}{5} \in (0, 1). \quad (87)$$

There is a constant $C_{20}(a, \tau)$ such that for every $g \in \mathcal{M}^{P, \tau}$ and every z ,

$$(I_a g)(z) \leq C_{20}(a, \tau) (\mathcal{M}g(z))^{1-\varkappa} \|g\|_{\mathcal{M}^{P, \tau}}^\varkappa. \quad (88)$$

Proof. Fix z and $R > 0$ and split $I_a g(z) = A_R + B_R$ with A_R the integral over $\{d_{\text{par}}(z, w) < R\} = \mathfrak{B}_R(z)$ and B_R the integral over the complement.

Near part. Decompose $\mathfrak{B}_R(z)$ into the dyadic shells $S_j = \mathfrak{B}_{2^{-j}R}(z) \setminus \mathfrak{B}_{2^{-j-1}R}(z)$, $j \geq 0$. On S_j we have $d_{\text{par}}(z, w)^{-(5-a)} \leq (2^{-j-1}R)^{-(5-a)}$, so

$$\begin{aligned} A_R &\leq \sum_{j \geq 0} (2^{-j-1}R)^{-(5-a)} \iint_{\mathfrak{B}_{2^{-j}R}(z)} |g| \\ &\leq \sum_{j \geq 0} (2^{-j-1}R)^{-(5-a)} \frac{8\pi}{3} (2^{-j}R)^5 \mathcal{M}g(z) \\ &= \frac{8\pi}{3} 2^{5-a} R^a \mathcal{M}g(z) \sum_{j \geq 0} 2^{-ja}, \end{aligned}$$

so $A_R \leq CR^a \mathcal{M}g(z)$ with $C = \frac{8\pi}{3} 2^{5-a} (1 - 2^{-a})^{-1}$.

Far part. Decompose the complement into $T_j = \mathfrak{B}_{2^{j+1}R}(z) \setminus \mathfrak{B}_{2^jR}(z)$, $j \geq 0$. On T_j , $d_{\text{par}}(z, w)^{-(5-a)} \leq (2^jR)^{-(5-a)}$. By Hölder’s inequality on $\mathfrak{B}_{2^{j+1}R}(z)$ with exponents P and P' , and Definition 5.2,

$$\begin{aligned} \iint_{\mathfrak{B}_{2^{j+1}R}(z)} |g| &\leq |\mathfrak{B}_{2^{j+1}R}(z)|^{1-\frac{1}{P}} \left(\iint_{\mathfrak{B}_{2^{j+1}R}(z)} |g|^P \right)^{1/P} \\ &\leq C' (2^{j+1}R)^{5(1-\frac{1}{P})} (2^{j+1}R)^{5(\frac{1}{P}-\frac{1}{\tau})} \|g\|_{\mathcal{M}^{P, \tau}} \\ &= C' (2^{j+1}R)^{5(1-\frac{1}{\tau})} \|g\|_{\mathcal{M}^{P, \tau}}, \end{aligned}$$

with $C' = (\frac{8\pi}{3})^{1-1/P}$. Hence

$$B_R \leq C' 2^5 \|g\|_{\mathcal{M}^{P,\tau}} \sum_{j \geq 0} (2^j R)^{a-\frac{5}{\tau}} = C'' R^{a-\frac{5}{\tau}} \|g\|_{\mathcal{M}^{P,\tau}},$$

the geometric series converging because $a - \frac{5}{\tau} < 0$ by hypothesis.

Optimisation. If $\mathcal{M}g(z) = 0$ then $g = 0$ a.e. and there is nothing to prove; if $\mathcal{M}g(z) = \infty$ the bound is trivial. Otherwise choose R with $R^{5/\tau} = \|g\|_{\mathcal{M}^{P,\tau}} / \mathcal{M}g(z)$, that is $R = (\|g\|_{\mathcal{M}^{P,\tau}} / \mathcal{M}g(z))^{\tau/5}$. Then $R^a \mathcal{M}g(z) = \|g\|_{\mathcal{M}^{P,\tau}}^{a\tau/5} \mathcal{M}g(z)^{1-a\tau/5}$ and $R^{a-5/\tau} \|g\|_{\mathcal{M}^{P,\tau}}$ equals the same quantity. Adding the two bounds gives (88) with $C_{20} = C + C''$. $\square$

Corollary 5.9 (Adams' theorem). *Let $1 < P \leq \tau < \infty$ and $0 < a < 5/\tau$, and set $\varkappa = a\tau/5$, $\lambda = 1 - \varkappa$, so that*

$$\frac{1}{\tau/\lambda} = \frac{1}{\tau} - \frac{a}{5}. \quad (89)$$

There is $C_{21}(a, P, \tau)$ such that

$$\|I_a g\|_{\mathcal{M}^{P/\lambda, \tau/\lambda}} \leq C_{21}(a, P, \tau) \|g\|_{\mathcal{M}^{P,\tau}} \quad \text{for all } g \in \mathcal{M}^{P,\tau}. \quad (90)$$

Proof. Write $\Lambda = \|g\|_{\mathcal{M}^{P,\tau}}$ and let $z_0, r > 0$. By Lemma 5.8, $|I_a g|^{P/\lambda} \leq C_{20}^{P/\lambda} (\mathcal{M}g)^P \Lambda^{\varkappa P/\lambda}$, since $(1 - \varkappa)P/\lambda = P$. Therefore

$$\iint_{\mathfrak{B}_r(z_0)} |I_a g|^{P/\lambda} \leq C_{20}^{P/\lambda} \Lambda^{\varkappa P/\lambda} \iint_{\mathfrak{B}_r(z_0)} (\mathcal{M}g)^P.$$

It therefore suffices to prove the Morrey bound

$$\iint_{\mathfrak{B}_r(z_0)} (\mathcal{M}g)^P \leq C r^{5(1-\frac{P}{\tau})} \Lambda^P. \quad (91)$$

Split $g = g_1 + g_2$ with $g_1 = g \mathbf{1}_{\mathfrak{B}_{2r}(z_0)}$. For g_1 , External Input 5.7 gives $\iint (\mathcal{M}g_1)^P \leq C_{19}(P)^P \iint |g_1|^P \leq C_{19}(P)^P (2r)^{5(1-\frac{P}{\tau})} \Lambda^P$. For g_2 and $z \in \mathfrak{B}_r(z_0)$, any ball $\mathfrak{B}_\varrho(z)$ meeting $\text{supp } g_2$ has $\varrho > r$, and then $\mathfrak{B}_\varrho(z) \subset \mathfrak{B}_{2\varrho}(z_0)$, so

$$\mathcal{M}g_2(z) \leq \sup_{\varrho > r} \frac{1}{\frac{8\pi}{3}\varrho^5} \iint_{\mathfrak{B}_{2\varrho}(z_0)} |g| \leq \sup_{\varrho > r} \frac{C}{\varrho^5} (2\varrho)^{5(1-\frac{1}{\tau})} \Lambda = C' r^{-5/\tau} \Lambda,$$

using the Hölder step of the previous proof. Hence $\iint_{\mathfrak{B}_r(z_0)} (\mathcal{M}g_2)^P \leq \frac{8\pi}{3} r^5 (C' r^{-5/\tau} \Lambda)^P = C'' r^{5(1-\frac{P}{\tau})} \Lambda^P$. Adding the two contributions, with the factor 2^{P-1} from $(s_1 + s_2)^P \leq 2^{P-1}(s_1^P + s_2^P)$ absorbed into C , gives (91), and therefore $\iint_{\mathfrak{B}_r(z_0)} |I_a g|^{P/\lambda} \leq C r^{5(1-\frac{P}{\tau})} \Lambda^{P/\lambda}$. Since $5(1 - \frac{P/\lambda}{\tau/\lambda}) = 5(1 - \frac{P}{\tau})$, this is exactly (90). $\square$

Remark 5.10. Lemma 5.8 and Corollary 5.9 are [11, Lemma 5.3 and Corollary 5.1, p. 93–94], stated there for a general space of homogeneous type of homogeneous dimension Q ; we have specialized to $Q = 5$ and written the proof out. Corollary 5.9 is due to Adams [1]. Note that only the Morrey exponent τ enters the condition $a\tau < 5$ and the gain (89); the integrability exponent P is carried along.

Lemma 5.11 (Parabolic Campanato lemma). *Let $P \in [1, \infty)$, $\gamma \in (0, 1)$ and $h \in L_{loc}^P(\mathbb{R}^3 \times \mathbb{R})$, and suppose*

$$H = \sup_z \sup_{r > 0} \frac{1}{r^\gamma} \left(\frac{1}{|\mathfrak{B}_r(z)|} \iint_{\mathfrak{B}_r(z)} |h - (h)_{\mathfrak{B}_r(z)}|^P \right)^{1/P} < \infty,$$

where $(h)_{\mathfrak{B}_r(z)}$ is the average of h over $\mathfrak{B}_r(z)$. Then h agrees almost everywhere with a function $\bar{h}$ with $|\bar{h}(z) - \bar{h}(z')| \leq C_{22}(\gamma, P) H d_{\text{par}}(z, z')^\gamma$ for all z, z' , and moreover, for every z and every $R > 0$,

$$\|\bar{h}\|_{L^\infty(\mathfrak{B}_R(z))} \leq 2^\gamma C_{22}(\gamma, P) H R^\gamma + \left(\frac{1}{|\mathfrak{B}_R(z)|} \iint_{\mathfrak{B}_R(z)} |h|^P \right)^{1/P}. \quad (92)$$

Conversely, if h satisfies such a Hölder bound with constant $[h]$ then $H \leq 2^\gamma [h]$.

Proof. The converse is immediate: for $w \in \mathfrak{B}_r(z)$,

$$|h(w) - (h)_{\mathfrak{B}_r(z)}| \leq |\mathfrak{B}_r(z)|^{-1} \iint_{\mathfrak{B}_r(z)} |h(w) - h(w')| dw' \leq [h] (2r)^\gamma,$$

since $d_{\text{par}}(w, w') < 2r$ for $w, w' \in \mathfrak{B}_r(z)$.

For the direct statement, fix $\Phi \in C_c^\infty(\mathbb{R}^3 \times \mathbb{R})$ supported in $\mathfrak{B}_1(0)$ with $\iint \Phi = 1$, and set $\Phi_\epsilon(x, t) = \epsilon^{-5} \Phi(x/\epsilon, t/\epsilon^2)$, so that Φ_ϵ is supported in $\mathfrak{B}_\epsilon(0)$, $\iint \Phi_\epsilon = 1$, $\|\Phi_\epsilon\|_\infty \leq \epsilon^{-5} \|\Phi\|_\infty$, $\|\nabla_x \Phi_\epsilon\|_\infty \leq \epsilon^{-6} \|\nabla \Phi\|_\infty$ and $\|\partial_t \Phi_\epsilon\|_\infty \leq \epsilon^{-7} \|\partial_t \Phi\|_\infty$.

(a) *Successive scales.* Since $\iint (\Phi_\epsilon - \Phi_{\epsilon/2}) = 0$ and both are supported in $\mathfrak{B}_\epsilon(0)$,

$$\Phi_\epsilon * h(z) - \Phi_{\epsilon/2} * h(z) = \iint_{\mathfrak{B}_\epsilon(z)} (\Phi_\epsilon(z-w) - \Phi_{\epsilon/2}(z-w)) (h(w) - (h)_{\mathfrak{B}_\epsilon(z)}) dw,$$

so, bounding the kernel by $C\epsilon^{-5} \leq C' |\mathfrak{B}_\epsilon(z)|^{-1}$ and using Hölder's inequality and the hypothesis,

$$|\Phi_\epsilon * h(z) - \Phi_{\epsilon/2} * h(z)| \leq CH\epsilon^\gamma. \quad (93)$$

(b) *Two points at one scale.* Let z, z' with $\varrho = d_{\text{par}}(z, z') \leq \epsilon$. Since $\iint (\Phi_\epsilon(z-\cdot) - \Phi_\epsilon(z'-\cdot)) = 0$ and both kernels are supported in $\mathfrak{B}_{\epsilon+\varrho}(z) \subset \mathfrak{B}_{2\epsilon}(z)$,

$$\Phi_\epsilon * h(z) - \Phi_\epsilon * h(z') = \iint_{\mathfrak{B}_{2\epsilon}(z)} (\Phi_\epsilon(z-w) - \Phi_\epsilon(z'-w)) (h(w) - (h)_{\mathfrak{B}_{2\epsilon}(z)}) dw.$$

Writing $z = (x, t)$, $z' = (x', t')$ and using the mean value theorem separately in x and in t ,

$$|\Phi_\epsilon(z-w) - \Phi_\epsilon(z'-w)| \leq \frac{|x-x'|}{\epsilon^6} \|\nabla \Phi\|_\infty + \frac{|t-t'|}{\epsilon^7} \|\partial_t \Phi\|_\infty \leq \frac{C}{\epsilon^5} \cdot \frac{\varrho}{\epsilon},$$

because $|x-x'| \leq \varrho$ and $|t-t'| \leq \varrho^2 \leq \varrho\epsilon$. Hence

$$|\Phi_\epsilon * h(z) - \Phi_\epsilon * h(z')| \leq CH\epsilon^\gamma \frac{\varrho}{\epsilon}. \quad (94)$$

(c) *A continuous representative.* By (93), for $0 < \epsilon' \leq \epsilon$ with $\epsilon' = 2^{-m}\epsilon$,

$$|\Phi_\epsilon * h(z) - \Phi_{\epsilon'} * h(z)| \leq CH \sum_{k=0}^{m-1} (2^{-k}\epsilon)^\gamma \leq \frac{CH}{1-2^{-\gamma}} \epsilon^\gamma, \quad (95)$$

uniformly in z , so $(\Phi_{2^{-k}\epsilon} * h)_{k \geq 0}$ has a uniform limit $\bar{h}$. Repeat part (a) with $\epsilon' \in [\epsilon/2, \epsilon]$ in place of $\epsilon/2$. Both kernels still have integral one, support in $\mathfrak{B}_\epsilon(0)$, and supremum bounded by $C\epsilon^{-5}$. Consequently

$$\sup_z |\Phi_\epsilon * h(z) - \Phi_{\epsilon'} * h(z)| \leq CH\epsilon^\gamma \quad (\epsilon/2 \leq \epsilon' \leq \epsilon).$$

For any small $s > 0$, choose k with $2^{-k-1} \leq s \leq 2^{-k}$ and compare $\Phi_s * h$ to $\Phi_{2^{-k}\epsilon} * h$. This comparison tends to zero uniformly as $s \downarrow 0$, proving $\bar{h} = \lim_{s \downarrow 0} \Phi_s * h$. Telescoping from an arbitrary initial scale ϵ now gives $\sup_z |\bar{h}(z) - \Phi_\epsilon * h(z)| \leq C'H\epsilon^\gamma$. The limit is continuous because each mollification is continuous. By the Lebesgue differentiation theorem on

the homogeneous space $(\mathbb{R}^3 \times \mathbb{R}, d_{\text{par}}, dz)$ (a consequence of External Input 5.7), $\Phi_\epsilon * h(z) \rightarrow h(z)$ at every Lebesgue point of h , hence $\bar{h} = h$ almost everywhere.

(d) *The Hölder bound.* Let $z \neq z'$ and $\varrho = d_{\text{par}}(z, z')$. By (95) with $\epsilon = \varrho$ and $\epsilon' \rightarrow 0$, $|\bar{h}(z) - \Phi_\varrho * h(z)| \leq C'H\varrho^\gamma$ and likewise at z' ; by (94) with $\epsilon = \varrho$, $|\Phi_\varrho * h(z) - \Phi_\varrho * h(z')| \leq CH\varrho^\gamma$. Adding the three gives $|\bar{h}(z) - \bar{h}(z')| \leq C_{22}(\gamma, P)H\varrho^\gamma$.

(e) *The supremum bound* (92). For $w \in \mathfrak{B}_R(z)$,

$$|\bar{h}(w) - (h)_{\mathfrak{B}_R(z)}| = \left| \frac{1}{|\mathfrak{B}_R(z)|} \iint_{\mathfrak{B}_R(z)} (\bar{h}(w) - \bar{h}(w')) dw' \right| \leq C_{22}(\gamma, P)H(2R)^\gamma,$$

because $d_{\text{par}}(w, w') < 2R$, and $|(h)_{\mathfrak{B}_R(z)}| \leq (|\mathfrak{B}_R(z)|^{-1} \iint_{\mathfrak{B}_R(z)} |h|^P)^{1/P}$ by Jensen's inequality. $\square$

Proposition 5.12 (Heat potentials of Morrey sources are Hölder). *Let $\gamma \in (0, 1)$ and define*

$$\frac{1}{\theta_0} = \frac{2-\gamma}{5}, \quad \frac{1}{\theta_1} = \frac{1-\gamma}{5}, \quad \text{so} \quad \theta_0 = \frac{5}{2-\gamma} > \frac{5}{2}, \quad \theta_1 = \frac{5}{1-\gamma} > 5. \quad (96)$$

Let $1 \leq P \leq \theta_0 < \theta_1 < \infty$, let $K \in \mathbb{N}$, let $\varsigma_1, \dots, \varsigma_K$ be smooth on $\mathbb{R}^3 \setminus \{0\}$ and homogeneous of degree 1, and let $\varsigma_k(D)$ be the associated Fourier multipliers, acting in the space variable only. Let $F \in \mathcal{M}^{P, \theta_0}$ and $G_1, \dots, G_K \in \mathcal{M}^{P, \theta_1}$, all supported in a fixed bounded set, and put

$$h = W_+ * \left(F + \sum_{k=1}^K \varsigma_k(D) G_k \right), \quad W_+(x, t) = \mathbf{1}_{t>0} (4\pi t)^{-3/2} e^{-|x|^2/(4t)}, \quad (97)$$

the convolution being taken in $\mathbb{R}^3 \times \mathbb{R}$ in the distributional, equivalently almost-everywhere kernel, sense. In particular, its slice formula is

$$\begin{aligned} h(x, t) &= \int_{\mathbb{R}} (W_+(\cdot, t-s) *_x F(\cdot, s))(x) ds \\ &\quad + \sum_k \int_{\mathbb{R}} (\varsigma_k(D)(W_+(\cdot, t-s) *_x G_k(\cdot, s)))(x) ds \end{aligned} \quad \text{a.e.}$$

The translated kernel products are absolutely integrable almost everywhere; this is a conclusion, not an additional source hypothesis. The distributional pairing remains an identity for every compactly supported smooth test. The finite constant C_{23} below is chosen from γ, P and the finite symbol family before the sources or their support. Then h agrees a.e. with a function $\bar{h}$ that is Hölder of exponent γ with respect to d_{par} , with

$$[\bar{h}]_{C^{\gamma, \gamma/2}} \leq C_{23}(\gamma, P, \varsigma_1, \dots, \varsigma_K) \left(\|F\|_{\mathcal{M}^{P, \theta_0}} + \sum_k \|G_k\|_{\mathcal{M}^{P, \theta_1}} \right) =: C_{23}\mathcal{N}, \quad (98)$$

and, for every z and $R > 0$,

$$\|\bar{h}\|_{L^\infty(\mathfrak{B}_R(z))} \leq 2^\gamma C_{23}\mathcal{N} R^\gamma + \left(\frac{1}{|\mathfrak{B}_R(z)|} \iint_{\mathfrak{B}_R(z)} |h|^P \right)^{1/P}. \quad (99)$$

Proof. By External Input A.11 the kernels satisfy, for $w = (y, s)$ with $s > 0$ and $\varrho = d_{\text{par}}(w, 0)$,

$$\begin{aligned} |W_+| &\leq C\varrho^{-3}, \quad |\nabla_y W_+| \leq C\varrho^{-4}, \quad |\partial_s W_+| \leq C\varrho^{-5}, \\ |\varsigma(D)W_+| &\leq C\varrho^{-4}, \quad |\nabla_y \varsigma(D)W_+| \leq C\varrho^{-5}, \quad |\partial_s \varsigma(D)W_+| \leq C\varrho^{-6}. \end{aligned} \quad (100)$$

The causal kernels vanish for $s < 0$; their values on $s = 0$ are set to zero and do not affect convolution integrals. A general multiplier kernel need not have zero right trace at $s = 0$, even when $y \neq 0$. Accordingly no classical time derivative or cross-zero mean-value estimate

is asserted there. By linearity it suffices to treat $K = 1$ and to sum the resulting bounds, so we write $\varsigma = \varsigma_1$, $G = G_1$. Note that $\varsigma(D)W_+$ is a locally integrable function by (100) ($\iint_{\{\varrho < R\}} \varrho^{-4} = \frac{40\pi}{3}R < \infty$), so $W_+ * \varsigma(D)G = (\varsigma(D)W_+) * G$.

Fix z and $r > 0$. Split $h = h^{\text{near}} + \sum_{j \geq 6} h_j$, where

$$h^{\text{near}} = W_+ * (F^{\text{near}} + \varsigma(D)G^{\text{near}}), \quad F^{\text{near}} = \mathbf{1}_{\mathfrak{B}_{2^6 r}(z)} F, \quad G^{\text{near}} = \mathbf{1}_{\mathfrak{B}_{2^6 r}(z)} G,$$

and, for $j \geq 6$, $\Gamma_j = \mathfrak{B}_{2^{j+1}r}(z) \setminus \mathfrak{B}_{2^j r}(z)$, $F_j = \mathbf{1}_{\Gamma_j} F$, $G_j = \mathbf{1}_{\Gamma_j} G$, $h_j = W_+ * (F_j + \varsigma(D)G_j)$. The sum over $j \geq 6$ is finite because F and G have bounded support. We estimate

$$\mathcal{E}_j(z, r) = \left(\frac{1}{|\mathfrak{B}_r(z)|^2} \iint_{\mathfrak{B}_r(z)} \iint_{\mathfrak{B}_r(z)} |h_j(w) - h_j(w')|^P dw dw' \right)^{1/P}. \quad (101)$$

Near part, treated as one block. No cancellation is used. For $w \in \mathfrak{B}_r(z)$ and $v \in \mathfrak{B}_{2^6 r}(z)$ we have $d_{\text{par}}(w, v) < 2^7 r$, so Young's inequality with the truncated kernels gives

$$\|h^{\text{near}}\|_{L^P(\mathfrak{B}_r(z))} \leq \|\mathbf{1}_{\varrho < 2^7 r} \varrho^{-3}\|_{L^1} \|F^{\text{near}}\|_{L^P} + C_\varsigma \|\mathbf{1}_{\varrho < 2^7 r} \varrho^{-4}\|_{L^1} \|G^{\text{near}}\|_{L^P}.$$

Since $|\mathfrak{B}_s| = \frac{8\pi}{3}s^5$, we have $\iint_{\{\varrho < R\}} \varrho^{-3} = \frac{40\pi}{3} \int_0^R s ds = \frac{20\pi}{3} R^2$ and $\iint_{\{\varrho < R\}} \varrho^{-4} = \frac{40\pi}{3} R$, so the two kernel norms are Cr^2 and Cr . By Definition 5.2, $\|F^{\text{near}}\|_{L^P} \leq (2^6 r)^{5(\frac{1}{P} - \frac{1}{\theta_0})} \|F\|_{\mathcal{M}^{P, \theta_0}}$ and $\|G^{\text{near}}\|_{L^P} \leq (2^6 r)^{5(\frac{1}{P} - \frac{1}{\theta_1})} \|G\|_{\mathcal{M}^{P, \theta_1}}$. Using $\mathcal{E}^{\text{near}} \leq 2|\mathfrak{B}_r|^{-1/P} \|h^{\text{near}}\|_{L^P(\mathfrak{B}_r(z))}$, $|\mathfrak{B}_r|^{-1/P} = Cr^{-5/P}$, and $2 - 5/\theta_0 = 1 - 5/\theta_1 = \gamma$ from (96),

$$\mathcal{E}^{\text{near}}(z, r) \leq Cr^\gamma (\|F\|_{\mathcal{M}^{P, \theta_0}} + \|G\|_{\mathcal{M}^{P, \theta_1}}).$$

No series and no constraint beyond $P \leq \theta_0$ is involved.

Far shells, $j \geq 6$. Write $L = 2^j r$, let $w, w' \in \mathfrak{B}_r(z)$ and split Γ_j into the regions where $t_w - s$ and $t_{w'} - s$ are both positive, both nonpositive, or have opposite signs. Spatial separation along the intervening segments, or time separation when it is the larger coordinate, is comparable to L . On the first region the separate space and time mean-value estimates give

$$\begin{aligned} |W_+(w - v) - W_+(w' - v)| &\leq C d_{\text{par}}(w, w') L^{-4}, \\ |(\varsigma(D)W_+)(w - v) - (\varsigma(D)W_+)(w' - v)| &\leq C d_{\text{par}}(w, w') L^{-5}. \end{aligned}$$

The contribution from the second region vanishes. By Hölder and the Morrey bounds, the first region contributes at most

$$CrL^{\gamma-1} (\|F\|_{\mathcal{M}^{P, \theta_0}} + \|G\|_{\mathcal{M}^{P, \theta_1}}).$$

Here $1 - 5/\theta_0 = \gamma - 1$ and $-5/\theta_1 = \gamma - 1$.

On the crossing region the source time s lies between t_w and $t_{w'}$, an interval of length at most $2r^2$. Since $j \geq 6$, membership in Γ_j then forces spatial separation comparable to L . Cover this spatial shell and time interval by at most $C(L/r)^3$ parabolic balls of radius Cr . Hölder on each such ball gives

$$\iint_{\text{crossing region}} |G| \leq C(L/r)^3 r^{5-5/\theta_1} \|G\|_{\mathcal{M}^{P, \theta_1}} = CL^3 r^{1+\gamma} \|G\|_{\mathcal{M}^{P, \theta_1}}.$$

The kernel size bound CL^{-4} therefore bounds its crossing contribution by $C(r/L)r^\gamma \|G\|_{\mathcal{M}^{P, \theta_1}}$. This estimate does not differentiate through time zero. For the Gaussian part, the zero extension and its time derivative vanish at $t = 0$ at nonzero spatial separation. Alternatively, $W_+(x, \tau) \leq C\tau L^{-5}$ for $|x| \asymp L$, $0 < \tau \leq 2r^2$, by the exponential formula. The same cover yields

$$\iint_{\text{crossing region}} |F| \leq CL^3 r^\gamma \|F\|_{\mathcal{M}^{P, \theta_0}},$$

so this contribution is at most $C(r/L)^2 r^\gamma \|F\|_{\mathcal{M}^{P,\theta_0}}$. Consequently

$$\mathcal{E}_j(z, r) \leq C r^\gamma (2^{j(\gamma-1)} + 2^{-j} + 2^{-2j}) (\|F\|_{\mathcal{M}^{P,\theta_0}} + \|G\|_{\mathcal{M}^{P,\theta_1}}).$$

All three shell series converge for $0 < \gamma < 1$.

For comparison with a pointwise shell-difference estimate, set $\delta = d_{\text{par}}(w, w')$. The case $\delta = 0$ is immediate. Otherwise repeat the crossing cover at radius $C\delta$ instead of Cr ; the time strip has length at most δ^2 . The two crossing contributions are bounded by $C\delta^{1+\gamma}L^{-1}\|G\|_{\mathcal{M}^{P,\theta_1}}$ and $C\delta^{2+\gamma}L^{-2}\|F\|_{\mathcal{M}^{P,\theta_0}}$. Since $\delta \leq 2r \leq L$, each is at most the corresponding source norm times $C\delta L^{\gamma-1}$. Together with the positive-time region, this gives

$$|h_j(w) - h_j(w')| \leq C\delta L^{\gamma-1} (\|F\|_{\mathcal{M}^{P,\theta_0}} + \|G\|_{\mathcal{M}^{P,\theta_1}}).$$

Conclusion. By the triangle inequality in $L^P(\mathfrak{B}_r \times \mathfrak{B}_r)$ and Jensen's inequality,

$$\left(\frac{1}{|\mathfrak{B}_r(z)|} \iint_{\mathfrak{B}_r(z)} |h - (h)_{\mathfrak{B}_r(z)}|^P \right)^{1/P} \leq \mathcal{E}^{\text{near}}(z, r) + \sum_{j \geq 6} \mathcal{E}_j(z, r) \leq C r^\gamma (\|F\|_{\mathcal{M}^{P,\theta_0}} + \|G\|_{\mathcal{M}^{P,\theta_1}}),$$

uniformly in z and r . Lemma 5.11 now gives (98) with $C_{23} = C_{22}(\gamma, P) \cdot C$, and (99) follows by averaging the resulting Hölder bound over $\mathfrak{B}_R(z)$, as in part (e) of that lemma. Its coefficient is $2^\gamma C_{23}$, since C_{23} already includes C_{22} . For $K > 1$ apply the above to each pair $(0, G_k)$ and to $(F, 0)$ and add the $K + 1$ resulting seminorm bounds. $\square$

Remark 5.13 (Averages and the continuous representative). There is also a direct dyadic proof of Lemma 5.11, avoiding mollification. Set $a_r(z) = (h)_{\mathfrak{B}_r(z)}$. Hölder's inequality and the volume ratio give

$$|a_{r/2}(z) - a_r(z)| \leq 2^{5/P} H r^\gamma.$$

Telescoping at dyadic radii gives a locally uniform limit of the averages, equal to h almost everywhere by differentiation. To compare two centres at distance r , compare both radius- r averages to the average on a radius- $3r$ ball containing both balls. This gives a bound CHr^γ ; the two telescoping tails give the same bound. The resulting single representative is Hölder everywhere. Averaging its Hölder estimate gives (92). This is an alternative to the mollification proof above; its constants may depend on (γ, P) .

5.2. Step 1: the local energy and pressure estimates. All two-scale estimates in this section are applied with the closure of the outer cylinder inside the solution domain. Cutoff coefficients are fixed before the solution. The absolute energy constants and the q -dependent force constants may be enlarged to dominate the finitely many explicit cutoff and norm-comparison constants, without changing their stated dependence. The iteration uses one fixed contraction κ chosen in (115); a bound proved only at that choice is not an assertion uniform over every contraction.

Lemma 5.14 (The backward Gaussian test function). *Let $G(x, \vartheta) = (4\pi\vartheta)^{-3/2} e^{-|x|^2/(4\vartheta)}$ and, for $r > 0$ and $z_0 = (x_0, t_0)$, set*

$$\psi_r(x, t) = r^2 G(x - x_0, r^2 - (t - t_0)), \quad (x, t) \in \mathbb{R}^3 \times (-\infty, t_0 + r^2). \quad (102)$$

Then ψ_r is smooth and positive there, and there is an absolute constant $C_{24} \geq 1$ such that, for all $0 < r \leq \rho/2$,

$$\partial_t \psi_r + \Delta \psi_r = 0 \quad \text{on } \mathbb{R}^3 \times (-\infty, t_0 + r^2), \quad (103)$$

$$\psi_r \geq C_{24}^{-1} r^{-1} \quad \text{on } Q_r(z_0), \quad (104)$$

$$\psi_r \leq C_{24} r^{-1}, \quad |\nabla \psi_r| \leq C_{24} r^{-2} \quad \text{on } \mathbb{R}^3 \times (-\infty, t_0], \quad (105)$$

$$\psi_r \leq C_{24} \frac{r^2}{\rho^3}, \quad |\nabla \psi_r| \leq C_{24} \frac{r^2}{\rho^4} \quad \text{on } Q_\rho(z_0) \setminus Q_{\rho/2}(z_0). \quad (106)$$

Proof. We may take $z_0 = (0, 0)$. Write $\vartheta = r^2 - t > 0$ for $t < r^2$ and $s = |x|^2/(4\vartheta)$. By External Input A.10, $\partial_\vartheta G = \Delta G$, so $\partial_t \psi_r = -r^2(\partial_\vartheta G)(x, \vartheta) = -r^2(\Delta G)(x, \vartheta) = -\Delta \psi_r$, which is (103).

(104). On Q_r we have $|x| < r$ and $t \in (-r^2, 0]$, so $\vartheta \in [r^2, 2r^2]$ and $s \leq r^2/(4r^2) = 1/4$. Hence $\psi_r \geq r^2(8\pi r^2)^{-3/2} e^{-1/4} = (8\pi)^{-3/2} e^{-1/4} r^{-1}$.

(105). For $t \leq 0$, $\vartheta \geq r^2$, so $\psi_r \leq r^2(4\pi\vartheta)^{-3/2} \leq (4\pi)^{-3/2} r^{-1}$. For the gradient, $\nabla \psi_r = -r^2 \frac{x}{2\vartheta} G(x, \vartheta)$, and $\frac{|x|}{2\vartheta} = \vartheta^{-1/2} s^{1/2}$, so

$$|\nabla \psi_r| = r^2(4\pi)^{-3/2} \vartheta^{-2} s^{1/2} e^{-s} \leq r^2(4\pi)^{-3/2} (2e)^{-1/2} \vartheta^{-2}, \quad (107)$$

using $\sup_{s \geq 0} s^{1/2} e^{-s} = (2e)^{-1/2}$. With $\vartheta \geq r^2$ this gives $|\nabla \psi_r| \leq (4\pi)^{-3/2} (2e)^{-1/2} r^{-2}$.

(106). Let $(x, t) \in Q_\rho \setminus Q_{\rho/2}$, so $t \in (-\rho^2, 0]$ and either (A) $t \leq -\rho^2/4$, or (B) $|x| \geq \rho/2$.

In case (A), $\vartheta = r^2 - t \geq \rho^2/4$, hence $\psi_r \leq r^2(4\pi\vartheta)^{-3/2} \leq r^2(\pi\rho^2)^{-3/2} = \pi^{-3/2} r^2 \rho^{-3}$, and by (107) $|\nabla \psi_r| \leq (4\pi)^{-3/2} (2e)^{-1/2} r^2 (4/\rho^2)^2 = 16(4\pi)^{-3/2} (2e)^{-1/2} r^2 \rho^{-4}$.

In case (B), put $\varsigma = \rho^2/(16\vartheta)$, so that $s = |x|^2/(4\vartheta) \geq \varsigma$ and $\vartheta = \rho^2/(16\varsigma)$. Then

$$\begin{aligned} \psi_r &\leq r^2(4\pi\vartheta)^{-3/2} e^{-\varsigma} = r^2(4\pi)^{-3/2} \left(\frac{16\varsigma}{\rho^2}\right)^{3/2} e^{-\varsigma} \\ &= 64(4\pi)^{-3/2} \frac{r^2}{\rho^3} \varsigma^{3/2} e^{-\varsigma} \leq 64(4\pi)^{-3/2} \left(\frac{3}{2}\right)^{3/2} e^{-3/2} \frac{r^2}{\rho^3}, \end{aligned}$$

using $\sup_{\varsigma \geq 0} \varsigma^{3/2} e^{-\varsigma} = (3/2)^{3/2} e^{-3/2}$. For the gradient, (107) and $s^{1/2} e^{-s} \leq e^{-1/2} e^{-s/2} \leq e^{-1/2} e^{-\varsigma/2}$ (valid because $\sup_s s^{1/2} e^{-s/2} = e^{-1/2}$) give

$$|\nabla \psi_r| \leq r^2(4\pi)^{-3/2} e^{-1/2} \left(\frac{16\varsigma}{\rho^2}\right)^2 e^{-\varsigma/2} = 256(4\pi)^{-3/2} e^{-1/2} \frac{r^2}{\rho^4} \varsigma^2 e^{-\varsigma/2} \leq C \frac{r^2}{\rho^4}.$$

Taking C_{24} to be the largest of the finitely many constants produced in (105)–(106) and of $((8\pi)^{-3/2} e^{-1/4})^{-1}$, which is what makes (104) read $\psi_r \geq C_{24}^{-1} r^{-1}$, completes the proof. $\square$

Lemma 5.15 (Caccioppoli inequality; [11, (13.30)]). *There is an absolute constant C_{25} and a constant $C_{26} = C_{26}(q)$ such that for every suitable weak solution (u, p) on $\mathcal{O}$ with force $f \in L^q_{loc}$, $q > 5/2$, every z_0 and $\rho > 0$ with $Q_\rho(z_0) \subset \mathcal{O}$, and every $0 < r \leq \rho/2$, writing $\kappa = r/\rho$ and suppressing the base point z_0 ,*

$$\begin{aligned} \alpha(r) + \beta(r) &\leq C_{25} \kappa \alpha(\rho) \\ &\quad + C_{25} \kappa^{-1} \alpha(\rho)^{1/2} \beta(\rho)^{1/2} \gamma(\rho)^{1/2} \\ &\quad + C_{25} \kappa^{-1} \delta(\rho) \gamma(\rho)^{1/2} + C_{26} \kappa^{-1/2} \gamma(\rho)^{1/2} \lambda(\rho)^{1/2}. \end{aligned} \quad (108)$$

Proof. Write $B_\varrho = B_\varrho(x_0)$ and $J_\rho = (t_0 - \rho^2, t_0)$. Since $t_0 \in I$ and I is open, there is $\epsilon_0 > 0$ with $[t_0 - \rho^2 + \frac{1}{2}\epsilon_0, t_0 + \epsilon_0] \subset I$; shrinking ϵ_0 we may and do also require

$$\epsilon_0 < \frac{1}{2} r^2, \quad (109)$$

which is legitimate because $r > 0$ is fixed before ϵ_0 is chosen and only smallness of ϵ_0 is used. By (109) the function ψ_r of Lemma 5.14 is smooth on a neighbourhood of $\overline{B_\rho(x_0)} \times [t_0 - \rho^2, t_0 + \epsilon_0]$, since that set lies in $\mathbb{R}^3 \times (-\infty, t_0 + r^2)$.

Step 1: the test function. Let $\eta_1 \in C_c^\infty(B_\rho)$ with $0 \leq \eta_1 \leq 1$, $\eta_1 \equiv 1$ on $B_{\rho/2}$, $\text{supp } \eta_1 \subset B_{3\rho/4}$ and $|\partial^a \eta_1| \leq C_{10}(|a|)\rho^{-|a|}$; this is Lemma 4.3 verbatim, up to renaming the two radii $13\rho/20 \rightsquigarrow \rho/2$ and $3\rho/4 \rightsquigarrow 3\rho/4$, which only changes the numerical constants. Let $\eta_2 \in C^\infty(\mathbb{R})$ with $0 \leq \eta_2 \leq 1$, $\eta_2 \equiv 0$ on $(-\infty, t_0 - \rho^2 + \frac{1}{4}\epsilon_0]$, $\eta_2 \equiv 1$ on $[t_0 - \rho^2/4, t_0 + \frac{1}{2}\epsilon_0]$, $\text{supp } \eta_2 \subset (t_0 - \rho^2 + \frac{1}{8}\epsilon_0, t_0 + \epsilon_0)$ and $|\eta_2'| + \rho^2 |\eta_2''| \leq C\rho^{-2}$ on $(-\infty, t_0]$, where the transition occupies the interval $[t_0 - \rho^2 + \frac{1}{4}\epsilon_0, t_0 - \rho^2/4]$ of length $\geq \rho^2/2$; such η_2 exists by mollifying a piecewise affine function as in Lemma 4.3. Put $\eta = \eta_1 \eta_2$ and

$$\phi = \psi_r \eta \in C_c^\infty(\mathcal{O}), \quad \phi \geq 0,$$

with ψ_r as in Lemma 5.14. Then $\eta \equiv 1$ on $Q_{\rho/2}(z_0)$, and $\nabla \eta$, $\partial_t \eta$, $\Delta \eta$ all vanish on $Q_{\rho/2}(z_0)$ and, for $t \leq t_0$, are supported in $Q_\rho(z_0) \setminus Q_{\rho/2}(z_0)$.

Step 2: the four terms. Apply Lemma 1.13 with this ϕ and $t \in (t_0 - r^2, t_0]$. Since $\partial_t \psi_r + \Delta \psi_r = 0$,

$$\partial_t \phi + \Delta \phi = (\partial_t \eta + \Delta \eta) \psi_r + 2 \nabla \eta \cdot \nabla \psi_r, \quad (110)$$

which vanishes on $Q_{\rho/2}(z_0)$ and, by (106) and the bounds on η , satisfies $|\partial_t \phi + \Delta \phi| \leq C\rho^{-2} \cdot C_{24}r^2\rho^{-3} + 2C\rho^{-1}C_{24}r^2\rho^{-4} \leq C'r^2\rho^{-5}$ on $Q_\rho(z_0)$, $t \leq t_0$. Also, by (105) and $\rho \geq 2r$,

$$|\nabla \phi| \leq \eta |\nabla \psi_r| + \psi_r |\nabla \eta| \leq C_{24}r^{-2} + C_{24}r^{-1}C\rho^{-1} \leq C'r^{-2} \quad \text{on } Q_\rho(z_0), \quad (111)$$

and $\phi \leq C_{24}r^{-1}$. Since $\text{supp } \phi(\cdot, s) \subset \overline{B_{3\rho/4}} =: E$ for every s , Lemma 1.12 supplies a single null set $N \subset J_\rho$ off which $\int_{B_\rho} u(\cdot, s) \cdot \nabla \psi = 0$ for all $\psi \in C_c^1(B_\rho)$ supported in E ; applying it to $\psi = \phi(\cdot, s)$ gives $\int_{B_\rho} \Gamma(s) u(\cdot, s) \cdot \nabla \phi(\cdot, s) dy = 0$ for $s \notin N$ and any function Γ of time alone. We use this with $\Gamma(s) = [|u|^2]_{B_\rho}(s)$. Therefore Lemma 1.13 gives, for almost every $t \in (t_0 - r^2, t_0]$,

$$\int |u(\cdot, t)|^2 \phi(\cdot, t) + 2 \int_{t_0 - \rho^2}^t \int |\nabla u|^2 \phi \leq I_1 + I_2 + I_3 + I_4,$$

where

$$\begin{aligned} I_1 &= \iint_{Q_\rho(z_0)} |u|^2 |\partial_s \phi + \Delta \phi|, \\ I_2 &= \iint_{Q_\rho(z_0)} \left| |u|^2 - [|u|^2]_{B_\rho} \right| |u| |\nabla \phi|, \\ I_3 &= 2 \iint_{Q_\rho(z_0)} |p| |u| |\nabla \phi|, \\ I_4 &= 2 \iint_{Q_\rho(z_0)} |f| |u| \phi. \end{aligned}$$

I_1 . By the bound on $\partial_s \phi + \Delta \phi$,

$$I_1 \leq C \frac{r^2}{\rho^5} \iint_{Q_\rho(z_0)} |u|^2 \leq C \frac{r^2}{\rho^5} \rho^2 \text{ess sup}_{J_\rho} \int_{B_\rho} |u|^2 = C \frac{r^2}{\rho^3} \rho \alpha(\rho)^2 = C \kappa^2 \alpha(\rho)^2.$$

I_2 . By (111) and Hölder's inequality with $\frac{2}{3} + \frac{1}{3} = 1$,

$$I_2 \leq \frac{C}{r^2} \left\| |u|^2 - [|u|^2]_{B_\rho} \right\|_{L^{3/2}(Q_\rho(z_0))} \|u\|_{L^3(Q_\rho(z_0))}.$$

For a.e. s , apply the squared-velocity Sobolev–Poincaré estimate recorded after External Input A.5. The closed outer-cylinder containment supplies a slightly larger spatial ball on which the slice has its H^1 weak gradient. The estimate gives $\| |u|^2 - [u^2]_{B_\rho}(s) \|_{L^{3/2}(B_\rho)} \leq 2C \|u(\cdot, s)\|_{L^2(B_\rho)} \|\nabla u(\cdot, s)\|_{L^2(B_\rho)}$. Taking the $L^{3/2}$ norm in s over J_ρ and using Hölder in time as in (64),

$$\begin{aligned} & \| |u|^2 - [u^2]_{B_\rho} \|_{L^{3/2}(Q_\rho(z_0))} \\ & \leq C(\text{ess sup } \mathbf{u}) |J_\rho|^{1/6} \|\mathbf{g}\|_{L^2(J_\rho)} \\ & \leq C\rho^{1/2}\alpha(\rho) \cdot \rho^{1/3} \cdot \rho^{1/2}\beta(\rho) = C\rho^{4/3}\alpha(\rho)\beta(\rho), \end{aligned}$$

with $\mathbf{u}, \mathbf{g}$ as in (62). Since $\|u\|_{L^3(Q_\rho(z_0))} = \rho^{2/3}\gamma(\rho)$,

$$I_2 \leq \frac{C}{r^2} \rho^{4/3} \alpha(\rho) \beta(\rho) \rho^{2/3} \gamma(\rho) = C \frac{\rho^2}{r^2} \alpha(\rho) \beta(\rho) \gamma(\rho) = C \kappa^{-2} \alpha(\rho) \beta(\rho) \gamma(\rho).$$

I_3 . By (111) and Hölder, $I_3 \leq Cr^{-2} \|p\|_{L^{3/2}(Q_\rho(z_0))} \|u\|_{L^3(Q_\rho(z_0))} = Cr^{-2} \rho^{4/3} \delta(\rho)^2 \rho^{2/3} \gamma(\rho) = C \kappa^{-2} \delta(\rho)^2 \gamma(\rho)$.

I_4 . Since $\phi \leq C_{24} r^{-1}$ and $q > 5/2$, Hölder with $\frac{1}{q} + \frac{1}{q'} = 1$ and then Hölder on $Q_\rho(z_0)$ (using $q' < \frac{5}{3} < 3$) give

$$\begin{aligned} I_4 & \leq \frac{C}{r} \|f\|_{L^q(Q_\rho(z_0))} \|u\|_{L^{q'}(Q_\rho(z_0))} \\ & \leq \frac{C}{r} \rho^{-\sigma} \lambda(\rho) \cdot |Q_\rho(z_0)|^{\frac{1}{q'} - \frac{1}{3}} \rho^{2/3} \gamma(\rho) \\ & \leq \frac{C'}{r} \rho^{\frac{5}{q} - 3} \lambda(\rho) \rho^{\frac{10}{3} - \frac{5}{q}} \rho^{2/3} \gamma(\rho) = C' \kappa^{-1} \gamma(\rho) \lambda(\rho), \end{aligned}$$

where we used $|Q_\rho(z_0)|^{\frac{1}{q'} - \frac{1}{3}} = C \rho^{5(\frac{2}{3} - \frac{1}{q})} = C \rho^{\frac{10}{3} - \frac{5}{q}}$ and $\frac{5}{q} - 3 - \sigma$ cancels.

Step 3: the left side. On $Q_r(z_0)$ we have $\eta = 1$ and $\psi_r \geq C_{24}^{-1} r^{-1}$, so $\int |u(\cdot, t)|^2 \phi(\cdot, t) \geq C_{24}^{-1} r^{-1} \int_{B_r} |u(\cdot, t)|^2$ for $t \in (t_0 - r^2, t_0]$, and $2 \int_{t_0 - \rho^2}^{t_0} \int |\nabla u|^2 \phi \geq 2C_{24}^{-1} r^{-1} \iint_{Q_r(z_0)} |\nabla u|^2 = 2C_{24}^{-1} \beta(r)^2$. Taking the essential supremum over the full-measure set of good times $t \in (t_0 - r^2, t_0)$ bounds the terminal energy term. For dissipation, choose good times $t_n \uparrow t_0$, discard the nonnegative terminal energy, and use monotone convergence on $\mathbf{1}_{(t_0 - \rho^2, t_n)} |\nabla u|^2 \phi$. This gives the integral up to t_0 without assuming that t_0 is a good time. Since $I_1, \dots, I_4$ do not depend on the terminal time,

$$\alpha(r)^2 + \beta(r)^2 \leq \frac{3}{2} C_{24} (I_1 + I_2 + I_3 + I_4) \leq C [\kappa^2 \alpha(\rho)^2 + \kappa^{-2} \alpha \beta \gamma + \kappa^{-2} \delta^2 \gamma + \kappa^{-1} \gamma \lambda](\rho).$$

Taking square roots and using $\alpha(r) + \beta(r) \leq \sqrt{2}(\alpha(r)^2 + \beta(r)^2)^{1/2}$ and $(s_1 + s_2 + s_3 + s_4)^{1/2} \leq \sum s_i^{1/2}$ gives (108). $\square$

Remark 5.16 (Comparison with the sources). Inequality (108) is [7, (2.18)] with the force term in the L^q form of [7, Theorem 2], and it is [11, (13.30)] after the substitutions $U_\rho = \rho\alpha(\rho)^2$, $V_\rho = \rho\beta(\rho)^2$, $W_\rho = \rho^2\gamma(\rho)^3$, $P_\rho = \rho^2\delta(\rho)^3$. We have used Kukavica’s test function $\psi_r \eta$ rather than Lemarié-Rieusset’s $r^3 \omega \theta H(4r^2 + t - s, x - y)$; the two differ by the normalizing power of r (his ψ is $\approx r$ times ours, matching his un-normalized U_r, V_r) and by the auxiliary cut-off θ in time, which is not needed here because Lemma 1.13 already provides the pointwise-in-time inequality. The four terms and their exponents agree.

5.3. Step 2: the scale iteration.

Lemma 5.17 (Combined decay inequality). *For $\kappa \in (0, \frac{1}{2}]$ and $Q_\rho(z) \subset \mathcal{O}$ set*

$$\theta(z, \rho) = \alpha(z, \rho) + \beta(z, \rho) + \kappa^{-4}\delta(z, \rho)^2. \quad (112)$$

There are an absolute constant C_{27} and a constant $C_{28} = C_{28}(q)$ such that for every such z, ρ, κ , suppressing z ,

$$\theta(\kappa\rho) \leq C_{27}\kappa^{2/3}\theta(\rho) + C_{27}\kappa^{-5}(\beta(\rho)^{1/2} + \beta(\rho))\theta(\rho) + C_{28}\kappa^{-1/2}\theta(\rho)^{1/2}\lambda(\rho)^{1/2} + C_{28}\kappa^{-3}\lambda(\rho), \quad (113)$$

and, if in addition $\theta(\rho) \leq 1$,

$$\theta(\kappa\rho) \leq C_{27}\kappa^{2/3}\theta(\rho) + 2C_{27}\kappa^{-5}\theta(\rho)^{1/2}\theta(\rho) + C_{28}\kappa^{-1/2}\theta(\rho)^{1/2}\lambda(\rho)^{1/2} + C_{28}\kappa^{-3}\lambda(\rho). \quad (114)$$

Proof. Write $r = \kappa\rho$ and suppress ρ on the right. All of α, β and $\kappa^{-4}\delta^2$ are $\leq \theta$ by definition.

Pressure part. By Theorem 4.7 and $(s_1 + s_2 + s_3)^2 \leq 3(s_1^2 + s_2^2 + s_3^2)$,

$$\kappa^{-4}\delta(r)^2 \leq 3C_{14}^2\kappa^{-5}\alpha\beta + 3C_{14}^2\kappa^{2/3}(\kappa^{-4}\delta^2) + 3C_{15}^2\kappa^{-3}\lambda \leq 3C_{14}^2\kappa^{-5}\beta\theta + 3C_{14}^2\kappa^{2/3}\theta + 3C_{15}^2\kappa^{-3}\lambda.$$

Energy part. By Corollary 3.8, $\gamma \leq C_9(\alpha^{1/2}\beta^{1/2} + \alpha) \leq C_9(\theta^{1/2}\beta^{1/2} + \theta)$ and hence $\gamma^{1/2} \leq C_9^{1/2}(\theta^{1/4}\beta^{1/4} + \theta^{1/2})$. Insert this into (108), term by term, using $\beta \leq \theta$ and $\delta \leq \kappa^2\theta^{1/2}$ (from $\kappa^{-4}\delta^2 \leq \theta$):

$$\begin{aligned} \kappa\alpha &\leq \kappa\theta; \\ \kappa^{-1}\alpha^{1/2}\beta^{1/2}\gamma^{1/2} &\leq C_9^{1/2}\kappa^{-1}\theta^{1/2}\beta^{1/2}(\theta^{1/4}\beta^{1/4} + \theta^{1/2}) \leq 2C_9^{1/2}\kappa^{-1}\beta^{1/2}\theta; \\ \kappa^{-1}\delta\gamma^{1/2} &\leq \kappa\theta^{1/2}C_9^{1/2}(\theta^{1/4}\beta^{1/4} + \theta^{1/2}) \leq 2C_9^{1/2}\kappa\theta; \\ \kappa^{-1/2}\gamma^{1/2}\lambda^{1/2} &\leq C_9^{1/2}\kappa^{-1/2}(\theta^{1/4}\beta^{1/4} + \theta^{1/2})\lambda^{1/2} \leq 2C_9^{1/2}\kappa^{-1/2}\theta^{1/2}\lambda^{1/2}, \end{aligned}$$

where in the second line we used $\beta^{1/4}\theta^{1/4} \leq \theta^{1/2}$ and in the third and fourth $\beta^{1/4}\theta^{1/4} \leq \theta^{1/2}$ again. Hence

$$\alpha(r) + \beta(r) \leq C_{25}(1 + 2C_9^{1/2})\kappa\theta + 2C_{25}C_9^{1/2}\kappa^{-1}\beta^{1/2}\theta + 2C_{26}C_9^{1/2}\kappa^{-1/2}\theta^{1/2}\lambda^{1/2}.$$

Adding the two parts and using $\kappa \leq \kappa^{2/3}$ and $\kappa^{-1} \leq \kappa^{-5}$, valid for $\kappa \leq \frac{1}{2}$, gives (113) with $C_{27} = 3C_{14}^2 + C_{25}(1 + 2C_9^{1/2}) + 2C_{25}C_9^{1/2}$ and $C_{28} = 3C_{15}^2 + 2C_{26}C_9^{1/2}$ (sums, not maxima, so that nothing is lost when the two groups are added). If $\theta(\rho) \leq 1$ then $\beta \leq \theta \leq \theta^{1/2}$, so $\beta^{1/2} + \beta \leq 2\theta^{1/2}$, which gives (114). $\square$

Convention 5.18 (The constants $\kappa, \eta, \varepsilon_*$). Recall $\varepsilon = 2/5$. Define, once and for all,

$$\kappa = \min\left\{\frac{1}{2}, (8C_{27})^{-\frac{1}{2/3-\varepsilon}}\right\}, \quad \eta = \min\left\{1, \left(\frac{\kappa^{5+\varepsilon}}{16C_{27}}\right)^2\right\}, \quad \varepsilon_* = \min\left\{1, \left(\frac{\kappa^{5+\varepsilon}}{32C_{27}}\right)^2\right\}, \quad (115)$$

so that

$$C_{27}\kappa^{2/3-\varepsilon} \leq \frac{1}{8}, \quad 2C_{27}\kappa^{-5-\varepsilon}\eta^{1/2} \leq \frac{1}{8}, \quad 2C_{27}\kappa^{-5-\varepsilon}(\varepsilon_*^{1/2} + \varepsilon_*) \leq \frac{1}{8}. \quad (116)$$

Since $\varepsilon = 2/5$ is fixed and C_{27} is absolute, κ, η and ε_* are *absolute constants*. Finally set

$$C_{29} = C_{29}(q) = 2C_{28}^2\kappa^{-1-2\varepsilon} + C_{28}\kappa^{-3-\varepsilon}, \quad \Lambda_0 = \Lambda_0(q) = \frac{\eta}{2C_{29}}. \quad (117)$$

Proposition 5.19 (The iteration). *Let $z \in \mathcal{O}$ and $r_5 > 0$ with $Q_{r_5}(z) \subset \mathcal{O}$, and suppose*

$$\theta(z, r_5) \leq \eta \quad \text{and} \quad \lambda(z, r_5) \leq \Lambda_0. \quad (118)$$

Then $\theta(z, \kappa^n r_5) \leq \eta \kappa^{n\varepsilon}$ for every $n \in \mathbb{N}_0$, and consequently

$$\max\{\alpha(z, r), \beta(z, r), \delta(z, r)^2\} \leq \theta(z, r) \leq M r^\varepsilon, \quad 0 < r \leq r_5, \quad M := \kappa^{-4/3-\varepsilon} \eta r_5^{-\varepsilon}. \quad (119)$$

Proof. Put $\rho_n = \kappa^n r_5$, $\theta_n = \theta(z, \rho_n)$, $\lambda_n = \lambda(z, \rho_n)$, $T_n = \theta_n \kappa^{-n\varepsilon}$ and $L_n = \lambda_n \kappa^{-n\varepsilon}$. By Lemma 3.9, $\lambda_n \leq \kappa^{n\sigma} \lambda_0$, and $\sigma > 1 > \varepsilon$ gives $L_n \leq \kappa^{n(\sigma-\varepsilon)} \lambda_0 \leq \Lambda_0$ for all n .

We prove $T_n \leq \eta$ by induction; $T_0 = \theta(z, r_5) \leq \eta$ by hypothesis. Assume $T_n \leq \eta$; then $\theta_n \leq \eta \leq 1$, so (114) applies with $\rho = \rho_n$. Dividing it by $\kappa^{(n+1)\varepsilon}$,

$$T_{n+1} \leq C_{27} \kappa^{2/3-\varepsilon} T_n + 2C_{27} \kappa^{-5-\varepsilon} \theta_n^{1/2} T_n + C_{28} \kappa^{-1/2-\varepsilon} T_n^{1/2} L_n^{1/2} + C_{28} \kappa^{-3-\varepsilon} L_n.$$

By (116) the first term is $\leq \frac{1}{8} T_n$ and, since $\theta_n^{1/2} \leq \eta^{1/2}$, the second is $\leq \frac{1}{8} T_n$. For the third, Young's inequality $ab \leq \frac{1}{8} a^2 + 2b^2$ with $a = T_n^{1/2}$ and $b = C_{28} \kappa^{-1/2-\varepsilon} L_n^{1/2}$ gives $\leq \frac{1}{8} T_n + 2C_{28}^2 \kappa^{-1-2\varepsilon} L_n$. Hence

$$T_{n+1} \leq \frac{3}{8} T_n + C_{29} L_n \leq \frac{3}{8} \eta + C_{29} \Lambda_0 = \frac{3}{8} \eta + \frac{1}{2} \eta = \frac{7}{8} \eta \leq \eta,$$

completing the induction.

For (119), let $0 < r \leq r_5$ and choose $n \in \mathbb{N}_0$ with $\kappa^{n+1} r_5 < r \leq \kappa^n r_5$. By Lemma 3.9 with $\rho_1 = r$, $\rho_2 = \rho_n$ and $\rho_n/r < \kappa^{-1}$,

$$\alpha(z, r) \leq \kappa^{-1/2} \alpha(z, \rho_n), \quad \beta(z, r) \leq \kappa^{-1/2} \beta(z, \rho_n), \quad \delta(z, r)^2 \leq \kappa^{-4/3} \delta(z, \rho_n)^2,$$

so $\theta(z, r) \leq \kappa^{-4/3} \theta(z, \rho_n) \leq \kappa^{-4/3} \eta \kappa^{n\varepsilon}$. Finally $\kappa^n < r/(\kappa r_5)$ gives $\kappa^{n\varepsilon} < \kappa^{-\varepsilon} r_5^{-\varepsilon} r^\varepsilon$, whence $\theta(z, r) \leq \kappa^{-4/3-\varepsilon} \eta r_5^{-\varepsilon} r^\varepsilon$. $\square$

Lemma 5.20 (Upper semicontinuity of α in the base point). *Let $0 < r < R$ and $t_1 < t_2$ be such that $\overline{B_R(x_0)} \times [t_1, t_2] \subset \mathcal{O}$ and $t_2 \in I$. Then*

$$\lim_{h \rightarrow 0^+} \operatorname{ess\,sup}_{s \in (t_1, t_2+h)} \int_{B_r(x_0)} |u(y, s)|^2 dy \leq \operatorname{ess\,sup}_{s \in (t_1, t_2)} \int_{B_R(x_0)} |u(y, s)|^2 dy. \quad (120)$$

Proof. Write S for the right side, which is finite by (S1). Choose $\epsilon_0 > 0$ with $[t_1, t_2 + \epsilon_0] \subset I$ and $\overline{B_R(x_0)} \times [t_1, t_2 + \epsilon_0] \subset \mathcal{O}$; this is possible because $\mathcal{O}$ is open and the left-hand set is compact. Let $\psi_1 \in C_c^\infty(B_R(x_0))$ with $0 \leq \psi_1 \leq 1$ and $\psi_1 \equiv 1$ on $B_r(x_0)$. Define

$$\Xi(a, b) = \int_a^b \int_{B_R(x_0)} \left[|u|^2 |\Delta \psi_1| + (|u|^2 + 2|p|) |u| |\nabla \psi_1| + 2|f| |u| \right] dy ds,$$

which is finite for $[a, b] \subset [t_1, t_2 + \epsilon_0]$ by the integrability checks in Step 2 of the proof of Lemma 1.13; by absolute continuity of the integral, $\Xi(a, b) \rightarrow 0$ as $b - a \rightarrow 0$.

Fix $0 < h < \min\{\epsilon_0/2, t_2 - t_1\}$ and let $T \in (t_2, t_2 + h)$. Let $T_0 \in (t_2 - h, t_2)$ and $h' \in (0, t_2 - T_0)$, and let $\chi = \chi_{T_0, h'}$ be a smooth nondecreasing function with $\chi \equiv 0$ on $(-\infty, T_0]$, $\chi \equiv 1$ on $[T_0 + h', \infty)$ and $0 \leq \chi' \leq 2/h'$ supported in $(T_0, T_0 + h')$; such χ is obtained by mollifying the piecewise affine function as in (20). Multiply χ by a smooth time cut-off equal to 1 on $(-\infty, t_2 + \epsilon_0/2]$ and vanishing near $t_2 + \epsilon_0$, which does not affect the computation below because we integrate only up to $T < t_2 + \epsilon_0/2$. Then $\phi(y, s) = \psi_1(y) \chi(s) \in C_c^\infty(\mathcal{O})$ is nonnegative and vanishes for s near t_1 , so Lemma 1.13 applies for almost every T (for each fixed cutoff). Since

$\chi(T) = 1$, $\partial_s \phi = \psi_1 \chi'$, $\Delta \phi = \chi \Delta \psi_1$, $\nabla \phi = \chi \nabla \psi_1$ and $0 \leq \chi \leq 1$, discarding the (nonnegative) dissipation term gives

$$\int_{B_r(x_0)} |u(y, T)|^2 dy \leq \int |u(y, T)|^2 \psi_1 dy \leq \int_{T_0}^{T_0+h'} \int_{B_R(x_0)} |u|^2 \chi' dy ds + \Xi(T_0, T).$$

The first term on the right is at most $\frac{2}{h'} \int_{T_0}^{T_0+h'} (\int_{B_R(x_0)} |u(\cdot, s)|^2) ds \leq 2S$ provided $(T_0, T_0+h') \subset (t_1, t_2)$. To remove the factor 2, choose instead T_0 to be a Lebesgue point in $(t_2 - h, t_2)$ of the function $s \mapsto \int_{B_R(x_0)} |u(\cdot, s)|^2$ (almost every point is one) and let $h' \rightarrow 0^+$ through a fixed sequence, intersecting the countably many full-measure sets of terminal times: the first term then converges to $\int_{B_R(x_0)} |u(\cdot, T_0)|^2 \leq S$ if T_0 is moreover a point where the essential supremum bound holds, which is the case for a.e. $T_0 \in (t_1, t_2)$. Hence

$$\int_{B_r(x_0)} |u(y, T)|^2 dy \leq S + \Xi(t_2 - h, t_2 + h) \quad \text{for a.e. } T \in (t_2, t_2 + h).$$

The same bound holds trivially for a.e. $T \in (t_1, t_2)$ since $B_r(x_0) \subset B_R(x_0)$. Taking the essential supremum over $T \in (t_1, t_2 + h)$ and letting $h \rightarrow 0^+$, so that $\Xi(t_2 - h, t_2 + h) \rightarrow 0$, gives (120). $\square$

Corollary 5.21. *Let $z_0 \in \mathcal{O}$ and $0 < \varrho < R$ with $\overline{B_R(x_0)} \times [t_0 - R^2, t_0] \subset \mathcal{O}$ and $t_0 \in I$. Then*

$$\limsup_{z \rightarrow z_0} \alpha(z, \varrho)^2 \leq \frac{R}{\varrho} \alpha(z_0, R)^2, \quad \lim_{z \rightarrow z_0} \beta(z, \varrho) = \beta(z_0, \varrho), \quad \lim_{z \rightarrow z_0} \delta(z, \varrho) = \delta(z_0, \varrho). \quad (121)$$

Proof. Let $z = (x, t) \rightarrow z_0$. For z close enough to z_0 we have $B_\varrho(x) \subset B_R(x_0)$ and $(t - \varrho^2, t) \subset (t_0 - R^2, t_0 + h)$ for any prescribed $h > 0$. Hence

$$\alpha(z, \varrho)^2 = \frac{1}{\varrho} \operatorname{ess\,sup}_{(t-\varrho^2, t)} \int_{B_\varrho(x)} |u|^2 \leq \frac{1}{\varrho} \operatorname{ess\,sup}_{(t_0-R^2, t_0+h)} \int_{B_R(x_0)} |u|^2,$$

Since $\varrho < R$ strictly, fix an intermediate radius r with $\varrho < r < R$. For z with $|x - x_0| < r - \varrho$ we have $B_\varrho(x) \subset B_r(x_0)$, so

$$\alpha(z, \varrho)^2 \leq \frac{1}{\varrho} \operatorname{ess\,sup}_{s \in (t_0-R^2, t_0+h)} \int_{B_r(x_0)} |u(y, s)|^2 dy.$$

Letting $z \rightarrow z_0$ and then $h \rightarrow 0^+$ and applying Lemma 5.20 with $t_1 = t_0 - R^2$, $t_2 = t_0$, inner radius r and outer radius R (so that $r < R$ as the lemma requires) gives $\limsup_{z \rightarrow z_0} \alpha(z, \varrho)^2 \leq \varrho^{-1} \operatorname{ess\,sup}_{(t_0-R^2, t_0)} \int_{B_R(x_0)} |u|^2 = (R/\varrho) \alpha(z_0, R)^2$. For β and δ , the maps $z \mapsto \iint_{Q_\varrho(z)} |\nabla u|^2$ and $z \mapsto \iint_{Q_\varrho(z)} |p|^{3/2}$ are continuous, because $|Q_\varrho(z) \triangle Q_\varrho(z_0)| \rightarrow 0$ as $z \rightarrow z_0$ and $|\nabla u|^2, |p|^{3/2} \in L^1$ near z_0 , so absolute continuity of the integral applies. $\square$

Proposition 5.22 (Step 2: Morrey decay on a neighbourhood). *Let $q > 5/2$, $\varepsilon = 2/5$, and let $\kappa, \eta, \varepsilon_*, \Lambda_0$ be as in Convention 5.18. Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force $f \in L^q_{loc}$, and let $z_0 \in \mathcal{O}$ satisfy*

$$\limsup_{r \rightarrow 0^+} \beta(z_0, r) < \varepsilon_*. \quad (122)$$

Then there are $r_2 > 0$ with $\mathfrak{B}_{2r_2}(z_0) \subset \mathcal{O}$ and $M \geq 1$ such that

$$\max\{\alpha(z, r), \beta(z, r), \delta(z, r)^2\} \leq Mr^\varepsilon \quad \text{for all } z \in \mathfrak{B}_{r_2}(z_0) \text{ and } 0 < r < r_2. \quad (123)$$

Proof. Step (a): descent at z_0 . Choose $r_4 > 0$ with $\overline{\mathfrak{B}_{4r_4}(z_0)} \subset \mathcal{O}$ and $\beta(z_0, \rho) \leq \varepsilon_*$ for all $\rho \leq r_4$; this is possible by (122). Put $M_f = \|f\|_{L^q(\mathfrak{B}_{4r_4}(z_0))}$, so that $\lambda(z, \rho) \leq M_f \rho^\sigma$ whenever $Q_\rho(z) \subset \mathfrak{B}_{4r_4}(z_0)$. Put $T_n = \theta(z_0, \kappa^n r_4) \kappa^{-n\varepsilon}$ and $L_n = \lambda(z_0, \kappa^n r_4) \kappa^{-n\varepsilon}$, so that $L_n \leq M_f r_4^\sigma \kappa^{n(\sigma-\varepsilon)}$. Apply (113) (the hypothesis $\theta(z_0, \cdot) \leq \eta$ of (114) is not yet known) at z_0 with $\rho = \kappa^n r_4$ and divide by $\kappa^{(n+1)\varepsilon}$:

$$\begin{aligned} T_{n+1} &\leq C_{27} \kappa^{2/3-\varepsilon} T_n \\ &\quad + C_{27} \kappa^{-5-\varepsilon} (\beta(z_0, \kappa^n r_4)^{1/2} + \beta(z_0, \kappa^n r_4)) T_n \\ &\quad + C_{28} \kappa^{-1/2-\varepsilon} T_n^{1/2} L_n^{1/2} + C_{28} \kappa^{-3-\varepsilon} L_n. \end{aligned}$$

Since $\kappa^n r_4 \leq r_4$ we have $\beta(z_0, \kappa^n r_4) \leq \varepsilon_*$ for every n , so the second term is at most

$$C_{27} \kappa^{-5-\varepsilon} (\varepsilon_*^{1/2} + \varepsilon_*) T_n \leq \frac{1}{16} T_n$$

by the third inequality of (116); the first is $\leq \frac{1}{8} T_n$ by the first inequality there; and the third is $\leq \frac{1}{8} T_n + 2C_{28}^2 \kappa^{-1-2\varepsilon} L_n$ by Young's inequality $ab \leq \frac{1}{8} a^2 + 2b^2$. Hence $T_{n+1} \leq \frac{5}{16} T_n + C_{29} L_n \leq \frac{3}{8} T_n + C_{29} L_n$, and

$$T_n \leq \left(\frac{3}{8}\right)^n T_0 + C_{29} M_f r_4^\sigma \sum_{k=0}^{n-1} \left(\frac{3}{8}\right)^{n-1-k} \kappa^{k(\sigma-\varepsilon)} \leq \left(\frac{3}{8}\right)^n T_0 + C M_f r_4^\sigma, \quad C := \frac{8}{5} C_{29},$$

because $\sum_{k=0}^{n-1} (3/8)^{n-1-k} \kappa^{k(\sigma-\varepsilon)} \leq \sum_{m \geq 0} (3/8)^m = \frac{8}{5}$ (each factor $\kappa^{k(\sigma-\varepsilon)} \leq 1$). Shrinking r_4 so that $C M_f r_4^\sigma \leq \eta/8$ and choosing n_0 with $(3/8)^{n_0} T_0 \leq \eta/8$ (possible since $T_0 = \theta(z_0, r_4) < \infty$ by (S1)), we obtain

$$\theta(z_0, R_0) \leq \frac{1}{4} \eta \kappa^{n_0 \varepsilon} \leq \frac{1}{4} \eta, \quad R_0 := \kappa^{n_0} r_4. \quad (124)$$

Step (b): transfer to nearby centres. Let $\varrho \in (0, R_0)$ with $(R_0/\varrho)^{4/3+\varepsilon} \leq 2$. By Lemma 3.9, $\beta(z_0, \varrho) \leq (R_0/\varrho)^{1/2} \beta(z_0, R_0)$ and $\delta(z_0, \varrho)^2 \leq (R_0/\varrho)^{4/3} \delta(z_0, R_0)^2$, and by Corollary 5.21 applied with $R = R_0$,

$$\limsup_{z \rightarrow z_0} \theta(z, \varrho) \leq \left(\frac{R_0}{\varrho}\right)^{4/3} \theta(z_0, R_0) \leq 2 \cdot \frac{1}{4} \eta = \frac{1}{2} \eta.$$

Hence there is $r_2 \in (0, \varrho/4)$ with $\mathfrak{B}_{2r_2}(z_0) \subset \mathcal{O}$ and

$$\theta(z, \varrho) \leq \eta \quad \text{for every } z \in \mathfrak{B}_{r_2}(z_0). \quad (125)$$

Shrinking r_4 further at the start if necessary, we may also assume $\lambda(z, \varrho) \leq M_f \varrho^\sigma \leq \Lambda_0$ for all such z .

Step (c): iterate at every centre. For each $z \in \mathfrak{B}_{r_2}(z_0)$, Proposition 5.19 with $r_5 = \varrho$ gives $\theta(z, r) \leq M r^\varepsilon$ for $0 < r \leq \varrho$, with coefficient $\kappa^{-4/3-\varepsilon} \eta \varrho^{-\varepsilon}$ independent of z . Set $M = \max\{1, \kappa^{-4/3-\varepsilon} \eta \varrho^{-\varepsilon}\}$; the estimate remains valid and $M \geq 1$ as asserted. Since $r_2 < \varrho$, this is (123). $\square$

Lemma 5.23 (Step 2 in Morrey form). *Let z_0, r_2, M be as in Proposition 5.22 and put $Q_2 = \mathfrak{B}_{r_2/4}(z_0)$. Then, with τ_2, τ_3, τ_p as in (83),*

$$\mathbf{1}_{Q_2} u \in \mathcal{M}^{3, \tau_2}, \quad \mathbf{1}_{Q_2} \nabla u \in \mathcal{M}^{2, \tau_3}, \quad \mathbf{1}_{Q_2} p \in \mathcal{M}^{3/2, \tau_p}, \quad (126)$$

with norms bounded by $C(M, r_2)$. Explicitly $\tau_2 = 25/3 > 5$ and $\tau_3 = \tau_p = 25/8 > 5/2$, and $\frac{1}{\tau_3} = \frac{1}{\tau_2} + \frac{1}{5}$.

Proof. By Lemma 5.3 it suffices to bound the integrals over one-sided cylinders $Q_\varrho(z)$, $z \in \mathbb{R}^3 \times \mathbb{R}$, $\varrho > 0$. If $Q_\varrho(z) \cap Q_2 = \emptyset$ there is nothing to prove.

Small radii, $\varrho \leq r_2/8$. Pick $z' = (x', t') \in Q_\varrho(z) \cap Q_2$, so that $Q_\varrho(z) \subset Q_{2\varrho}((x', t' + \varrho^2))$ by the triangle inequality for d_{par} and Lemma 1.3. The centre $z'' = (x', t' + \varrho^2)$ satisfies $d_{\text{par}}(z'', z_0) \leq d_{\text{par}}(z', z_0) + \varrho < r_2/4 + r_2/8 < r_2$, so $z'' \in \mathfrak{B}_{r_2}(z_0)$, and $\varrho' = 2\varrho < r_2$. By Proposition 5.22 and Corollary 3.8,

$$\iint_{Q_\varrho(z)} |u|^3 \leq \iint_{Q_{\varrho'}(z'')} |u|^3 = \varrho'^2 \gamma(z'', \varrho')^3 \leq \varrho'^2 (2C_9 M \varrho'^\varepsilon)^3 = CM^3 \varrho'^{2+3\varepsilon},$$

and $2 + 3\varepsilon = 16/5 = 5(1 - 3/\tau_2)$ with $\tau_2 = 25/3$, which is the Morrey condition for $\mathcal{M}^{3, \tau_2}$. Similarly

$$\iint_{Q_\varrho(z)} |\nabla u|^2 \leq \varrho' \beta(z'', \varrho')^2 \leq M^2 \varrho'^{1+2\varepsilon}, \quad \iint_{Q_\varrho(z)} |p|^{3/2} \leq \varrho'^2 \delta(z'', \varrho')^3 \leq M^{3/2} \varrho'^{2+\frac{3}{2}\varepsilon},$$

and $1 + 2\varepsilon = 9/5 = 5(1 - 2/\tau_3)$ with $\tau_3 = 25/8$, while $2 + \frac{3}{2}\varepsilon = 13/5 = 5(1 - \frac{3/2}{\tau_p})$ with $\tau_p = 25/8$. Finally $\frac{1}{\tau_3} = \frac{8}{25} = \frac{3}{25} + \frac{5}{25} = \frac{1}{\tau_2} + \frac{1}{5}$.

Large radii, $\varrho > r_2/8$. Cover the bounded set Q_2 by finitely many cylinders of radius $r_2/8$ and apply the small-radius case to each: this gives $M_0 := \iint_{Q_2} |u|^3 < \infty$, and likewise $M'_0 := \iint_{Q_2} |\nabla u|^2 < \infty$ and $M''_0 := \iint_{Q_2} |p|^{3/2} < \infty$ (all three are finite already by (S1)). For $\varrho > r_2/8$,

$$\iint_{Q_\varrho(z)} |\mathbf{1}_{Q_2} u|^3 \leq M_0 = M_0 \left(\frac{r_2}{8}\right)^{-16/5} \left(\frac{r_2}{8}\right)^{16/5} \leq M_0 \left(\frac{r_2}{8}\right)^{-16/5} \varrho^{16/5},$$

which is the required Morrey bound with the constant enlarged by $M_0(r_2/8)^{-16/5}$; identically for ∇u with the exponent $9/5$ and for p with the exponent $13/5$. $\square$

Remark 5.24 (Comparison with [11, Lemma 13.4]). Lemarié-Rieusset states the conclusion of Step 2 as $\mathbf{1}_{Q_2} u \in \mathcal{M}^{3, \tau_2}$ and $\mathbf{1}_{Q_2} p \in \mathcal{M}^{q_0, \tau_2/2}$ under the restriction $1 < \tau_2/5 < \min\{q_0, 2\}$, equivalently $5 < \tau_2 < 5 \min\{q_0, 2\}$. This is the condition printed in his Lemma 13.4. In particular $\tau_2 < 5q_0$, so both terms in $\lambda = \min\{10/\tau_2, 10q_0/\tau_2 - 2\}$ are positive, as required by his iteration on p. 437. With $q_0 = 3/2$ that restriction forces $\tau_2 < 15/2$, i.e. $\varepsilon < 1/3$. We instead iterate Kukavica's functional $\theta = \alpha + \beta + \kappa^{-4} \delta^2$, in which the pressure is normalized by δ^2 rather than by P_τ ; this allows $\varepsilon = 2/5$ and hence $\tau_2 = 25/3$ for the velocity, at the price of the weaker pressure exponent $\tau_p = 25/8$ instead of $\tau_2/2 = 25/6$. The weaker exponent is sufficient: the pressure Morrey norm is used below only through fixed-scale $L^{3/2}$ integrability in Lemma 5.26, never through its Morrey exponent. The gradient bound $\mathbf{1}_{Q_2} \nabla u \in \mathcal{M}^{2, \tau_3}$, $\frac{1}{\tau_3} = \frac{1}{\tau_2} + \frac{1}{5}$, is the one extracted in [11, p. 439] from the same iteration. We use this gradient information explicitly in the pressure-gradient estimate and the velocity bootstrap below.

5.4. Step 3: the local representation and the Morrey bootstrap. Throughout §5.4–§5.5, z_0, r_2, M are as in Proposition 5.22, $Q_2 = \mathfrak{B}_{r_2/4}(z_0)$, and $\varphi \in C_c^\infty(\mathbb{R}^3 \times \mathbb{R})$ satisfies

$$\varphi \equiv 1 \text{ on } Q_3 := \mathfrak{B}_{r_3}(z_0), \quad \text{supp } \varphi \subset Q_2. \quad (127)$$

with $0 < r_3 < r_2/4$. We write $N(y) = -\frac{1}{4\pi|y|}$ as in Proposition 4.4, W_+ for the forward heat kernel of (97) (unit viscosity, as throughout), and I_1, I_2 for the parabolic Riesz potentials of Definition 5.6.

We keep the pressure gradient in the undifferentiated source of the localized heat equation. The pressure estimate below uses a fixed spatial localization and one measurable weak gradient; it does not require a commutator expansion.

Lemma 5.25 (The localized equation). *Set $v = \varphi u$. Then, in $\mathcal{D}'(\mathbb{R}^3 \times \mathbb{R})$,*

$$\partial_t v - \Delta v = \mathbf{g} + \sum_{i=1}^3 \partial_i \mathbf{h}_i - \varphi \nabla p, \quad \mathbf{g} = (\partial_t \varphi)u + (\Delta \varphi)u - \varphi u \cdot \nabla u + \varphi f, \quad \mathbf{h}_i = -2(\partial_i \varphi)u, \quad (128)$$

and the compact-support adjoint identity proved below gives

$$v = W_+ * \left(\mathbf{g} + \sum_i \partial_i \mathbf{h}_i - \varphi \nabla p \right). \quad (129)$$

Proof. By (S3), $\partial_t u - \Delta u = -u \cdot \nabla u - \nabla p + f$ in $\mathcal{D}'(\mathcal{O})$, where $u \cdot \nabla u = \operatorname{div}(u \otimes u)$ makes sense as a distribution because $u \otimes u \in L_{loc}^{5/3}$. Multiplying by φ and using the Leibniz rules $\partial_t(\varphi u) = (\partial_t \varphi)u + \varphi \partial_t u$ and $\Delta(\varphi u) = (\Delta \varphi)u + 2\nabla \varphi \cdot \nabla u + \varphi \Delta u$, together with $2\nabla \varphi \cdot \nabla u = \sum_i \partial_i(2(\partial_i \varphi)u) - 2(\Delta \varphi)u$, gives (128). For (129), write $\mathcal{F}$ for the compactly supported distribution on the right-hand side of (128). Given a compactly supported smooth vector test ψ , let

$$\Psi(y, s) = \int_s^\infty \int_{\mathbb{R}^3} W_+(x - y, t - s) \psi(x, t) dx dt.$$

Then $-\partial_s \Psi - \Delta_y \Psi = \psi$. Choose a smooth compact cutoff equal to one on a neighbourhood of the supports of v and $\mathcal{F}$. Testing the localized equation with this cutoff times Ψ gives

$$\langle v, \psi \rangle = \langle \mathcal{F}, \Psi \rangle = \langle W_+ * \mathcal{F}, \psi \rangle.$$

The last equality is the defining adjoint convolution identity; for the locally integrable source terms it also follows from Fubini. The source support makes the cutoff immaterial in the middle pairing. Thus (129) holds distributionally, and almost everywhere whenever the source potential is represented by a locally integrable function. This proof requires neither an algebra of arbitrary causal distributions nor unrestricted uniqueness for the heat equation. $\square$

Lemma 5.26 (Localized pressure-gradient Morrey bound). *Let $\mathfrak{B}_{2R}(z) \subset \mathcal{O}$, $R > 0$, and $25/3 \leq \tau \leq 25$. If $\mathbf{1}_{\mathfrak{B}_R(z)} u \in \mathcal{M}^{3, \tau}$ and $\mathbf{1}_{\mathfrak{B}_R(z)} \nabla u \in \mathcal{M}^{2, 25/8}$, then the distributional spatial gradient of p has a jointly measurable representative Dp on $\mathfrak{B}_{R/2}(z)$, and*

$$\mathbf{1}_{\mathfrak{B}_{R/2}(z)} Dp \in \mathcal{M}^{6/5, \varkappa}, \quad \varkappa = \min \left\{ q, \left(\frac{1}{\tau} + \frac{8}{25} \right)^{-1} \right\}. \quad (130)$$

In particular, for every smooth test ψ compactly supported there, $\iint p \partial_i \psi = -\iint D_i p \psi$. The bounds are finite functions of the incoming Morrey norms, the local energy, pressure and force norms on a fixed surrounding cylinder, q, τ , and the localization radii. No bound is required on a shrinking-cell norm of the unnormalized pressure.

Proof. Write $z = (x, t)$ and fix the cylinder $Q = B_R(x) \times (t - 3R^2/4, t + R^2/4]$, which lies in $\mathfrak{B}_R(z)$ and whose closure lies in $\mathfrak{B}_{2R}(z)$. Let η be the spatial cutoff of Lemma 4.3, and let $c(s) = [u(\cdot, s)]_{B_R(x)}$. In the pressure decomposition of Proposition 4.4 use the singly centred tensor $U_{ij} = -u_i(u_j - c_j)$. Its differentiated, localized source is

$$V_i = \sum_j [\eta(u_j - c_j) \partial_j u_i + (\partial_j \eta) u_i (u_j - c_j)].$$

All sources in this proof are extended by zero outside the fixed time window of Q . The weak product rule, incompressibility and the completed second Riesz transforms give, on the inner

cylinder,

$$D_i p = - \sum_j \partial_i \partial_j \Delta^{-1} (\mathbf{1}_Q V_j) + \sum_j \partial_i \partial_j \Delta^{-1} (\mathbf{1}_Q \eta f_j) + H_i, \quad H = \nabla(p_{\text{har}} + p_8). \quad (131)$$

Here $p_{\text{har}} = p_2 + \dots + p_6$ is the smooth harmonic remainder on the inner ball; it agrees almost everywhere there with $p - p_1 - p_7 - p_8$. The gradient in H is taken of this smooth representative and the smooth annular potential p_8 , not of an arbitrary representative of p . The operators are their $L^{6/5}$ extensions, not pointwise differentiation of a possibly undefined singular integral. Spatial mollification and almost-everywhere limits select jointly measurable representatives; the slice weak-derivative identities, followed by Fubini, give the asserted space–time pairing. Uniqueness of locally integrable weak derivatives identifies these representatives on overlapping cylinders.

We give the estimates on the three terms. The fixed spatial mean c is essentially bounded in time by the local energy bound. Thus its restriction to Q belongs to every Morrey class used below. Hölder's inequality gives $u \nabla u \in \mathcal{M}^{6/5, (1/\tau + 8/25)^{-1}}$. The remaining products in V have at least this regularity: for example, $u(u - c)$ has integrability $3/2$ and Morrey exponent $\tau/2$ before lowering the two exponents. The bounded cutoff derivatives do not change membership. Since $q > 5/2$, the force term also belongs to $\mathcal{M}^{6/5, \varkappa}$. These facts use only the fixed support and Hölder's inequality, with its volume factors.

For completeness, the spatial Riesz operators preserve this parabolic Morrey bound for the spatially supported sources just used. Split each source into $B_{2r}(y)$ and the spatial annuli of radii $2^k r$, on the time interval of a testing cylinder. The near term is bounded by the $L^{6/5}$ Calderón–Zygmund estimate. For the annuli use the kernel bound $C|x - y|^{-3}$, Hölder in space and time, and the source Morrey bound on the larger cylinders. After normalization, the series is bounded by a constant times

$$\sum_{k \geq 0} 2^{k(5/3 - 5/\varkappa)} \leq \sum_{k \geq 0} 2^{-2k/15} < \infty, \quad \varkappa \leq 25/9.$$

The constants here depend only on $\varkappa$ and the fixed-exponent Calderón–Zygmund constant, not the number of annuli or the source. The Newtonian kernel derivatives used in these estimates include, for $y \neq 0$,

$$4\pi \partial_i \partial_j \partial_l N(y) = -3\delta_{jl} \frac{y_i}{|y|^5} - 3\delta_{ij} \frac{y_l}{|y|^5} - 3\delta_{il} \frac{y_j}{|y|^5} + 15 \frac{y_i y_j y_l}{|y|^7}, \quad (132)$$

obtained by differentiating $N(y) = -(4\pi|y|)^{-1}$.

For the remainder, harmonic interior estimates on the fixed ball and the separated support of the density of p_8 give, for almost every s ,

$$\sup_{B_{R/2}(x)} |H(\cdot, s)| \leq M_R(s), \quad \int_{t-3R^2/4}^{t+R^2/4} M_R(s)^{3/2} ds < \infty.$$

More explicitly one may take, with absolute constants,

$$M_R(s) = CR^{-3} \left(\alpha((x, t + R^2/4), R)^2 + \|p(\cdot, s)\|_{L^{3/2}(B_R(x))} \right) + CR^{-2} \|f(\cdot, s)\|_{L^{3/2}(B_R(x))}.$$

The pressure term is integrated at this fixed scale; finiteness follows from (S1) and $q > 5/2$. Hölder in time shows that on any testing cylinder of radius r the $6/5$ -power integral of the restricted remainder is at most $Cr^{17/5} \|M_R\|_{L^{3/2}}^{6/5}$. For small r this is stronger than the required power $r^{5-6/\varkappa}$, since $\varkappa \leq 25/9$; for large r use the finite integral on its bounded support. Adding the three bounds proves (130). $\square$

We record three elementary facts about parabolic Morrey spaces.

Lemma 5.27 (Hölder, embeddings, and Minkowski). *Let all functions be measurable on $\mathbb{R}^3 \times \mathbb{R}$.*

(i) (Hölder.) *If $g_1 \in \mathcal{M}^{P_1, \tau_1}$, $g_2 \in \mathcal{M}^{P_2, \tau_2}$, $\frac{1}{P} = \frac{1}{P_1} + \frac{1}{P_2}$ and $\frac{1}{\tau} = \frac{1}{\tau_1} + \frac{1}{\tau_2}$, then*

$$\|g_1 g_2\|_{\mathcal{M}^{P, \tau}} \leq \|g_1\|_{\mathcal{M}^{P_1, \tau_1}} \|g_2\|_{\mathcal{M}^{P_2, \tau_2}}. \quad (133)$$

(ii) (Lowering the integrability exponent.) *If $1 \leq P' \leq P \leq \tau$ then*

$$\|g\|_{\mathcal{M}^{P', \tau}} \leq \left(\frac{8\pi}{3}\right)^{\frac{1}{P'} - \frac{1}{P}} \|g\|_{\mathcal{M}^{P, \tau}},$$

with no support hypothesis.

(iii) (Lowering the Morrey exponent.) *If $P \leq \kappa \leq \tau$ and $\text{supp } g \subset E$ with $d_{\text{par}}\text{-diam } E \leq D$, then*

$$\|g\|_{\mathcal{M}^{P, \kappa}} \leq \max\{1, D^{5(\frac{1}{\kappa} - \frac{1}{\tau})}\} \|g\|_{\mathcal{M}^{P, \tau}}. \quad (134)$$

(iv) (Minkowski.) *If $k \in L^1(\mathbb{R}^3)$ and $g \in \mathcal{M}^{P, \tau}$ then $\|\int_{\mathbb{R}^3} k(y) g(\cdot - y, \cdot) dy\|_{\mathcal{M}^{P, \tau}} \leq \|k\|_{L^1} \|g\|_{\mathcal{M}^{P, \tau}}$; the same holds for a convolution in all four space-time variables with $k \in L^1(\mathbb{R}^3 \times \mathbb{R})$.*

Proof. (i) Hölder's inequality on each ball $\mathfrak{B}_r(z)$ with exponents P_1/P and P_2/P , and $5(\frac{1}{P} - \frac{1}{\tau}) = 5(\frac{1}{P_1} - \frac{1}{\tau_1}) + 5(\frac{1}{P_2} - \frac{1}{\tau_2})$.

(ii) Hölder on $\mathfrak{B}_r(z)$ against $|\mathfrak{B}_r|^{\frac{1}{P'} - \frac{1}{P}} = (\frac{8\pi}{3})^{\frac{1}{P'} - \frac{1}{P}} r^{5(\frac{1}{P'} - \frac{1}{P})}$, and $5(\frac{1}{P'} - \frac{1}{P}) + 5(\frac{1}{P} - \frac{1}{\tau}) = 5(\frac{1}{P'} - \frac{1}{\tau})$.

(iii) If $D = 0$, then E contains at most one point and $g = 0$ almost everywhere, so the assertion is immediate. Assume $D > 0$. Let $r > 0$ and z . If $r \leq D$ then $r^{-5(\frac{1}{P} - \frac{1}{\kappa})} = r^{5(\frac{1}{\kappa} - \frac{1}{\tau})} r^{-5(\frac{1}{P} - \frac{1}{\tau})} \leq D^{5(\frac{1}{\kappa} - \frac{1}{\tau})} r^{-5(\frac{1}{P} - \frac{1}{\tau})}$, giving the bound. If $r > D$ then E is contained in the closed ball of radius D centred at any point of E . Apply the Morrey bound on the open ball of radius $D + \delta$ and let $\delta \downarrow 0$ (the empty-set case is immediate). This gives $(\iint_{\mathfrak{B}_r(z)} |g|^P)^{1/P} \leq D^{5(\frac{1}{P} - \frac{1}{\tau})} \|g\|_{\mathcal{M}^{P, \tau}} \leq r^{5(\frac{1}{P} - \frac{1}{\kappa})} D^{5(\frac{1}{\kappa} - \frac{1}{\tau})} \|g\|_{\mathcal{M}^{P, \tau}}$, again as claimed.

(iv) Minkowski's integral inequality in $L^P(\mathfrak{B}_r(z))$ together with the translation invariance of $\|\cdot\|_{\mathcal{M}^{P, \tau}}$:

$$\begin{aligned} & \left(\iint_{\mathfrak{B}_r(z)} \left| \int k(y) g(\cdot - y) dy \right|^P \right)^{1/P} \\ & \leq \int |k(y)| \left(\iint_{\mathfrak{B}_r(z - (y, 0))} |g|^P \right)^{1/P} dy \\ & \leq \|k\|_{L^1} r^{5(\frac{1}{P} - \frac{1}{\tau})} \|g\|_{\mathcal{M}^{P, \tau}} \end{aligned}$$

□

Proposition 5.28 (Step 3: one bootstrap round). *Assume $q > 5/2$ and the conclusions (126) of Step 2. Let $\tau > 5$ be such that $\mathbf{1}_{Q_2} u \in \mathcal{M}^{3, \tau}$ and*

$$\frac{1}{\tau_3} + \frac{1}{\tau} > \frac{2}{5}. \quad (135)$$

Since $\tau_3 = 25/8$, this last condition is equivalent, for $\tau > 0$, to $\tau < 25/2$. It is a hypothesis, not a consequence of $\tau > 5$ alone. Then $\mathbf{1}_{Q_3} u \in \mathcal{M}^{3, \varsigma}$ where

$$\frac{1}{\varsigma} = \frac{1}{\tau} - \varpi, \quad \varpi := \frac{1}{5} - \frac{1}{\tau_2} = \frac{\varepsilon}{5} = \frac{2}{25}. \quad (136)$$

Proof. First suppose $25/3 \leq \tau < 25/2$. We work on a small symmetric parabolic ball $\mathfrak{B}_R(z)$ with $\mathfrak{B}_{2R}(z) \subset Q_2$, and choose a smooth cutoff φ equal to one on $\mathfrak{B}_{R/4}(z)$, supported in $\mathfrak{B}_{R/2}(z)$. Its derivatives are bounded by constants depending only on the fixed radius. Set

$$\frac{1}{\kappa} = \frac{1}{\tau} + \frac{8}{25}, \quad \frac{1}{\theta} = \frac{1}{\tau} + \frac{3}{25}, \quad \frac{1}{\varsigma} = \frac{1}{\tau} - \frac{2}{25}.$$

Then $3/2 < \kappa < 5/2$, $3 < \theta < 5$ and $\varsigma > 5$. Since $\kappa < 5/2 < q$, Lemma 5.26 gives $\mathbf{1}_{\mathfrak{B}_{R/2}(z)} Dp \in \mathcal{M}^{6/5, \kappa}$.

The undifferentiated source. With the notation of Lemma 5.25, put

$$F = (\partial_t \varphi + \Delta \varphi)u - \varphi u \cdot \nabla u + \varphi f - \varphi Dp, \quad G_i = -2(\partial_i \varphi)u.$$

By Morrey Hölder, $\varphi u \cdot \nabla u \in \mathcal{M}^{6/5, \kappa}$ because $5/6 = 1/3 + 1/2$ and $1/\kappa = 1/\tau + 1/\tau_3$. The cutoff-linear terms have this same membership by lowering exponents on their bounded support. The force term does too, since $f \in L_{\text{loc}}^q$, $6/5 < \kappa < q$. Together with the pressure-gradient estimate, this proves $F \in \mathcal{M}^{6/5, \kappa}$.

The differentiated sources. The fields G_i initially belong to $\mathcal{M}^{3, \tau}$. Since $3 < \theta < \tau$ and they have bounded support, Lemma 5.27(iii) yields $G_i \in \mathcal{M}^{3, \theta}$.

The two potential estimates. The localized equation and its causal representation give

$$v = W_+ * F + \sum_i (\partial_i W_+) * G_i, \quad |v| \leq C \left(I_2 |F| + \sum_i I_1 |G_i| \right) \quad \text{almost everywhere.}$$

The second inequality uses only the heat kernel and its spatial derivatives, with the maximum gauge of Definition 5.6. Its constant is absolute and chosen before the sources. The same proof may first use the sum gauge $d_\Sigma = |x - y| + |t - s|^{1/2}$: for $a = 1, 2$,

$$I_a^\Sigma g \leq I_a g \leq 2^{5-a} I_a^\Sigma g.$$

These absolute comparison factors transfer both the pointwise bound and the Morrey estimates to the maximum gauge; the two potentials are not identified literally. Corollary 5.9 supplies both the almost-everywhere finiteness needed for these convolutions and the bounds

$$I_2 |F| \in \mathcal{M}^{P_F, \varsigma}, \quad P_F = \frac{6/5}{1 - 2\kappa/5}, \quad I_1 |G_i| \in \mathcal{M}^{P_G, \varsigma}, \quad P_G = \frac{3}{1 - \theta/5},$$

because

$$\frac{1}{\varsigma} = \frac{1}{\kappa} - \frac{2}{5} = \frac{1}{\theta} - \frac{1}{5} = \frac{1}{\tau} - \frac{2}{25}.$$

Both P_F and P_G are at least 3 and at most ς . Lower their integrability exponents to 3 using Lemma 5.27(ii), and add the finitely many terms. Since $v = u$ on $\mathfrak{B}_{R/4}(z)$, this proves the desired local velocity bound. All constants are finite functions of the incoming bounds and the cutoff radius; no uniform bound independent of the solution's incoming norms is asserted by this proposition.

Choose R from the positive gap $r_2/4 - r_3$ and cover $\overline{Q_3}$ by finitely many such inner balls with their doubled balls in Q_2 . Restriction and the triangle inequality for Morrey norms give the conclusion on Q_3 .

Finally, if $5 < \tau < 25/3$, use the Step 2 velocity exponent $25/3$ instead of τ . The case just proved gives exponent 25. Here $(1/\tau - 2/25)^{-1} < 25$, so lowering the Morrey exponent on the bounded set Q_3 gives exactly (136). $\square$

Corollary 5.29 (The bootstrap terminates after one round). *With $\varepsilon = 2/5$, Proposition 5.28 applies once with $\tau = \tau_2 = 25/3$ and yields $\mathbf{1}_{Q_3}u \in \mathcal{M}^{3,\varsigma}$ with $\varsigma = 25$; and then*

$$\frac{1}{\varsigma} + \frac{1}{\tau_2} = \frac{1}{25} + \frac{3}{25} = \frac{4}{25} < \frac{1}{5}. \quad (137)$$

A second round is not available, since $\frac{1}{\tau_3} + \frac{1}{\varsigma} = \frac{8}{25} + \frac{1}{25} = \frac{9}{25} < \frac{2}{5}$ violates (135).

Proof. Condition (135) for $\tau = \tau_2$ reads $\frac{8}{25} + \frac{3}{25} = \frac{11}{25} > \frac{10}{25}$, which holds. The gain is $\frac{1}{\varsigma} = \frac{3}{25} - \frac{2}{25} = \frac{1}{25}$. The two displayed computations are immediate. $\square$

5.5. Step 4: Hölder continuity.

Theorem 5.30 (Endgame). *Let $q > 5/2$ and let z_0, r_2, M be as in Proposition 5.22 with $\varepsilon = 2/5$. Then u agrees almost everywhere on $Q_3 = \mathfrak{B}_{r_3}(z_0)$, for every $r_3 < r_2/4$, with a function that is Hölder continuous with respect to d_{par} of exponent*

$$\gamma_0 = \min\left\{2 - \frac{5}{q}, \frac{1}{5}\right\} > 0, \quad (138)$$

with norm bounded in terms of q, M, r_2, r_3 and the norms $\|u\|_{L^{10/3}}, \|p\|_{L^{3/2}}, \|f\|_{L^q}$ on $\mathfrak{B}_{r_2}(z_0)$. In particular every point of Q_3 is a regular point of u in the sense of Definition 2.2.

Proof. Choose an intermediate symmetric ball Q'_2 with $\overline{Q_3} \subset Q'_2 \Subset Q_2$. Corollary 5.29, applied with Q'_2 as its inner set, gives $\mathbf{1}_{Q'_2}u \in \mathcal{M}^{3,25}$; Step 2 also gives $\mathbf{1}_{Q'_2}u \in \mathcal{M}^{3,25/3}$. Choose a new cutoff $\varphi \in C_c^\infty(Q'_2)$ with $\varphi = 1$ near $\overline{Q_3}$. Thus every localized source below is supported where the improved velocity exponent is available. Apply Lemma 5.26 on finitely many balls whose doubled balls lie in Q'_2 and whose inner balls cover $\text{supp } \varphi$. Weak-derivative uniqueness identifies the pressure gradients on overlaps. Since $1/25 + 8/25 = 9/25$, the resulting pressure gradient belongs locally to $\mathcal{M}^{6/5, \min\{q, 25/9\}}$. Use the same sources $F = \mathbf{g} - \varphi Dp$ and $G_i = \mathbf{h}_i$ as in the proof of Proposition 5.28. Morrey Hölder and bounded-support lowering now give

$$F \in \mathcal{M}^{6/5, \theta_0}, \quad G_i \in \mathcal{M}^{3, \theta_1}, \quad \theta_0 = \min\{q, 25/9\}, \quad \theta_1 = 25/3.$$

The pressure is retained in F , not converted into a differentiated quadratic source. The local equation therefore has the representation $v = W_+ * (F + \sum_i \partial_i G_i)$. Lower the integrability of the G_i to $6/5$, and, on the fixed bounded supports, lower the Morrey exponents to the values required by Proposition 5.12. The conditions are

$$\gamma \leq 2 - \frac{5}{\theta_0} = \min\{2 - 5/q, 1/5\}, \quad \gamma \leq 1 - \frac{5}{\theta_1} = \frac{2}{5}.$$

They hold for $\gamma = \gamma_0 > 0$ in (138). The resulting single representative agrees almost everywhere with u on Q_3 . On overlapping local balls the continuous representatives coincide, since they agree almost everywhere; a finite cover gives the claimed Hölder representative and norm bound.

For the quantitative bound, fix the cutoff family from the radii before the solution and retain the source-norm bounds in the preceding estimates. The fixed-scale harmonic remainder is controlled by the local energy and pressure bounds; the local energy inequalities bound these quantities in terms of the Step 2 bound and the listed surrounding norms. The Calderón–Zygmund constants occur only at fixed exponents, and the finite cover and cutoff derivatives depend only on the radii. Together with the heat-potential estimate this gives exactly the dependence asserted in the theorem. $\square$

Lemma 5.31 (One-sided pressure-gradient estimate). *Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force f and exponent $q > 5/2$. Write $E_R = B_R(0) \times (-R^2, 0]$. Suppose $\overline{E_R} \subset \mathcal{O}$, $0 < r < R$, and $25/3 \leq \tau \leq 25$. If*

$$\mathbf{1}_{E_R} u \in \mathcal{M}^{3, \tau}, \quad \mathbf{1}_{E_R} \nabla u \in \mathcal{M}^{2, 25/8},$$

then the weak spatial gradient of p has a jointly measurable, locally integrable representative Dp near $\overline{E_r}$, and

$$\mathbf{1}_{E_r} Dp \in \mathcal{M}^{6/5, \varkappa}, \quad \varkappa = \min \left\{ q, \left(\frac{1}{\tau} + \frac{8}{25} \right)^{-1} \right\}.$$

Its Morrey norm is bounded in terms of the two displayed norms, q, τ , the radii, and the following past-time quantities only:

$$A_R = \operatorname{ess\,sup}_{-R^2 < s < 0} \int_{B_R} |u(y, s)|^2 dy, \quad \|p\|_{L^{3/2}(E_R)}, \quad \|f\|_{L^q(E_R)}.$$

No Morrey estimate or quantitative norm at positive times is assumed.

Proof. Choose $r < a < b < R$ and a fixed spatial cutoff equal to one on B_b and supported in B_R . At almost every time $s \in (-a^2, 0)$, apply the spatial pressure decomposition with $c(s) = [u]_{B_R}(s)$ and the singly centred tensor $-u_i(u_j - c_j)$. The differentiated source is

$$V_i = \sum_j [\eta(u_j - c_j) \partial_j u_i + (\partial_j \eta) u_i (u_j - c_j)].$$

Set all sources to zero outside the time interval $(-a^2, 0]$. The spatial operators do not change this time support. On E_r , the same slice weak-derivative identity used in Lemma 5.26 is

$$D_i p = - \sum_j \partial_i \partial_j \Delta^{-1} (\mathbf{1}_{(-a^2, 0]} V_j) + \sum_j \partial_i \partial_j \Delta^{-1} (\mathbf{1}_{(-a^2, 0]} \eta f_j) + H_i.$$

All cutoffs in this identity are spatial; no time derivative of the indicator is taken. The mean satisfies $|c(s)| \leq |B_R|^{-1/2} A_R^{1/2}$ for a.e. past time. Morrey Hölder gives $\mathbf{1}_{(-a^2, 0]} \eta u \nabla u \in \mathcal{M}^{6/5, (1/\tau + 8/25)^{-1}}$. The terms containing c or a cutoff derivative have at least this membership after bounded-support lowering: $u(u - c)$ first has integrability $3/2$ and Morrey exponent $\tau/2$. The restricted force belongs to $\mathcal{M}^{6/5, \varkappa}$ by Hölder and bounded-support lowering from L^q .

For clarity, the Riesz estimate remains valid across the truncation time. On a testing cylinder of radius ℓ , split a source spatially into the ball of radius 2ℓ and dyadic annuli of radii $L = 2^k \ell$, while keeping its time interval unchanged. At exponent $P = 6/5$, the near term follows from the spatial L^P estimate integrated in time. For an annulus the kernel bound and spatial Hölder give

$$\|Tg_k\|_{L^P(B_\ell \times J_\ell)} \leq C \ell^{3/P} L^{-3/P} \|g\|_{L^P(B_{CL} \times J_\ell)} \leq C \ell^{3/P} L^{2/P - 5/\varkappa} \|g\|_{\mathcal{M}^{P, \varkappa}}.$$

Here the last norm is estimated on a parabolic ball of radius CL containing the spatial annulus and the original time interval. After division by $\ell^{5/P - 5/\varkappa}$, the annular factor is $C 2^{k(5/3 - 5/\varkappa)}$. Since $\varkappa \leq 25/9$, it is summable, with exponent at most $-2/15$. The argument applies directly to the zero-extended past sources; it never asks for their future values.

Harmonic interior estimates and separation from the annular force source give a majorant on B_r of the form

$$\sup_{B_r} |H(\cdot, s)| \leq m(s), \quad m(s) \leq C_{r,R} (A_R + \|p(\cdot, s)\|_{L^{3/2}(B_R)} + \|f(\cdot, s)\|_{L^{3/2}(B_R)}).$$

Thus $m \in L^{3/2}(-a^2, 0)$ with a bound depending only on the stated past quantities. For a testing cylinder of radius ℓ , Hölder in time bounds the $6/5$ -power integral of the restricted

remainder by $C\ell^{17/5}\|m\|_{L^{3/2}}^{6/5}$. This is sufficient at small radii, since $17/5 > 5 - 6/\varkappa$; bounded support handles large radii. This proves the asserted Morrey estimate.

To obtain a representative on an open neighbourhood as well, use the same spatial construction on a short time collar of $[-a^2, 0]$ lying inside the solution domain. Local suitability gives $u\nabla u \in L_{\text{loc}}^{5/4}$ there by $u \in L_{\text{loc}}^{10/3}$ and $\nabla u \in L_{\text{loc}}^2$. The spatial $L^{5/4}$ Riesz estimate and the harmonic remainder therefore give a locally integrable gradient on that collar. Spatial mollification and a.e. limits give joint measurability; the slice identities and Fubini give the spacetime weak-derivative identity. Weak-derivative uniqueness identifies it with the past representative already estimated. The collar is used only for this qualitative identity, not for a quantitative Morrey bound. $\square$

Proposition 5.32 (One-sided bootstrap and endgame). *Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force f . Let $E_R = B_R(0) \times (-R^2, 0]$ with $\overline{E_R} \subset \mathcal{O}$, and assume the finite past bounds*

$$\begin{aligned} \mathbf{1}_{E_R} u &\in \mathcal{M}^{3,25/3}, & \mathbf{1}_{E_R} \nabla u &\in \mathcal{M}^{2,25/8}, \\ A_R &< \infty, & p &\in L^{3/2}(E_R), \quad f \in L^q(E_R), \quad q > 5/2. \end{aligned}$$

For every $0 < r < R$ one has $\mathbf{1}_{E_r} u \in \mathcal{M}^{3,25}$. For every $0 < \rho < r$, there is a globally defined parabolically Hölder function $\bar{v}$ of exponent $\gamma_0 = \min\{2 - 5/q, 1/5\} > 0$, agreeing a.e. with u on E_ρ . Its Hölder seminorm and its supremum on $\overline{E_\rho}$ are bounded in terms of the displayed past bounds, q , and R, r, ρ . No conclusion of regularity on the top face in the open-neighbourhood sense is asserted.

Proof. First choose $r < a < b < R$. By Lemma 5.31, $\mathbf{1}_{E_b} Dp \in \mathcal{M}^{6/5,25/11}$, because $1/(25/3) + 8/25 = 11/25$ and $25/11 < q$. Choose a smooth product cutoff φ equal to one near $\overline{E_r}$, whose past support lies in E_a and whose full support lies in the open solution domain and inside the neighbourhood where Dp was constructed. Explicitly, the spatial cutoff is supported in B_a and is one on B_r ; the temporal cutoff vanishes before $-a^2$ and is one on $[-r^2, 0]$, remaining one for a short future interval before vanishing inside the available time collar. All derivatives at times $t \leq 0$ are bounded in terms of the fixed radii. The future cutoff may depend on the collar, but its derivatives occur only at $t > 0$.

Define the full localized sources

$$\begin{aligned} F &= (\partial_t \varphi + \Delta \varphi)u - \varphi u \cdot \nabla u + \varphi f - \varphi Dp, & G_i &= -2(\partial_i \varphi)u, \\ F^- &= \mathbf{1}_{\{t \leq 0\}} F, & G_i^- &= \mathbf{1}_{\{t \leq 0\}} G_i. \end{aligned}$$

The local equation and its adjoint-test proof give the full distributional representation for φu . Splitting the source in that representation, rather than differentiating $\mathbf{1}_{\{t \leq 0\}} \varphi u$, introduces no terminal-time distribution. The kernels W_+ and $\partial_i W_+$ vanish at negative time differences. Hence

$$\varphi u = W_+ * F^- + \sum_i (\partial_i W_+) * G_i^- \quad \text{a.e. on } \{t \leq 0\}.$$

All future sources make zero contribution on this set; the single time slice $t = 0$ has spacetime measure zero.

Morrey Hölder and bounded-support lowering, applied only to the past sources, give

$$F^- \in \mathcal{M}^{6/5,25/11}, \quad G_i^- \in \mathcal{M}^{3,25/6}.$$

For the latter use $25/6 < 25/3$; for the former use the pressure estimate above and $u\nabla u$ with reciprocal Morrey exponent $11/25$. Adams' estimate with potential orders two and one gives respectively

$$I_2|F^-| \in \mathcal{M}^{66/5,25}, \quad I_1|G_i^-| \in \mathcal{M}^{18,25}.$$

Lower the integrability exponents to three. The heat-kernel pointwise bound and the a.e. identity on E_r now prove $\mathbf{1}_{E_r} u \in \mathcal{M}^{3,25}$, with constants depending only on the past bounds and the chosen radii.

For the final step choose new intermediate radii $\rho < a' < b' < r$ and a new cutoff equal to one near $\overline{E_\rho}$ whose past support lies in $E_{a'}$. Its sources are therefore supported where the improved velocity bound is available. Apply Lemma 5.31 with outer radius r and inner radius b' at $\tau = 25$. The resulting past pressure-gradient exponent is $\theta_0 = \min\{q, 25/9\}$. The same source calculation gives

$$F^- \in \mathcal{M}^{6/5, \theta_0}, \quad G_i^- \in \mathcal{M}^{3, 25/3}.$$

Here the nonlinear term uses the improved exponent 25, not $25/3$; all other past bounds on E_r follow by restriction from E_R . Lower the integrability of G_i^- to $6/5$ and their Morrey exponents, on their bounded supports, to those prescribed by Proposition 5.12 for $\gamma = \gamma_0$. The requisite inequalities are

$$\gamma_0 \leq 2 - 5/\theta_0 = \min\{2 - 5/q, 1/5\}, \quad \gamma_0 \leq 1 - 5/(25/3) = 2/5.$$

That proposition gives a global Hölder representative of the potential of these past sources. By causality it agrees a.e. with u on E_ρ . Finally, for $z \in \overline{E_\rho}$ average its Hölder inequality over $y \in E_\rho$ to obtain

$$|\bar{v}(z)| \leq |E_\rho|^{-1/3} \|u\|_{L^3(E_\rho)} + [\bar{v}]_{C^{\gamma_0, \gamma_0/2}} \text{diam}_{d_{\text{par}}}(\overline{E_\rho})^{\gamma_0}.$$

This controls the supremum using past data only and proves the proposition. $\square$

Remark 5.33 (The exponent delivered by this route). With the fixed choice $\varepsilon = 2/5$, the single bootstrap round gives velocity exponent 25, source exponents $\theta_0 = \min\{q, 25/9\}$ and $\theta_1 = 25/3$, and hence $\gamma_0 = \min\{2 - 5/q, 1/5\}$. No optimization over a variable ε is asserted here: the pressure-gradient estimates above were used only in the stated range $25/3 \leq \tau \leq 25$. The ceiling $2/3$ appearing in the blow-up route [8, Lemma 2.4], [21, Remark 6.6] is not attained by this argument; only the positivity of γ_0 is used in Theorems 2.7–2.10.

5.6. Proof of Theorems 2.7 and 2.9.

Proof of Theorem 2.9. Let $\varepsilon_1 = \varepsilon_*$, the absolute constant of Convention 5.18; note that the hypothesis (34) of Theorem 2.9 is stated for β^2 against ε_1^2 , whereas Proposition 5.22 is stated for β against ε_* , so the two thresholds are the same number and no squaring discrepancy arises. If $\limsup_{r \rightarrow 0^+} \beta(z_0, r)^2 < \varepsilon_1^2$ then $\limsup_{r \rightarrow 0^+} \beta(z_0, r) < \varepsilon_*$, so Proposition 5.22 provides r_2 and M , and Lemma 5.23, Proposition 5.28, Corollary 5.29 and Theorem 5.30 show that u agrees a.e. on the open set $\mathfrak{B}_{r_2/8}(z_0)$ with a function that is Hölder continuous of exponent γ_0 with respect to d_{par} . Since $z_0 \in \mathfrak{B}_{r_2/8}(z_0)$, Definition 2.2 is satisfied and z_0 is a regular point. $\square$

Lemma 5.34 (Caccioppoli inequality, γ -form). *Under the hypotheses of Lemma 5.15, and with the same notation,*

$$\alpha(r) + \beta(r) \leq C'_{25} \left[\kappa \gamma(\rho) + \kappa^{-1} \gamma(\rho)^{3/2} + \kappa^{-1} \delta(\rho) \gamma(\rho)^{1/2} \right] + C'_{26} \kappa^{-1/2} \gamma(\rho)^{1/2} \lambda(\rho)^{1/2}, \quad (139)$$

with C'_{25} absolute and $C'_{26} = C'_{26}(q)$.

Proof. Run the proof of Lemma 5.15 verbatim, changing only the bounds of I_1 and I_2 .

I_1 . By Hölder on $Q_\rho(z_0)$ with $1 = \frac{2}{3} + \frac{1}{3}$, $\iint_{Q_\rho(z_0)} |u|^2 \leq |Q_\rho(z_0)|^{1/3} \|u\|_{L^3(Q_\rho(z_0))}^2 = C \rho^{5/3} \rho^{4/3} \gamma(\rho)^2 = C \rho^3 \gamma(\rho)^2$, so $I_1 \leq C r^2 \rho^{-5} \cdot \rho^3 \gamma(\rho)^2 = C \kappa^2 \gamma(\rho)^2$.

I_2 . Instead of the Poincaré step, use the spatial integral bound

$$\int_{B_\rho} ||u|^2 - [|u|^2]_{B_\rho}| |u| \leq 2 \int_{B_\rho} |u|^3,$$

valid at almost every time by Hölder's inequality on the mean term. Thus

$$I_2 \leq Cr^{-2} \iint_{Q_\rho(z_0)} |u|^3 = Cr^{-2} \rho^2 \gamma(\rho)^3 = C\kappa^{-2} \gamma(\rho)^3.$$

The bounds on I_3 and I_4 are unchanged. Adding and taking square roots as in Step 3 of that proof gives (139). $\square$

Lemma 5.35 (Smallness at one scale propagates to all centres). *There is $\varepsilon_0 = \varepsilon_0(q) > 0$ such that: if (u, p) is a suitable weak solution on an open set containing $\overline{Q_1}$ with force $f \in L^q(Q_1)$ and (33) holds, then, with $r_5 := \kappa/4$,*

$$\theta(z, r_5) \leq \eta \quad \text{and} \quad \lambda(z, r_5) \leq \Lambda_0 \quad \text{for every } z \in Q_{3/4}.$$

Proof. Let $z = (x, t) \in Q_{3/4}$, so $|x| < 3/4$ and $t \in (-\frac{9}{16}, 0]$. Then $Q_{1/4}(z) \subset Q_1$: indeed $B_{1/4}(x) \subset B_1(0)$ and $(t - \frac{1}{16}, t] \subset (-\frac{9}{16} - \frac{1}{16}, 0] \subset (-1, 0]$. By (33),

$$\gamma(z, \tfrac{1}{4})^3 = 16 \iint_{Q_{1/4}(z)} |u|^3 \leq 16\varepsilon_0, \quad \delta(z, \tfrac{1}{4})^3 \leq 16\varepsilon_0, \quad \lambda(z, \tfrac{1}{4}) = \left(\tfrac{1}{4}\right)^\sigma \|f\|_{L^q(Q_{1/4}(z))} \leq \varepsilon_0^{1/q}.$$

Apply Lemma 5.34 with $\rho = 1/4$ and $r = r_5 = \kappa/4$, and Proposition 4.11(ii-b). For a.e. time s , write $c(s) = [u]_{B_{1/4}(x)}(s)$. The inequalities $|a - b|^3 \leq 4(|a|^3 + |b|^3)$ and Jensen's inequality give

$$\int_{B_{1/4}(x)} |u(y, s) - c(s)|^3 dy \leq 4 \int_{B_{1/4}(x)} |u(y, s)|^3 dy + 4|B_{1/4}| |c(s)|^3 \leq 8 \int_{B_{1/4}(x)} |u(y, s)|^3 dy.$$

Integrating in time and multiplying by $(1/4)^{-2}$ proves $\widehat{C}(z, \frac{1}{4}) \leq 8\gamma(z, \frac{1}{4})^3$ for the spatial mean used here. Taking cube roots in the pressure estimate gives

$$\delta(z, r_5) \leq C_{32}(q)^{1/3} [2\kappa^{-2/3} \gamma(z, \tfrac{1}{4}) + \kappa^{1/3} \delta(z, \tfrac{1}{4}) + \kappa^{1/2} \lambda(z, \tfrac{1}{4})^{1/2}].$$

Every term on the right of both displays tends to 0 as $\varepsilon_0 \rightarrow 0$, with κ fixed. Choose $\varepsilon_0(q)$ so small that $\theta(z, r_5) = \alpha(z, r_5) + \beta(z, r_5) + \kappa^{-4} \delta(z, r_5)^2 \leq \eta$; and $\lambda(z, r_5) \leq \lambda(z, \frac{1}{4}) \leq \varepsilon_0^{1/q} \leq \Lambda_0$ after shrinking ε_0 once more. $\square$

Proof of Theorem 2.7. Let $\varepsilon_0(q)$ be as in Lemma 5.35 and $r_5 = \kappa/4$. The set of admissible centres is the *one-sided* cylinder $Q_{3/4}$, so the centre-shift device of Lemma 5.23, which moves a centre forward in time by ϱ^2 , is not available as it stands; and $\overline{Q_{1/2}}$ contains its top face, so no cut-off supported in $\{t \leq 0\}$ can equal 1 there. Three modifications repair this.

Step 1: the scale iteration at every centre of $Q_{3/4}$. By Lemma 5.35, $\theta(z, r_5) \leq \eta$ and $\lambda(z, r_5) \leq \Lambda_0$ for every $z \in Q_{3/4}$, with r_5 fixed. Proposition 5.19 therefore gives

$$\max\{\alpha(z, r), \beta(z, r), \delta(z, r)^2\} \leq M r^\varepsilon \quad \text{for all } z \in Q_{3/4}, \ 0 < r \leq r_5, \quad (140)$$

with $M = \kappa^{-4/3-\varepsilon} \eta r_5^{-\varepsilon}$ an absolute constant.

Step 2: a one-sided Morrey transfer. Put $Q_2^\sharp = B_{5/8}(0) \times (-\frac{25}{64}, 0]$, a one-sided cylinder with $\overline{Q_{1/2}} \cap \{t \leq 0\} \subset Q_2^\sharp \subset Q_{3/4}$. We claim that (126) holds with Q_2 replaced by $Q_2^\sharp$. By Lemma 5.3

it suffices to bound the integrals over one-sided cylinders $Q_\varrho(w)$. Since $Q_2^\sharp \subset \{t \leq 0\}$, only $Q_\varrho(w) \cap \{t \leq 0\}$ contributes. If that set meets $Q_2^\sharp$, pick $w' \in Q_\varrho(w) \cap Q_2^\sharp$ and set

$$w'' = (x_{w'}, \min\{t_w + \varrho^2, 0\}),$$

the *truncated* centre shift. Then $w'' \in Q_{3/4}$, because $|x_{w'}| < 5/8 < 3/4$ and $t_{w''} \leq 0$ while $t_{w''} \geq t_{w'} > -\frac{25}{64} > -\frac{9}{16}$. Moreover $Q_\varrho(w) \cap \{t \leq 0\} \subset Q_{2\varrho}(w'')$: for the spatial part, $|x - x_{w''}| \leq |x - x_w| + |x_w - x_{w'}| < 2\varrho$; for the upper time end, $s \leq \min\{t_w, 0\} \leq t_{w''}$; and for the lower end, $s > t_w - \varrho^2 \geq t_{w''} - 3\varrho^2 > t_{w''} - 4\varrho^2$. The computation of Lemma 5.23 now applies with w'' in place of z'' and (140) in place of (123), for $\varrho \leq r_5/2$; the large-radius case is unchanged. This gives $\mathbf{1}_{Q_2^\sharp} u \in \mathcal{M}^{3, \tau_2}$, $\mathbf{1}_{Q_2^\sharp} \nabla u \in \mathcal{M}^{2, \tau_3}$, $\mathbf{1}_{Q_2^\sharp} p \in \mathcal{M}^{3/2, \tau_p}$, with norms bounded in terms of M and r_5 , hence by absolute constants.

Step 3: the one-sided bootstrap and endgame. We apply Proposition 5.32 with $R = 5/8$, $r = 9/16$ and $\rho = 1/2$. The velocity and gradient Morrey hypotheses on $E_R = Q_2^\sharp$ are precisely those proved in Step 2. Its pressure Morrey bound also controls $\|p\|_{L^{3/2}(E_R)}$, and $\|f\|_{L^q(E_R)} \leq \varepsilon_0^{1/q}$ by the smallness assumption. For the remaining past energy bound, choose a fixed $0 < \ell < \min\{r_5, 3/4 - R\}/4$ and cover B_R by finitely many balls $B_\ell(x_j)$ with $x_j \in B_R$. Cover $(-R^2, 0)$ by finitely many time intervals $(t_k - \ell^2, t_k)$ with $-R^2 < t_k \leq 0$. Their cylinders have centres in $Q_{3/4}$, so Step 1 gives

$$\operatorname{ess\,sup}_{t_k - \ell^2 < s < t_k} \int_{B_\ell(x_j)} |u(y, s)|^2 dy \leq \ell \alpha((x_j, t_k), \ell)^2 \leq M^2 \ell^{1+2\varepsilon}.$$

Summing over the finite spatial cover and taking the maximum over the time intervals bounds A_R by a constant depending only on M, r_5 and the fixed radii. Thus every quantitative hypothesis of Proposition 5.32 is controlled by the past data.

The first conclusion of that proposition upgrades the velocity to $\mathbf{1}_{E_{9/16}} u \in \mathcal{M}^{3, 25}$ before its final pressure-gradient estimate and Hölder step are applied. It supplies a single function $\bar{v}$, of exponent $\gamma_0 = \min\{2 - 5/q, 1/5\}$, agreeing a.e. with u on $Q_{1/2}$ and satisfying $[\bar{v}]_{C^{\gamma_0, \gamma_0/2}} \leq C(q)$. The causal source splitting in that proposition requires no positive-time Morrey bound. To control its value as well as its oscillation, put $E = Q_{1/2}$ and average over this same past cylinder:

$$\sup_{z \in \bar{E}} |\bar{v}(z)| \leq |E|^{-1/3} \|u\|_{L^3(E)} + [\bar{v}]_{C^{\gamma_0, \gamma_0/2}} \operatorname{diam}_{d_{\text{par}}}(\bar{E})^{\gamma_0} \leq C\varepsilon_0^{1/3} + C(q).$$

Taking $w = \bar{v}|_{\bar{Q}_{1/2}}$ gives the required bound $\|w\|_{C^{\gamma_0, \gamma_0/2}(\bar{Q}_{1/2})} \leq C_4(q)$. Every point of the open interior has an open neighbourhood within $Q_{1/2}$ and is therefore regular. The argument makes no assertion of open-neighbourhood regularity on its top face. $\square$

6. FROM THE GRADIENT CRITERION TO THE PARTIAL REGULARITY THEOREM

This section proves Theorem 2.10 assuming Theorem 2.9.

Lemma 6.1 ($\mathcal{P}^1$ finite implies Lebesgue null). *Let $E \subset \mathbb{R}^3 \times \mathbb{R}$ with $\mathcal{P}^1(E) < \infty$. Then E has Lebesgue measure zero in $\mathbb{R}^4$.*

Proof. Let $M = \mathcal{P}^1(E)$ and let $\delta \in (0, 1]$. Since $\mathcal{P}_\delta^1(E) \leq \mathcal{P}^1(E) = M$, there are z_j and $r_j \in (0, \delta]$ with $E \subset \bigcup_j \mathfrak{B}_{r_j}(z_j)$ and $\sum_j r_j \leq M + 1$. By (3), $|\mathfrak{B}_r(z)| = |B_r(x)| \cdot 2r^2 = \frac{8\pi}{3} r^5$, so

$$|E| \leq \sum_j |\mathfrak{B}_{r_j}(z_j)| = \frac{8\pi}{3} \sum_j r_j^5 \leq \frac{8\pi}{3} \delta^4 \sum_j r_j \leq \frac{8\pi}{3} \delta^4 (M + 1).$$

Letting $\delta \rightarrow 0^+$ gives $|E| = 0$. Equivalently, the same cover gives zero five-dimensional parabolic Hausdorff measure, which dominates Lebesgue measure up to an absolute normalization. This comparison gives an alternative proof of the lemma. $\square$

Proof of Theorem 2.10. Let $\varepsilon_1 = \varepsilon_1(q)$ be the constant of Theorem 2.9 and set $\varepsilon_{\text{sing}} = \varepsilon_1^2 > 0$.

Step 1: the defect inequality at singular points. Let $z \in S$. By Theorem 2.9 (in contrapositive form), (34) must fail, so

$$\limsup_{r \rightarrow 0^+} \frac{1}{r} \iint_{Q_r(z)} |\nabla u|^2 dz' \geq \varepsilon_{\text{sing}}. \quad (141)$$

Consequently, for every $\tau > 0$ there exists $r \in (0, \tau)$ with $Q_r(z) \subset \mathcal{O}$ and

$$\iint_{Q_r(z)} |\nabla u|^2 dz' > \frac{1}{2} \varepsilon_{\text{sing}} r. \quad (142)$$

Step 2: reduction to relatively compact sets. Equivalently, one can cover the domain by countably many relatively compact local boxes, run the following argument on each closed smaller box, and use countable subadditivity. This uses only second countability and local finiteness of the Dirichlet energy. We give the compact-exhaustion version for comparison. $\mathcal{O}$ is open in $\mathbb{R}^4$, so there are compact sets $K_1 \subset K_2 \subset \dots$ with $\bigcup_n K_n = \mathcal{O}$ and $K_n \subset \text{int } K_{n+1}$; for instance $K_n = \{z \in \mathcal{O} : |z| \leq n, \text{dist}(z, \mathbb{R}^4 \setminus \mathcal{O}) \geq 1/n\}$ when $\mathcal{O} \neq \mathbb{R}^4$, and $K_n = \{|z| \leq n\}$ otherwise. Since $\mathcal{P}^1$ is an outer measure (External Input A.8) and $S = \bigcup_n (S \cap K_n)$, it suffices to prove $\mathcal{P}^1(S \cap K_n) = 0$ for each n . Fix n , write $K = K_n$ and $V_0 = \text{int } K_{n+1}$, so that $K \subset V_0$, $\overline{V_0} \subset K_{n+2} \subset \mathcal{O}$ is compact, and by (S1)

$$\mathcal{E}_0 := \iint_{V_0} |\nabla u|^2 dz' < \infty. \quad (143)$$

By Lemma 2.3, S is relatively closed in $\mathcal{O}$, so $S \cap K$ is compact.

Step 3: the covering. Let V be any open set with $S \cap K \subset V \subset V_0$, and let $\delta > 0$. For each $z \in S \cap K$ choose, by Step 1, a radius $r_z \in (0, \delta)$ such that $\mathfrak{B}_{r_z}(z) \subset V$ and (142) holds with $r = r_z$; this is possible because V is open and $z \in V$, so $\mathfrak{B}_r(z) \subset V$ for all small r . The family $\mathcal{F} = \{\mathfrak{B}_{r_z}(z) : z \in S \cap K\}$ consists of balls of the metric space $(\mathbb{R}^4, d_{\text{par}})$ with radii bounded by δ . That space is separable: $\mathbb{Q}^4$ is a countable d_{par} -dense subset, because $d_{\text{par}} \leq |x - x'| + |t - t'|^{1/2}$ and both terms can be made small by a rational approximation. By External Input A.7 there is a countable subfamily $\{\mathfrak{B}_{r_j}(z_j)\}_{j \in J}$, $J \subset \mathbb{N}$, $r_j = r_{z_j}$, which is pairwise disjoint and such that every member of $\mathcal{F}$ is contained in $\mathfrak{B}_{5r_j}(z_j)$ for some j . Since each $z \in S \cap K$ lies in $\mathfrak{B}_{r_z}(z) \in \mathcal{F}$,

$$S \cap K \subset \bigcup_{j \in J} \mathfrak{B}_{5r_j}(z_j), \quad 5r_j < 5\delta. \quad (144)$$

Step 4: the measure estimate. By Lemma 1.3, $Q_{r_j}(z_j) \subset \mathfrak{B}_{r_j}(z_j) \subset V$, and the cylinders $Q_{r_j}(z_j)$ are pairwise disjoint because the balls $\mathfrak{B}_{r_j}(z_j)$ are. Hence, by (142),

$$\sum_j r_j \leq \frac{2}{\varepsilon_{\text{sing}}} \sum_j \iint_{Q_{r_j}(z_j)} |\nabla u|^2 \leq \frac{2}{\varepsilon_{\text{sing}}} \iint_V |\nabla u|^2, \quad (145)$$

and therefore, by (144) and Definition 2.4,

$$\mathcal{P}_{5\delta}^1(S \cap K) \leq \sum_j 5r_j \leq \frac{10}{\varepsilon_{\text{sing}}} \iint_V |\nabla u|^2. \quad (146)$$

Step 5: $S \cap K$ is Lebesgue null. Apply (146) with $V = V_0$: for every $\delta > 0$, $\mathcal{P}_{5\delta}^1(S \cap K) \leq 10\varepsilon_{\text{sing}}^{-1}\mathcal{E}_0$, whence $\mathcal{P}^1(S \cap K) = \lim_{\delta \rightarrow 0} \mathcal{P}_{5\delta}^1(S \cap K) \leq 10\varepsilon_{\text{sing}}^{-1}\mathcal{E}_0 < \infty$ by (143). By Lemma 6.1, $|S \cap K| = 0$.

Step 6: conclusion. Let $\varepsilon > 0$. Since $|\nabla u|^2 \in L^1(V_0)$ by (143), absolute continuity of the integral gives $\tau > 0$ such that $\iint_W |\nabla u|^2 < \varepsilon$ for every measurable $W \subset V_0$ with $|W| < \tau$. By Step 5 and outer regularity of Lebesgue measure there is an open $V' \supset S \cap K$ with $|V'| < \tau$; replacing V' by $V = V' \cap V_0$, which is open and still contains $S \cap K$, we get $|V| < \tau$ and hence $\iint_V |\nabla u|^2 < \varepsilon$. Now (146) gives $\mathcal{P}_{5\delta}^1(S \cap K) \leq 10\varepsilon_{\text{sing}}^{-1}\varepsilon$ for every $\delta > 0$, so $\mathcal{P}^1(S \cap K) \leq 10\varepsilon_{\text{sing}}^{-1}\varepsilon$. As $\varepsilon > 0$ was arbitrary, $\mathcal{P}^1(S \cap K) = 0$. By Step 2, $\mathcal{P}^1(S) \leq \sum_n \mathcal{P}^1(S \cap K_n) = 0$. $\square$

Remark 6.2 (Why parabolic balls rather than cylinders in the covering). The Vitali lemma with the $5r$ enlargement is a statement about balls of a metric space, and the parabolic cylinders $Q_r(z)$ are not balls of d_{par} . Two intersecting backward cylinders need not be contained in one another's enlargements about the original top centres: a point later than one top time remains outside every cylinder with that top time. We therefore run the covering with the genuine d_{par} -balls $\mathfrak{B}_r(z)$, and recover the statement about cylinders through Lemma 2.6. The defect inequality (142) is still stated on cylinders, which is where the local energy inequality lives, and the inclusion $Q_r(z) \subset \mathfrak{B}_r(z)$ of Lemma 1.3 is what links the two. An alternative used in [21, §6.2, pp. 96–97] uses the shifted cylinders $Q^*(z, r) = B_r(x) \times (t - 7r^2/8, t + r^2/8)$, which contain both z and $Q_{r/2}(z)$.

APPENDIX A. EXTERNAL INPUTS

The analytic inputs and comparison results are collected below with their precise scope. Several inputs are proved internally in the formal development; their designation here does not mean that they are additional axioms. The Campanato comparison and Aubin–Lions compactness entries are not used by the direct A/B/C proof route. Constants named in this appendix are absolute unless a dependence is displayed.

External input A.1 (Calderón–Zygmund bounds for the Riesz transforms). *For $i \in \{1, 2, 3\}$ let R_i denote the i -th Riesz transform on $\mathbb{R}^3$, that is, for the Fourier convention with kernel $e^{-ix \cdot \xi}$, the multiplier operator with symbol $-i\xi_i/|\xi|$; equivalently $R_i = -\partial_i(-\Delta)^{-1/2}$, so that $R_i R_j = \partial_i \partial_j (-\Delta)^{-1}$. For every $s \in (1, \infty)$ there is a constant $C_{11} = C_{11}(s)$ such that*

$$\|R_i R_j g\|_{L^s(\mathbb{R}^3)} \leq C_{11}(s) \|g\|_{L^s(\mathbb{R}^3)} \quad \text{for all } g \in L^s(\mathbb{R}^3) \text{ and all } i, j.$$

For the pressure itself we write $C_{11} = C_{11}(3/2)$; the weak-gradient route also uses the fixed exponent $6/5$. These fixed-exponent constants are absolute. The operators are the bounded L^s extensions of the second Riesz transforms from a dense class, not absolutely convergent global integrals of the Hessian kernel. Literal kernel formulas are used only off the source support, or with their proper singular-integral interpretation. Interpolation is first performed on $L^2 \cap L^s$ and then passed to the completed L^s operator. Reference: [18, Ch. II–III].

External input A.2 (Hardy–Littlewood–Sobolev for the Riesz potential of order 1). *Let $N(y) = -\frac{1}{4\pi|y|}$, so that $|\nabla N(y)| = \frac{1}{4\pi|y|^2}$. There is an absolute finite constant C'_{11} such that, for every $g \in L^{5/2}(\mathbb{R}^3)$ and every $j \in \{1, 2, 3\}$, the literal convolution $\partial_j N * g$ is absolutely convergent for almost every x and*

$$\|\partial_j N * g\|_{L^{15}(\mathbb{R}^3)} \leq C'_{11} \|g\|_{L^{5/2}(\mathbb{R}^3)}.$$

This is the specialization $1/15 = 1/(5/2) - 1/3$ needed here; the full Hardy–Littlewood–Sobolev exponent family is not part of this input. No pointwise measurability or compact-support assumption is added to the $L^{5/2}$ datum. Reference: [18, Ch. V, Thm. 1].

External input A.3 (Newtonian representation of a compactly supported solution). Let $N(y) = -\frac{1}{4\pi|y|}$, so that $\Delta N = \delta_0$ in $\mathcal{D}'(\mathbb{R}^3)$. Let $v \in L^{3/2}(\mathbb{R}^3)$ have compact support and suppose that in $\mathcal{D}'(\mathbb{R}^3)$

$$\Delta v = \partial_i \partial_j G_{ij} + \partial_j H_j + K$$

where $G \in L^{3/2}(\mathbb{R}^3; \mathbb{R}^{3 \times 3})$, $H \in L^m(\mathbb{R}^3; \mathbb{R}^3)$ for some $m \in (1, 3)$, and $K \in L^{m'}(\mathbb{R}^3)$ for some $m' \in (1, 3/2)$, all with compact support. Then

$$v = -R_i R_j G_{ij} + \partial_j N * H_j + N * K \quad \text{almost everywhere in } \mathbb{R}^3.$$

Reference: [6, Ch. 2 and Ch. 4] for the Newtonian potential and [18, Ch. III, V] for the identification $\partial_i \partial_j N * G_{ij} = -R_i R_j G_{ij}$; the uniqueness statement is Liouville's theorem applied to the difference, which is harmonic on $\mathbb{R}^3$ and tends to 0 in the $L^{3/2}$ -average sense at infinity.

External input A.4 (Sobolev embeddings).

- (i) There is an absolute constant such that $\|g\|_{L^6(\mathbb{R}^3)} \leq C \|\nabla g\|_{L^2(\mathbb{R}^3)}$ for all $g \in H^1(\mathbb{R}^3)$; consequently $\|g\|_{L^6(\mathbb{R}^3)} \leq C \|g\|_{H^1(\mathbb{R}^3)}$.
- (ii) There is an absolute constant C such that for every Euclidean ball $B_r(x)$, $r > 0$, and every $g \in H^1(B_r(x))$,

$$\|g\|_{L^6(B_r(x))} \leq C(\|\nabla g\|_{L^2(B_r(x))} + r^{-1} \|g\|_{L^2(B_r(x))}).$$

In particular the unscaled H^1 estimate has an absolute constant on B_1 . No general Lipschitz-domain extension theorem is required here.

- (iii) There is an absolute constant with $\|g\|_{L^\infty(\mathbb{R}^3)} \leq C \|g\|_{H^2(\mathbb{R}^3)}$ for all $g \in H^2(\mathbb{R}^3)$.
- (iv) Let J be an interval and $U = B_r(x)$, $r > 0$. If $g \in L^\infty(J; L^2(U)) \cap L^2(J; H^1(U))$, then $g \in L^{10/3}(U \times J)$ and, with one absolute constant C ,

$$\|g\|_{L^{10/3}(U \times J)}^{10/3} \leq C(\|g\|_{L^\infty(J; L^2(U))}^{4/3} \|\nabla g\|_{L^2(U \times J)}^2 + r^{-2} \|g\|_{L^\infty(J; L^2(U))}^{10/3} |J|).$$

For unbounded J , the membership conclusion is asserted separately; an infinite right side is not a proof of membership. The displayed bound is (44) integrated in time with the slice L^2 mass bounded by the essential supremum. No embedding on an arbitrary bounded open set, or sharper public actual-mass estimate, is asserted here.

References: [6, Ch. 7], [18, Ch. V].

External input A.5 (Poincaré inequalities on the unit ball).

- (i) There is an absolute constant with $\|g - [g]_{B_1}\|_{L^1(B_1)} \leq C \|\nabla g\|_{L^1(B_1)}$ for all $g \in W^{1,1}(B_1)$.
- (ii) There is an absolute constant with $\|g - [g]_{B_1}\|_{L^6(B_1)} \leq C \|\nabla g\|_{L^2(B_1)}$ for all $g \in H^1(B_1)$.
- (iii) Additional classical statement. There is an absolute constant with $\|g - [g]_{B_1}\|_{L^{3/2}(B_1)} \leq C \|\nabla g\|_{L^1(B_1)}$ for all $g \in W^{1,1}(B_1)$. Rescaling gives the same coefficient on any Euclidean ball, since both sides have scaling degree 2. This arbitrary $W^{1,1}$ version is recorded as classical paper material, not as an available formal export; the formal scalar estimate presently treats C^1 data.

Reference: [6, Ch. 7].

The formalized Caccioppoli application uses a separate squared-velocity estimate: for $0 < r < R$, $B_R(x_0) \subset U$ with U open, and $u \in H^1(U; \mathbb{R}^3)$ with its weak gradient,

$$\| |u|^2 - [u]^2 \|_{B_r(x_0)} \|_{L^{3/2}(B_r(x_0))} \leq C \|u\|_{L^2(B_r(x_0))} \|\nabla u\|_{L^2(B_r(x_0))}.$$

The constant is absolute. Approximation on nested balls proves this particular weak-slice estimate; it does not by itself export clause (iii) for arbitrary $W^{1,1}$ functions.

External input A.6 (Interior estimates for harmonic functions). *Let h be harmonic on an open $U \subset \mathbb{R}^3$.*

- (i) (Mean value property.) *If $\overline{B_s(x)} \subset U$ then $h(x) = |B_s(x)|^{-1} \int_{B_s(x)} h$.*
- (ii) (Gradient estimate.) *If $\overline{B_s(x)} \subset U$ then $|\nabla h(x)| \leq \frac{3}{s} \sup_{B_s(x)} |h|$.*

Reference: [6, Ch. 2].

External input A.7 (Vitali covering lemma, $5r$ form). *Let (X, d) be a separable metric space and let $\mathcal{F}$ be a family of open balls $\mathfrak{B}_{r_a}(z_a)$, $a \in \mathcal{I}$, with $\sup_a r_a < \infty$. Then there is a countable set $\mathcal{J} \subset \mathcal{I}$ such that the balls $\{\mathfrak{B}_{r_j}(z_j)\}_{j \in \mathcal{J}}$ are pairwise disjoint and*

$$\bigcup_{a \in \mathcal{I}} \mathfrak{B}_{r_a}(z_a) \subset \bigcup_{j \in \mathcal{J}} \mathfrak{B}_{5r_j}(z_j).$$

In particular every $\mathfrak{B}_{r_a}(z_a)$ is contained in some $\mathfrak{B}_{5r_j}(z_j)$. Separability is needed for the subfamily to be countable: without it the standard Zorn's-lemma argument produces only a maximal disjoint subfamily, which may be uncountable. The open-ball version follows directly from the usual $5r$ argument. Choose $R > \sup_a r_a$ and successively choose a maximal disjoint family among balls of radii in $(2^{-n-1}R, 2^{-n}R]$ that avoid the earlier choices. Every unchosen ball meets a chosen ball of radius at least half its own, and is therefore contained in that ball's fivefold enlargement. The chosen balls are countable because each contains a distinct point of a fixed countable dense set. This also proves the individual containment assertion. For the covering argument in its standard form see [13, Thm. 2.1].

External input A.8 (Hausdorff measures of a metric space). *Let (X, d) be a metric space and $\alpha \geq 0$. The set functions $\mathcal{P}_\delta^\alpha$ and $\mathcal{P}^\alpha$ of Definition 2.4 are outer measures on X ; in particular $\mathcal{P}^\alpha$ is countably subadditive and monotone, and $\mathcal{P}^\alpha(E) = \lim_{\delta \rightarrow 0^+} \mathcal{P}_\delta^\alpha(E) = \sup_{\delta > 0} \mathcal{P}_\delta^\alpha(E)$, the family $\delta \mapsto \mathcal{P}_\delta^\alpha(E)$ being nonincreasing in δ . Reference: [13, Ch. 4].*

External input A.9 (Parabolic Campanato characterization of Hölder continuity). *Let $\alpha \in (0, 1)$, $R > 0$, $z_0 \in \mathbb{R}^3 \times \mathbb{R}$, and $w \in L^3(Q_{2R}(z_0); \mathbb{R}^m)$. Suppose*

$$\Lambda^3 := \sup \left\{ \frac{1}{r^{5+3\alpha}} \iint_{Q_r(z)} |w - (w)_{Q_r(z)}|^3 dz' : z \in Q_R(z_0), 0 < r \leq R \right\} < \infty.$$

Then w agrees almost everywhere on $Q_R(z_0)$ with a function $\bar{w} \in C^{\alpha, \alpha/2}(\overline{Q_R(z_0)}; \mathbb{R}^m)$, and there is a constant $C(\alpha)$ with

$$[\bar{w}]_{C^{\alpha, \alpha/2}(Q_R(z_0))} \leq C(\alpha) \Lambda, \quad \|\bar{w}\|_{L^\infty(Q_R(z_0))} \leq C(\alpha) \left(\Lambda R^\alpha + R^{-5/3} \|w\|_{L^3(Q_{2R}(z_0))} \right).$$

The mean $(w)_{Q_r(z)}$ is the genuine space-time average (5). Note that the normalization is $r^{-5-3\alpha}$, so that in terms of the excess (30) the hypothesis reads $\tilde{C}(z, r)^{1/3} \leq \Lambda r^{1+\alpha}$. Reference: [9, §4.2], [5, §III.1] for the elliptic model case, and [21, after (6.24)] for the form used here. This input is recorded for comparison with the blow-up route only: Section 5 uses instead the self-contained Lemma 5.11, and the exponent it delivers is $\gamma_0 = \min\{2 - 5/q, 1/5\}$ (Remark 5.33).

External input A.10 (Heat kernel). *Let $G(x, \tau) = (4\pi\tau)^{-3/2} \exp(-|x|^2/(4\tau))$ for $x \in \mathbb{R}^3$, $\tau > 0$. Then $G \in C^\infty(\mathbb{R}^3 \times (0, \infty))$, $G > 0$, $\int_{\mathbb{R}^3} G(x, \tau) dx = 1$ for every $\tau > 0$, and $\partial_\tau G = \Delta G$ on $\mathbb{R}^3 \times (0, \infty)$. Consequently, for $r > 0$, the function $\psi_r(x, t) = r^2 G(x, r^2 - t)$ is smooth and positive on $\mathbb{R}^3 \times (-\infty, r^2)$ and satisfies $\partial_t \psi_r + \Delta \psi_r = 0$ there. Reference: [21, Ch. 2].*

External input A.11 (Heat kernel: size and derivative bounds). *Let*

$$W_+(x, t) = \mathbf{1}_{t>0} (4\pi t)^{-3/2} e^{-|x|^2/(4t)}.$$

Then $W_+ \in L^1_{loc}(\mathbb{R}^3 \times \mathbb{R})$, $\text{supp } W_+ \subset \{t \geq 0\}$, $(\partial_t - \Delta)W_+ = \delta_{(0,0)}$ in $\mathcal{D}'(\mathbb{R}^3 \times \mathbb{R})$, and convolution with a compactly supported distribution is well-defined. Causal uniqueness, if used, must be restricted to tempered distributions: a tempered d supported in $\{t \geq c\}$ and satisfying $(\partial_t - \Delta)d = 0$ is zero. No such assertion holds for arbitrary distributions, and half-space support does not define a universal convolution algebra. The proof below uses the adjoint-test identity in Lemma 5.25, not this uniqueness assertion. There is an absolute constant C such that, with $\varrho = \max\{|x|, |t|^{1/2}\}$, for $t > 0$,

$$|W_+| \leq C\varrho^{-3}, \quad |\nabla_x W_+| \leq C\varrho^{-4}, \quad |\partial_t W_+| \leq C\varrho^{-5},$$

and, for every Fourier multiplier $\varsigma(D)$ with ς smooth on $\mathbb{R}^3 \setminus \{0\}$ and homogeneous of degree 1,

$$|\varsigma(D)W_+| \leq C_\varsigma \varrho^{-4}, \quad |\nabla_x \varsigma(D)W_+| \leq C_\varsigma \varrho^{-5}, \quad |\partial_t \varsigma(D)W_+| \leq C_\varsigma \varrho^{-6}.$$

The causal kernels vanish for $t < 0$. Their size estimates hold almost everywhere on spacetime, but no time derivative at $t = 0$ is asserted for the general multiplier kernels. Reference: [9, §IV.11], [11, Ch. 5].

External input A.12 (Aubin–Lions compactness). *Let $X \subset\subset Y \subset Z$ be Banach spaces with X reflexive, the embedding $X \hookrightarrow Y$ compact and $Y \hookrightarrow Z$ continuous. Let J be a bounded interval, $1 < s_0 \leq \infty$, $1 < s_1 < \infty$, and let (g_k) be a sequence with $\sup_k (\|g_k\|_{L^{s_0}(J; X)} + \|\partial_t g_k\|_{L^{s_1}(J; Z)}) < \infty$. Then (g_k) is relatively compact in $L^{s_0}(J; Y)$ (and in $C(\bar{J}; Y)$ if $s_0 = \infty$ and $s_1 > 1$). For comparison with a compactness proof one may take $X = H^1(B)$, $Y = L^2(B)$, $Z = Z_B$ of (6), $s_0 = 2$, $s_1 = 3/2$. This input is not consumed in the direct proof here. Reference: [17, Corollary 4], including the time-continuity endpoint.*

External input A.13 (Pettis measurability theorem). *Let X be a separable Banach space, $J \subset \mathbb{R}$ an interval, and $w : J \rightarrow X$ such that $t \mapsto \langle w(t), \varphi \rangle$ is Lebesgue measurable for every $\varphi \in X^*$. Then w is strongly (Bochner) measurable, i.e. an a.e. limit of countably valued measurable functions. In particular, if X is a separable Hilbert space or the dual of a separable Hilbert space, weak measurability implies strong measurability. Reference: [4, Ch. II].*

External input A.14 (Vector-valued du Bois-Reymond lemma). *Let X be a Banach space, $J \subset \mathbb{R}$ an open interval and $w \in L^1_{loc}(J; X)$ with $\int_J w(t) \theta'(t) dt = 0$ in X for every $\theta \in C^\infty_c(J)$. Then there is $\xi \in X$ with $w(t) = \xi$ for almost every $t \in J$. Reference: [20, Ch. III, Lemma 1.1].*

External input A.15 (Mollification and density). *Let $\zeta \in C^\infty_c(B_1)$ with $\zeta \geq 0$ and $\int_{\mathbb{R}^3} \zeta = 1$, and $\zeta_\varepsilon(y) = \varepsilon^{-3} \zeta(y/\varepsilon)$. For $g \in L^s(\mathbb{R}^3)$, $1 \leq s < \infty$, one has $g * \zeta_\varepsilon \in C^\infty(\mathbb{R}^3)$, $\|g * \zeta_\varepsilon\|_{L^s} \leq \|g\|_{L^s}$ and $g * \zeta_\varepsilon \rightarrow g$ in $L^s(\mathbb{R}^3)$; the analogous statements hold in one time variable. Moreover $C^\infty_c(U)$ is dense in $H^2_0(U)$ by definition of the latter, and $C^\infty_c(U)$ is dense in $L^s(U)$ for $1 \leq s < \infty$ and $U \subset \mathbb{R}^3$ open. Reference: [6, Ch. 7].*

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